



2F-85

2F-140

Original Notice

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Robotiq 2F-85 & 2F-140
for CB-Series Universal Robots



Instruction Manual

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Revisions

Robotiq may modify this product without notice, when necessary, due to product improvements, modifications or changes in specifications. If such modification is made, the manual will also be revised, see revision information. See the latest version of this manual online at support.robotiq.com.

Revision 2018/11/07

New major revision of the Gripper hardware

- All external visual supports changed to reflect the hardware changes
- Updated Section 1
- Updated Section 3
- Updated Section 4
- Updated Section 6
- Updated Section 7
- Updated Section 8
- Updated Section 9

Revision 2018/09/11

- Updated Section 3 according to the robot used (CB-Series vs. e-Series)
- Updated Section 4
 - Input & Output Registers
 - Control over Universal Robots depending on the robot software used (PolyScope 3.6 and later version vs. PolyScope 5.0 and later versions)
 - Added LED and move functions to the list of available functions
- Updated Section 8 with new spare parts, kits and accessories

Revision 2018/05/23

- Major update following the release of UCG-1.2.0
 - New subsections in Section 4: Control
 - Gripper Dashboard
 - Gripper Calibration menu and Calibration wizard for object validation

Revision 2017/06/06

Added section:

- 4.8.2.1 Multiple Grippers

Revised sections:

- 3.8.1 Installation procedure for URCaps
- 4.8.1 Gripper Toolbar



Revision 2016/07/04

Major revision : Updated for URCaps release Section added :

- 3.8 URCaps Package
- 3.9 UR Package without URCaps
- 4.8 Control over Universal Robots with URCaps
- 4.9 Control over Universal Robots without URCaps

Revision 2015/09/15

Major revision : Updated for 2-Finger 140

Revision 2015/07/21

Section added :

- 4.5 Picking features : Force control, re-grasp and object detection.
- A. Harmonized standards, declarations and certificates

Minor modifications :

- Section 1. General Presentation
- Section 3.7 Universal Robots package

Revision 2014/11/05

Modification for Robotiq 2-Finger 85 Adaptive Robot Gripper version 3

Revision 2014/07/22

Modification for use on Robotiq Universal Controller

Minor modifications : User Interface section, maintenance section

Revision 2013/02/06

Section added : Couplings ISO models and Baxter robots Minor modifications

Revision 2013/02/06

Section added : Communication with UR robots Minor modifications

Revision 2012/10/18

Official release

Revision 2012/03/02

Beta release



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1. General Presentation

The terms "Gripper", "Adaptive Gripper", "Robotiq Gripper", "Robotiq Adaptive Gripper", "2-Finger 85", "2-Finger 140", "2F-85" and "2F-140" used in the following manual all refer to the Robotiq 2-Finger Adaptive Robot Gripper. The Robotiq 2-Finger Adaptive Gripper has two versions, 85 and 140. The 2-Finger version will change finger opening dimensions, which will be 85 mm (2F-85) or 140 mm (2F-140). Both versions use the same base, installation and control will be exactly the same. The 2-Finger Gripper is a robotic peripheral that is designed for industrial applications. Its design makes it a unique robotic end-of-arm tool to quickly pick, place and handle a large range of objects of varying sizes and shapes.

Info

Unless specified, information in this manual applies to both the 85 and the 140 mm version of the 2-Finger Adaptive Robot Gripper.

Info

The following manual uses the metric system, unless specified, **all dimensions are in millimeters**.

Info

The following section presents the key features of the grasp-type gripper and must not be considered as appropriate to the gripper operation, each feature is detailed in the appropriate section of the manual. Safety guidelines must be read and understood before any operation is attempted with the grasp-type gripper.



1.1. Gripper nomenclature

The 2-Finger Gripper has two articulated fingers that each have two joints (two phalanxes per finger), as shown in the figure below. The grasp-type gripper can engage up to five points of contact with an object (two on each of the phalanges plus the palm). The fingers are under-actuated, meaning they have fewer motors than the total number of joints. This configuration allows the fingers to automatically adapt to the shape of the object they grasp and it also simplifies the control of the grasp-type gripper.

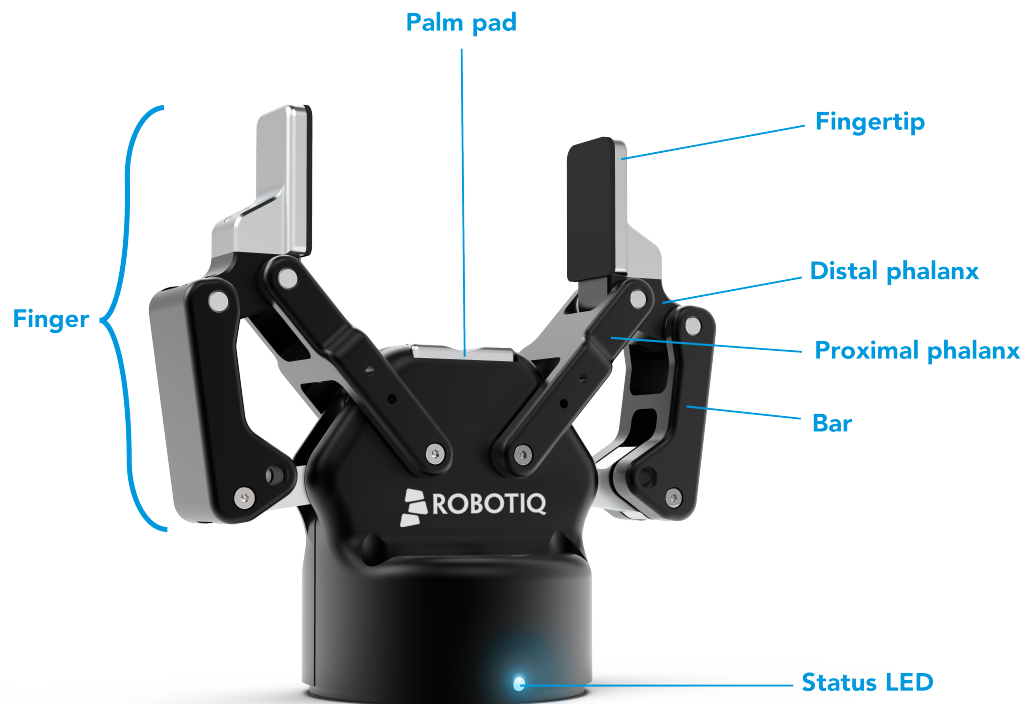


Fig. 1-1: Robotiq 2-Finger Adaptive Gripper.

Please refer to the for details on standard and optional parts.

The status LED presented in the figure above will be :

- solid blue/red when booting
- solid blue when powered with no errors (while communication is active)
- solid red if minor fault occurs, see status details in the **Control** section.
- blinking red/blue if major fault occurs, see status details in the **Control** section.

1.2. 2F-85 vs. 2F-140

The 2-Finger Gripper comes with either 85 mm opening (2-Finger 85) or 140 mm opening (2-Finger 140) according to the figure below. The chassis will remain the same, only the fingers will change. Please refer to the **Mechanical Installation** section for installation instructions. Finger kits are available in the Spare Parts and Accessories section.

Info

Details on the 2-Finger 85 and 2-Finger 140 (dimensions and specifications) can be found in the **Specifications** section.

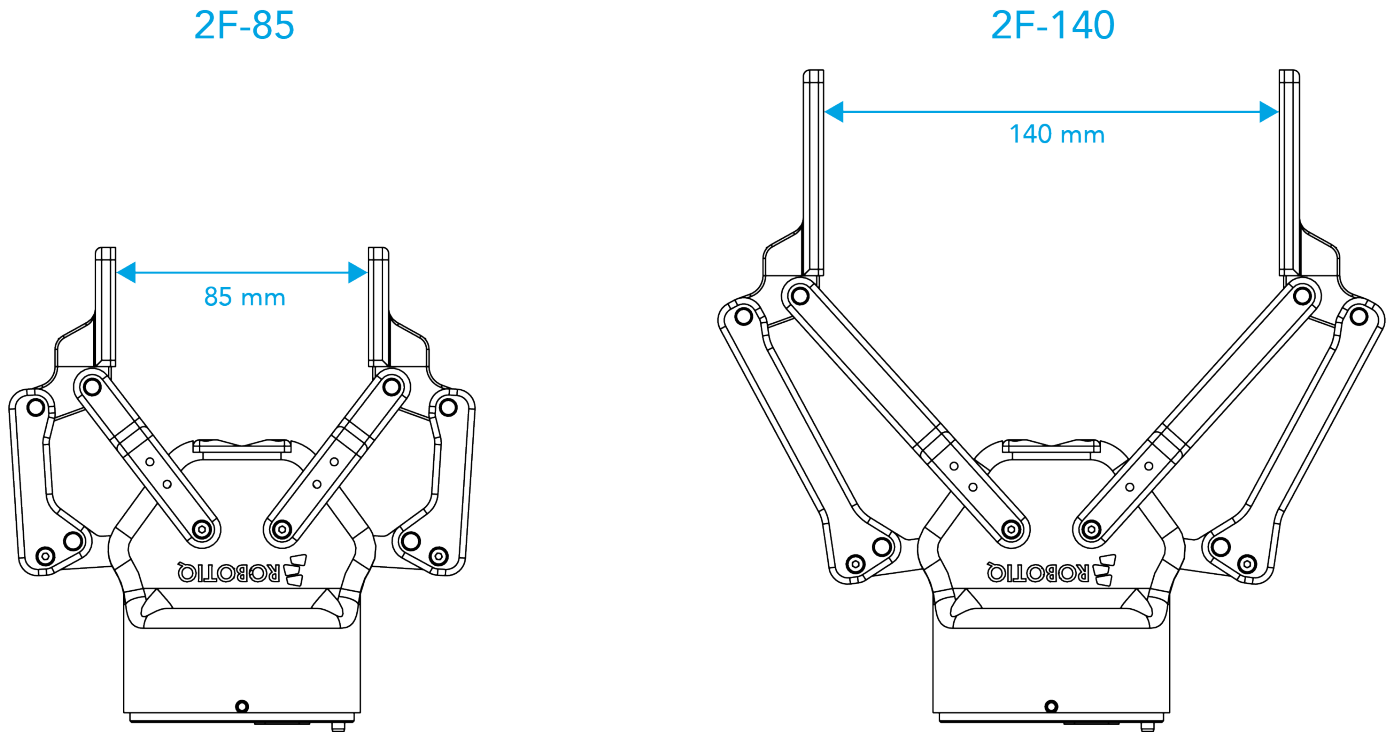


Fig. 1-2: The 2-Finger 85 and 140 mm versions.



1.3. Object picking

The 2-Finger Gripper has a single actuator for opening and closing the fingers, the fingers automatically adapt to the shape of the object manipulated.

Fingers will adopt either a parallel grasp or encompassing grasp as shown in the figure below.

Info

Closing or opening is done via the "Go to requested position" command and is input to the Gripper **Whether the fingers close to produce an encompassing or fingertip grasp is decided at the Gripper level automatically.** It will depend on:

- The objects's geometry;
- The relative position of the object with respect to the Gripper.

In other words, picking the same object could result in either an encompassing or fingertip grasp based on an object's position and geometry.

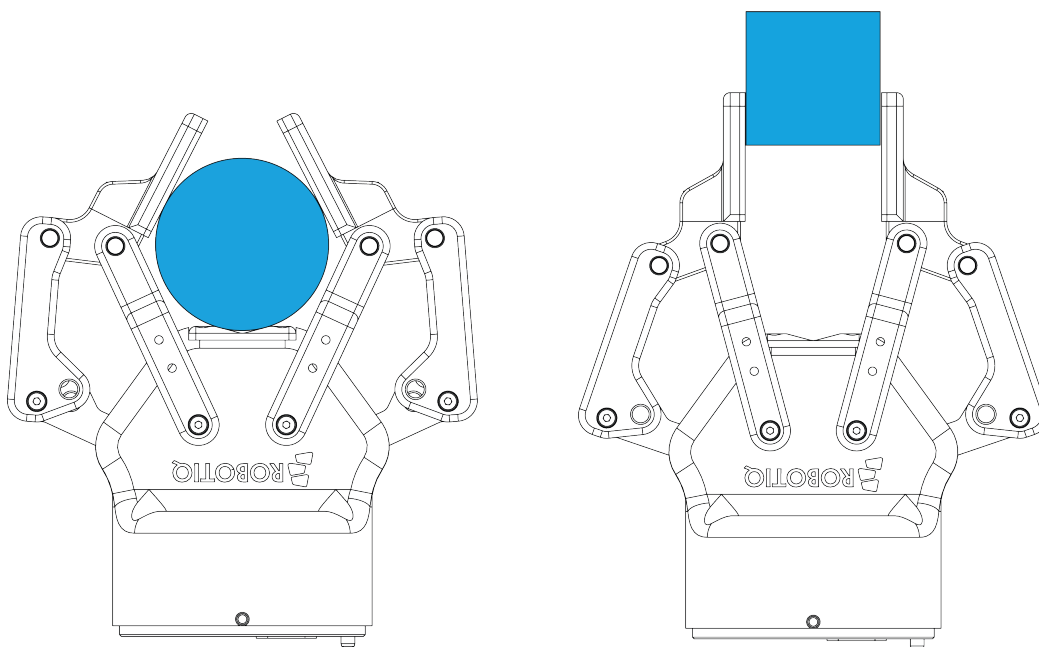


Fig. 1-3: 2-Finger parallel and encompassing grips.

Info

It is important to note that a fingertip grasp can only be performed when the fingers touch the object with the upper section of the distal phalanxes first. Inversely, for an encompassing grip, the fingers must touch the object with the proximal or the lower section of the distal phalanxes first. Also, to ensure stability, the object should be held against the Gripper palm while performing an encompassing grip. Refer to the figure below for a visual representation of the parallel and encompassing grasp regions on the distal phalanx of the 2-Finger Gripper.



The 2-Finger Adaptive Robot Gripper also allows for internal grasping. The fingers can pick hollow objects from the inside by applying pressure with the outside of the fingers. Refer to the figure below for a visual representation and to the **Picking Features** section for details on the possible position commands of your Gripper.

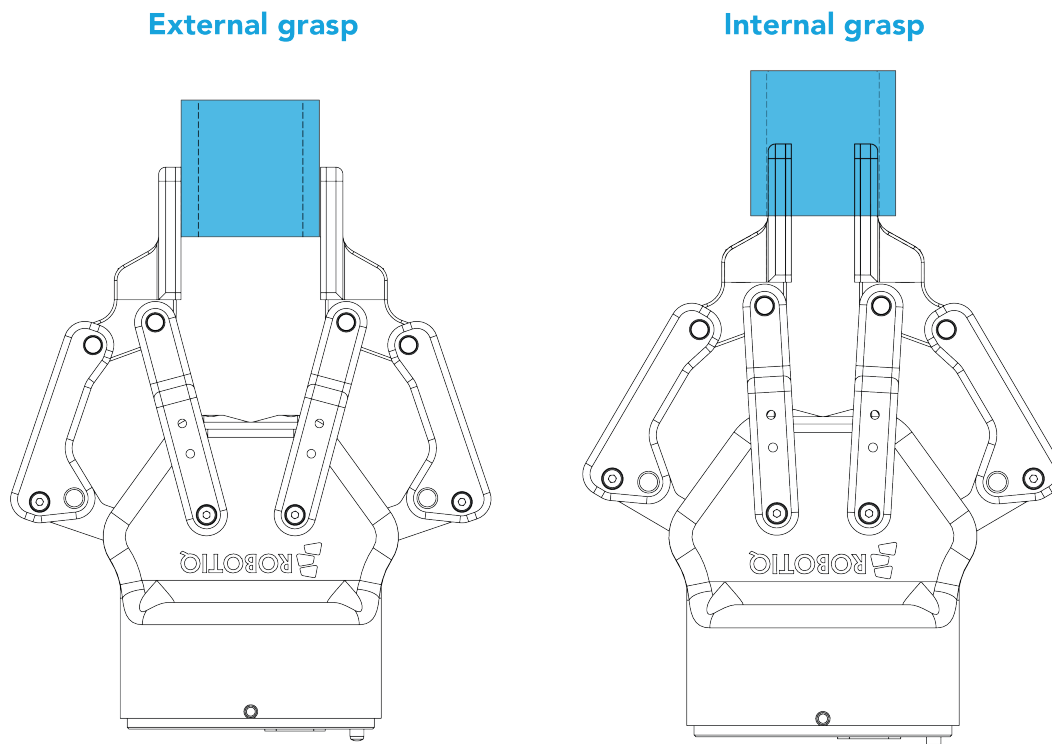


Fig. 1-4: Finger internal and external grasping.

The Gripper equilibrium line is the grasping region that separates the encompassing grasp from the parallel grip. When grasping an object close enough to the inside (palm) of the Gripper, the encompassing grasp will occur (unless the object size or shape is not adequate) and the fingers will close around the object.

If grasped above the equilibrium line, the same object will be picked up in a parallel grasp by the fingertips and the fingers will close with a parallel motion. The figure below shows the **encompassing grasp region**, the **equilibrium line**, and the **parallel grasp region** on the 2-Finger Adaptive Robot Gripper.

Info

The details of the equilibrium line relation between opening angle and the related position d can be found in the **Mechanical specifications** section.

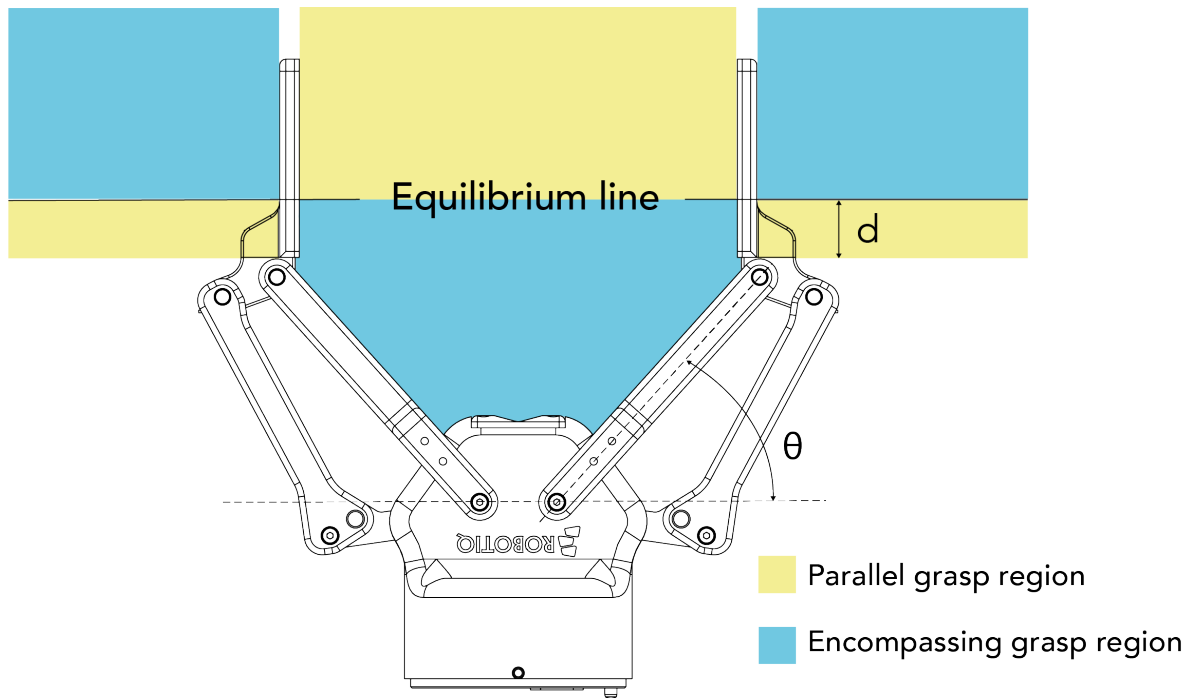


Fig. 1-5: Equilibrium line on the 2-Finger, shown with no fingertip pads.

Tip

Grasping an object that could be grasped by an encompassing grasp (a cylinder for example) on the equilibrium line is not recommended, as slight variations on the position will switch the grasp from parallel to encompassing and vice versa. Robot programming should be done so that the grasping mode will be predetermined.

1.4. Setup and control

The Gripper is powered and controlled directly via a single Device Cable that carries a 24V DC supply and Modbus RTU communication over RS-485, see Section 3.5 for wiring information and Section 4 for control of the Gripper (various software packages are available for control via various robot controllers).

Info

Robotiq Universal Controller is available when industrial communication protocols are required (other than Modbus RTU over serial).

Gripper Coupling is required for 2-Finger usage, the Coupling will provide mechanical and electrical connectivity. Please refer to the **Mechanical Installation** section for installation of the Coupling, to the **Specifications** section for technical drawings, and to the **Spare Parts, Kits and Accessories** section for available couplings.

The 2-Finger has an embedded object detection feature using indirect sensing methods. When picking an object via the "go to" command, the Gripper status will allow you to know if an object is picked or not via a simple object detection bit (0 or 1). When an object is detected, the Gripper will stop. If the object is being dropped, the Gripper will automatically close to keep the object until the "go to" command limit is attained. For details on object detection, see Control section.



2. Safety

Warning

The operator must have read and understood all of the instructions in the following manual before handling the Robotiq 2-Finger Adaptive Robot Gripper.

Caution

The term "operator" refers to anyone responsible for any of the following operations on the 2-Finger Adaptive Robot Gripper:

- Installation
- Control
- Maintenance
- Inspection
- Calibration
- Programming
- Decommissioning

This documentation explains the various components of the 2-Finger and general operations regarding the whole life-cycle of the product from installation to operation and decommissioning.

The drawings and photos in this documentation are representative examples and differences may exist between them and the delivered product.



2.1. Warning

Caution

Any use of the Gripper in noncompliance of these warnings is inappropriate and may cause injury or damage.

Warning

- The Gripper needs to be properly secured before operating the robot.
- Do not install or operate a Gripper that is damaged or lacking parts.
- Never supply the Gripper with an alternative current source.
- Make sure all cord sets are always secured at both ends, at the Gripper and at the robot.
- Always satisfy the recommended keying for electrical connections.
- Be sure no one is in the robot and/or Gripper path before initializing the robot's routine.
- Always satisfy the Gripper payload.
- Set the Gripper pinch force and speed accordingly, based on your application.
- Keep fingers and clothes away from the Gripper while the power is on.
- Do not use the Gripper on people or animals.
- For welding applications, make sure there are no Gripper parts on the ground path of the welding power source.

Any use of the Gripper in noncompliance of these warnings is inappropriate and may cause injury or damage.

2.1.1. Risk assessment and final application:

The Robotiq 2-Finger Adaptive Gripper is meant to be used on an industrial robot. The robot, Gripper and any other equipment used in the final application must be evaluated with a risk assessment. It is the robot integrator's duty to ensure that all local safety measures and regulations are respected. Depending on the application, there may be risks that need additional protection/safety measures, for example, the work-piece the Gripper is manipulating may be inherently dangerous to the operator.



2.2. Intended Use

The Gripper unit is designed for grasping and temporarily securing or holding objects.

Caution

The Gripper is NOT intended for applying force against objects or surfaces.

The product is intended for installation on a robot or other automated machinery and equipment.

Info

Always comply with local and/or national laws, regulations and directives on automation safety and general machine safety.

The unit may be used only within the range of its technical data. Any other use of the product is deemed improper and unintended use. Robotiq will not be liable for any damages resulting from any improper or unintended use.



3. Installation

The following subsections will guide you through the installation and general setup of your Robotiq 2-Finger Adaptive Robot Gripper.

- The **Scope of Delivery** section
- The **Required Tools and Equipment** section
- The **Environmental and Operating Conditions** section
- The **Mechanical Installation** section
- The **Electrical Setup** section
- The **Testing the Gripper with the Robotiq User Interface (RUI)** section.
- The **Installation for Universal Robots** section
- The **URCap Package** section

Warning

Before installing:

- Read and understand the safety instructions related to the 2-Finger Adaptive Robot Gripper.
- Verify your package according to the Scope of delivery and your order.
- Have the required parts, equipment and tools listed in the requirements readily available

Warning

When installing:

- Satisfy the environmental conditions.
- Do not operate the Gripper, or even turn on the power supply, before it is firmly anchored and the danger zone is cleared. The fingers of the Gripper may move and cause injury or damage.



3.1. Scope of Delivery

3.1.1. UR Kit

Robotiq 2-Finger Adaptive Gripper 85 (AGC-UR-KIT-002)	Robotiq 2-Finger Adaptive Gripper 140 (AGC-UR-KIT-140)
Standard upon delivery: - Robotiq 2-Finger Adaptive Gripper 85 complete unit: AGC-GRP-002 85 mm opening fingers without fingertips or pads (these are bought separately unless specified) - Palm pad - USB to RS485 signal converter: ACC-ADT-USB-RS485	Standard upon delivery: - Robotiq 2-Finger Adaptive Gripper 140 complete unit: AGC-GRP-140 140 mm opening fingers without fingertips or pads (these are bought separately unless specified) - Palm pad - USB to RS485 signal converter: ACC-ADT-USB-RS485
Coupling according to your robot bolt pattern: GRP-CPL-062 Info Please refer to the Spare Parts, Kits and Accessories section for a list of available couplings.	Coupling according to your robot bolt pattern: GRP-CPL-062 Info Please refer to the Spare Parts, Kits and Accessories section for a list of available couplings.
Robotiq device cable: CBL-COM-2065-10-HF for 10 meters cable	Robotiq device cable: CBL-COM-2065-10-HF for 10 meters cable

Info

The following are not included in standard delivery:

- Options such as adapter plates or couplings for mounting on various industrial robots, fingertips or finger pads.
- Hardware required for options; accessories or fixtures for the 2-Finger Adaptive Robot Gripper, unless specified.
- Power supply units, power supply wiring or fuses.

Info

When bought as a kit, the 2-Finger 85 or 140 will come in a package with the appropriate coupling, fingertips or finger pads and cabling. Please refer to the **Spare Parts, Kits and Accessories** section.

3.2. Required Tools and Equipment

The following tools are required to install the 2-Finger Adaptive Gripper:

- 4 mm hex key to mount the Gripper onto its coupling.
- Metric hex key according to your coupling to mount the coupling onto the robot.

Optional tools if installing finger kits: **AGC-FIN-KIT-085** or **AGC-FIN-KIT-140**:

- 2 mm hex key



Optional tools if installing other fingertips: AGC-TIP-204-085, AGC-TIP-205-085, AGC-TIP-420-140, AGC-TIP-420-140

- 4 mm hex key

The following parts are required for setup :

- Power supply (see below).
- Fuse (if applicable), see information below.
- Emergency stop is not provided, but its use is strongly advised.

The Gripper needs to be supplied by a DC voltage source. This power supply is not included with the Gripper. Required power supply must match the Robotiq device. The following table shows the specifications with regards to the power supply required to operate the Gripper and the optional Robotiq Controller.

SPECIFICATION	VALUE
Output voltage	24 V DC $\pm 10\%$
Output current	1 A
Overcurrent	Recommended power supply with internal protection, otherwise fusing is required. 2 A fuse at 25°C [77°F] ¹

Table 3-1: 2-Finger power supply requirements.

Info

¹ Suggested fuse is a: Phoenix Contact # 0916605 2 A thermal, use AWG #20 wiring.

Warning

If your power supply could exceed the specified regulation, over-voltage protection is required.

Robotiq recommends the use of the following power supplies:

- For the 1A output current: TDK-Lambda DPP Series, 100W Single Output DIN Rail Mount Power Supply: **DPP30-24**.

Tip

Optional Robotiq Universal Controller can use the same power supply.

3.3. Environmental and Operating Conditions

CONDITION	VALUE
Minimum storage/transit temperature	-30°C [-22°F]
Maximum storage/transit temperature	60°C [140°F]
Minimum operating temperature	-10°C [14°F]



CONDITION	VALUE
Maximum operating temperature	50°C [122°F]
Humidity (non-condensing)	20-80% RH
Vibration	< 0.5G
Other	• Free from dust, soot or fluids • Free from corrosive liquids or gases • Free from explosive liquids or gases • Free from powerful electromagnetic interference

Table 3-2: Environmental and operating conditions of the 2-Finger Adaptive Gripper.

3.4. Mechanical Installation

3.4.1. Installing fingers on the Gripper

Depending on your order, you may or may not have fingers already mounted on the Gripper. The first step of installation should be to install the fingers. Refer to the figure below for finger placement. To do so :

1. Align fingers with chassis axes. To do so, the slot on the finger bar should be aligned correctly with the corresponding chassis axis.
2. Insert the finger shaft in the finger bar bracket hole and through the chassis axis (top hole is for parallel locking while bottom hole is for finger installation)..
3. Apply medium strength threadlocker on the provided screws, align the finger bar and fasten to the chassis/chassis axis.

PROVIDED PARTS:

	AGC-FIN-KIT-85	AGC-FIN-KIT-140
①	85 mm fingers	140 mm fingers
②	5 mm stainless steel finger shaft	
③	Optional parallel locking kit with one 5 mm stainless steel finger shaft and two M3 X 5 mm socket low head screw	
④	M3 X 5 mm socket low head screws	

TOOLS YOU NEED:

- 3 mm hex key
- Medium strength threadlocker

EXPLODED VIEW SHOWN WITH ONE 85 MM FINGER AND ONE 140 MM.

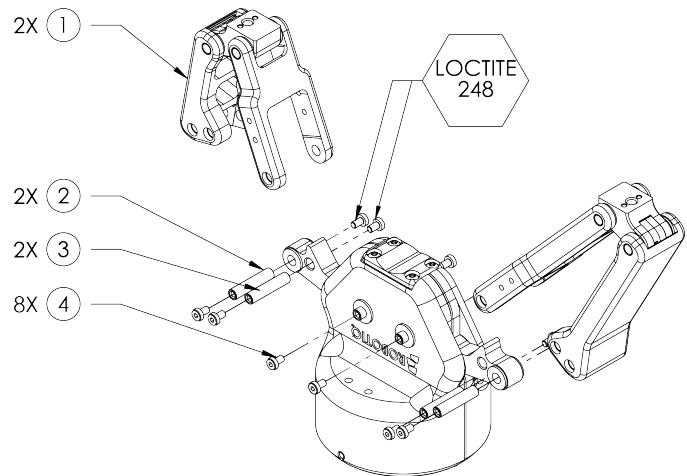


Fig. 3-1: 2-Finger Adaptive Gripper installation.

3.4.2. Installing the fingertips on the Gripper

Depending on your options, you may have fingertips to install. The second step of the installation should be to install the fingertips. To do so:

1. Align the fingertip indexing pins with the finger dowel holes.
2. Insert the M5 X 10 low head cap screws and screw on after applying low strength threadlocker.

PROVIDED PARTS:

	AGC-TIP-XXX-002	AGC-TIP-XXX-140
①	(85 mm fingertip option)	(140 mm fingertip option)
②	M5 X 10 mm head cap screw	
③	M2 X 6 mm indexing pins (M6 tolerance)	

TOOLS YOU NEED:

- 4 mm hex key
- Low strength threadlocker

Quantity shown for a single kit (1 pad)

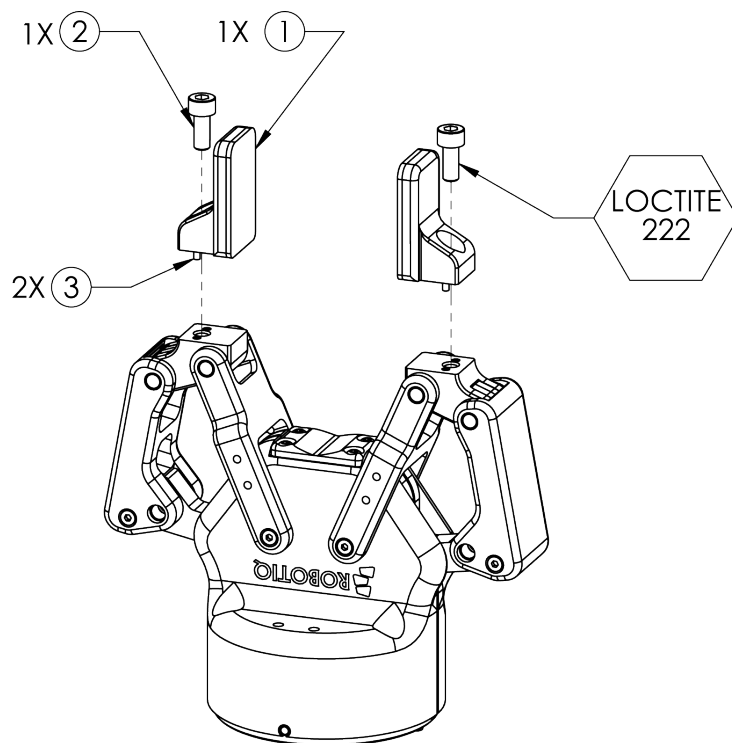


Fig. 3-2: Installing the fingertips on the Gripper

3.4.3. Installing a Protector Kit on the Gripper Fingers

An optional protector kit (AGC-PRO-KIT-V4) can be ordered to cover the fingers of the 2F-85 and therefore protect users and assets against pinch points.

You can install them using eight (8) M3 screws.

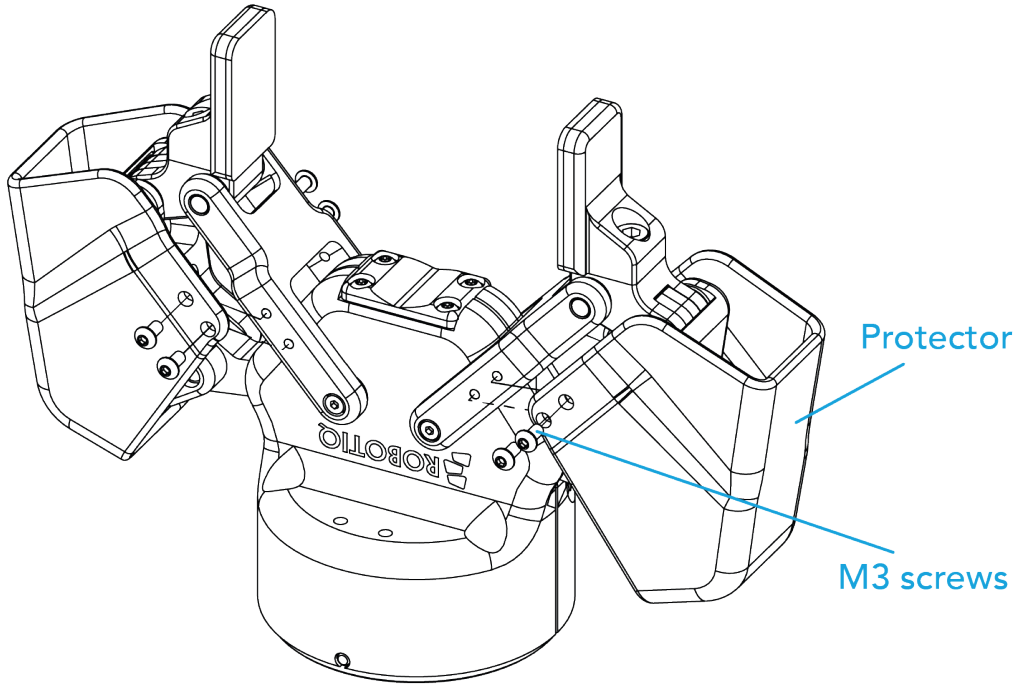


Fig. 3-3: Protector Kit Installation

3.4.4. Installing the Gripper onto the robot

Single Gripper

You must use a coupling to attach the Gripper to the robot. Be sure to use the coupling related to your robot model. If there is no coupling for your robot, you can modify a blank coupling or Robotiq can create a custom version for you. Some couplings may require an additional adapter plate. To create your own coupling or adapter plate you can refer to the Coupling section. To see the details of the available couplings and adapter plates, please refer to the **Spare Parts, Kits and Accessories** section.

Here are the steps to follow to mount the Gripper to your robot (exploded view in the figure below). Note that all screws must be locked in place using medium strength threadlocker.

1. Screw the adapter plate or the coupling to the robot (if adapter plate is required).
2. Screw the coupling to the adapter plate (if applicable).
3. Screw the Gripper onto its coupling.

PROVIDED PARTS:

	COUPLING
①	M5 x 25 mm socket head cap screw
②	M5 tooth washer
③	M6 x 5 mm low head cap screw
④	M6 tooth washer
⑤	2-Finger Gripper coupling option

	ADAPTER PLATE (optional)
⑥	M6 x 12 mm indexing pin (H8 tolerance)
⑦	2-Finger Gripper adapter plate option

TOOLS YOU NEED:

- 4 mm hex head key
- Medium strength Loctite

Shown with optional adapter plate.
Robot side screws are available only for specific robot kits.

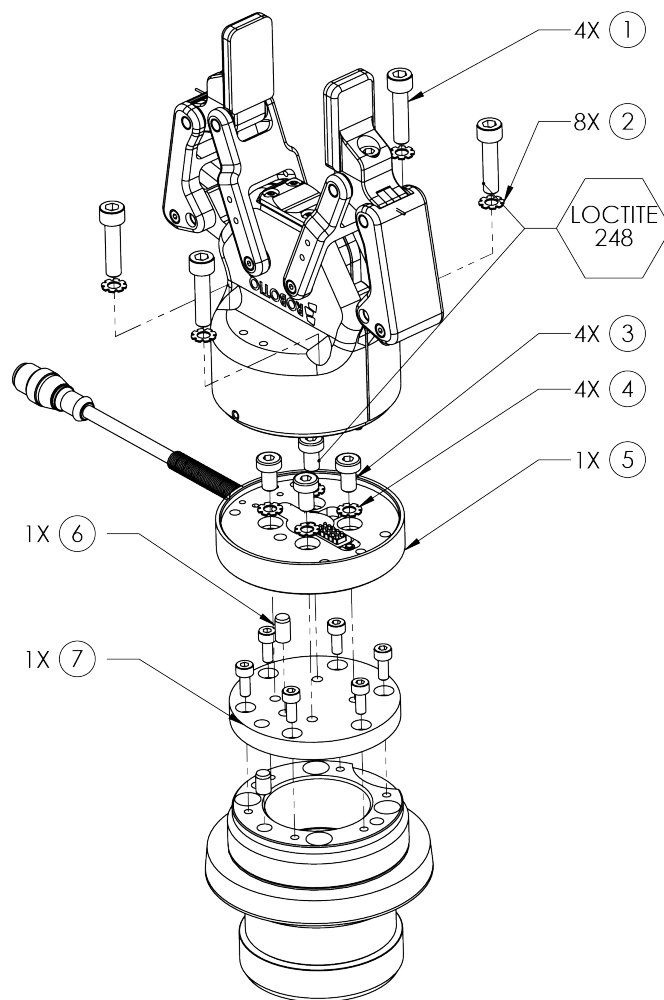


Fig. 3-4: Installing the Gripper to a robot using an adapter plate and coupling.



Multiple Grippers

When installing multiple grippers on one robot, every gripper must have its own coupling.

1. Install a mounting plate (not provided) first on the robot arm (AGC-APL-159-02).
2. Mount the gripper couplings on the mounting plate using the provided M6 X 10 mm screws.
3. Mount the grippers onto their coupling using the provided M5 X 35 mm screws.

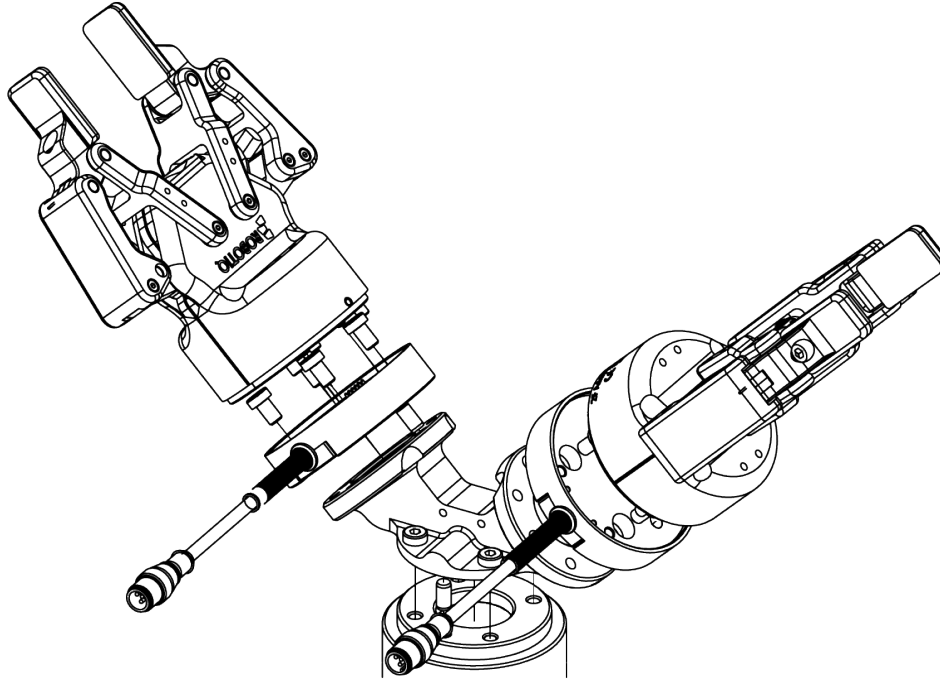


Fig. 3-5: Exploded View of a Dual Gripper Setup

Fig. 3-6: Exploded view of a dual gripper setup

3.5. Electrical Setup

Power and communication are established with the 2-Finger Adaptive Robot Gripper via a single Device Cable. The Device Cable provides a 24V power supply to the Gripper and enables serial RS485 communication to the robot controller. An optional Robotiq Universal Controller may be used between the Gripper and the network / robot controller if fieldbus communication is required.

Info

RS485 signals (485+, 485- and 485 GND) are isolated from the main 24V power supply. 4 GND can be connected to any other ground reference as long as the voltage potential between the grounds does not exceed 250V. Grounding reference is at the user's discretion.

Gripper grounding is optional and is done via the robot ground. The coupling indexing pin (dowel) is the ground connector. Gripper coupling, chassis and proximal phalanx are linked as illustrated in the figure below. They link through the coupling indexing pin to the robot ground. Proximal bars, distal phalanx, fingertip base and fingertips are isolated.

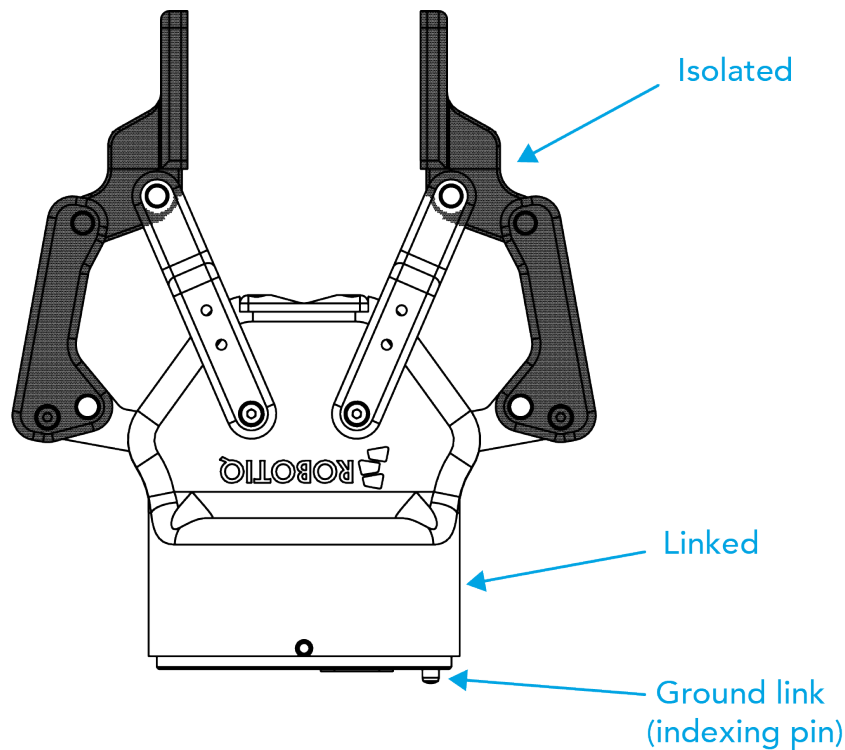
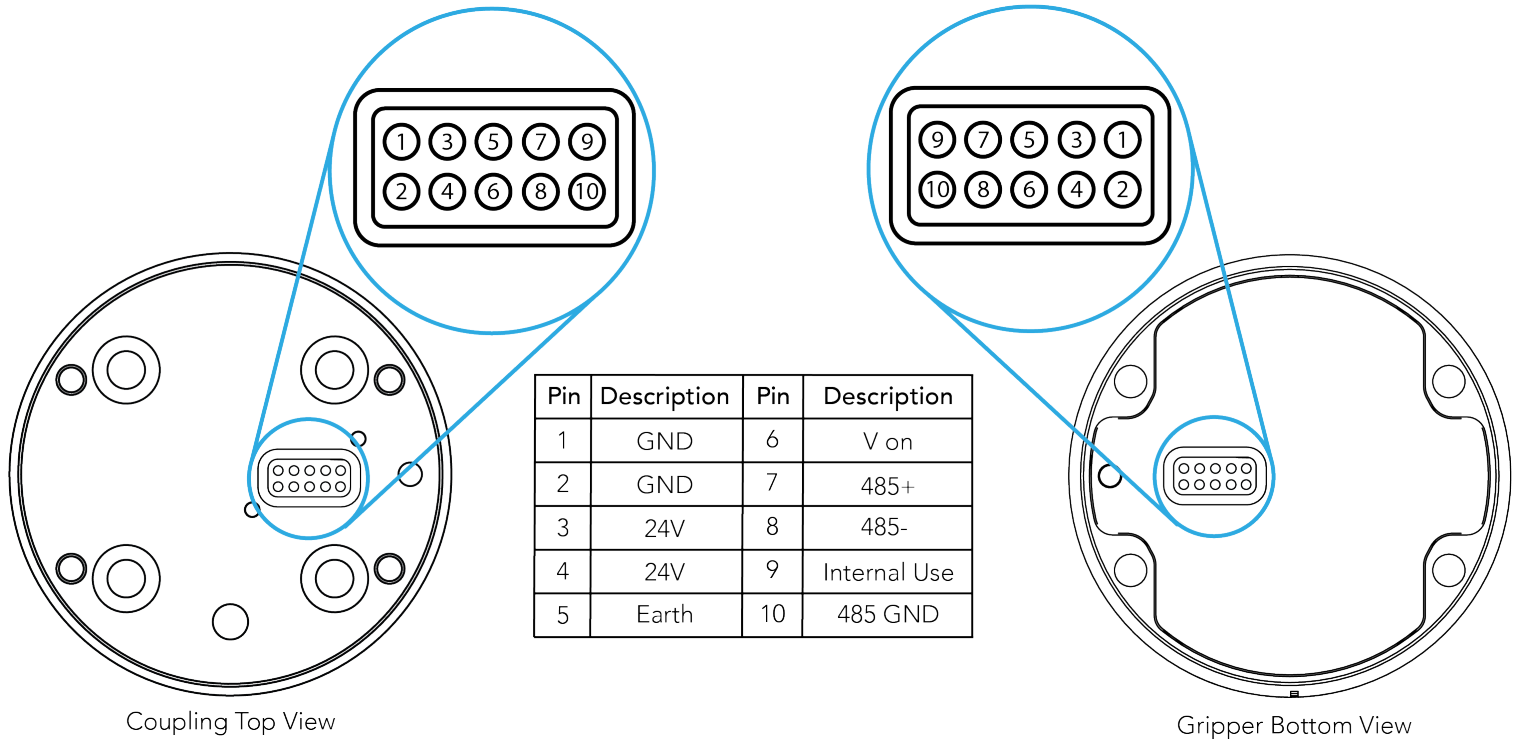


Fig. 3-7: Robotiq 2-Finger electrical isolation / grounding.



3.5.1. Pinout Interface

The Gripper interfaces with its coupling via a 10-spring pin connector located on its outer surface.



Info

The coupling used in the figure above is used for reference only and corresponds to bolt pattern ISO 9409-1-50-4-M6.

3.5.2. Coupling to controller

If a Robotiq Universal Controller is used, please refer to the Robotiq Universal Controller manual. The figure below represents the wiring schematic of the 2-Finger with device cable, power supply, fuse (please refer to the **Required Tools and Equipment** section) and grounding.

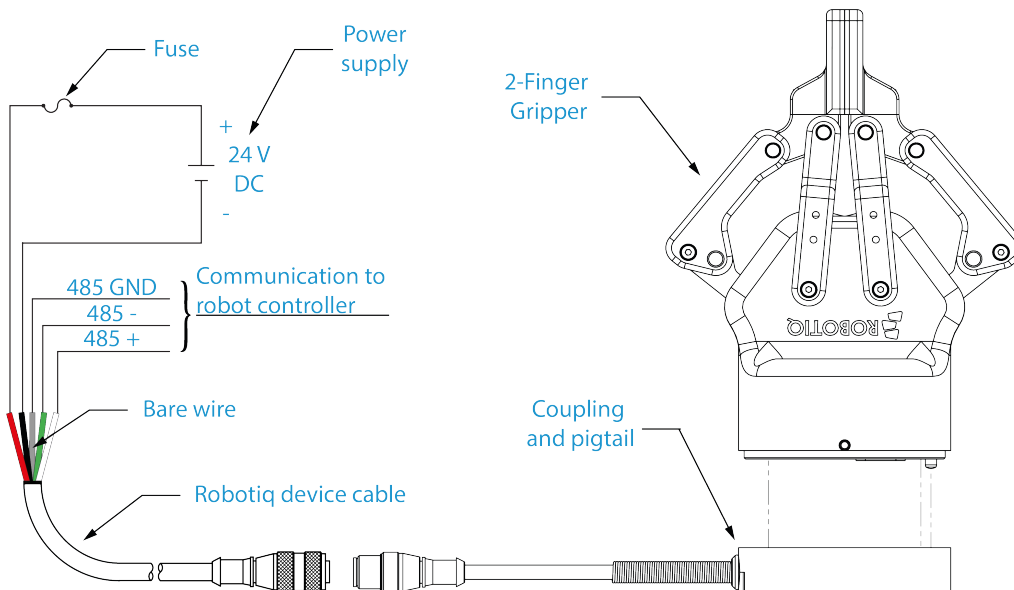


Fig. 3-8: Robotiq 2-Finger with pigtail cable and device cable wiring schematic.

Warning

Use proper cabling management. Be sure to have enough forgiveness in the cabling to allow movement of the Gripper along all axes without pulling out the connectors. Always protect the controller-side (robot side) connector of the cable with a strain relief cable clamp.

The figure below represents the 2-Finger pigtail connector from the coupling (**AGC-CPL-XXX**), device cable - robot side (**CBL-COM-2065-XX**) and their associated pinout.

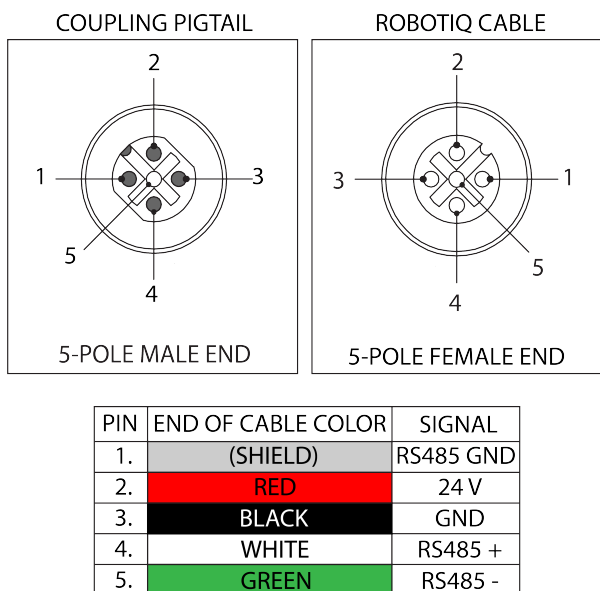


Fig. 3-9: Pinout of the 2-Finger pigtail and device cable.

If additional cable is used, suggested cable specifications are as follows:

Power supply, fusing:

- minimum #22 AWG TEW, 300 V or 600 V

RS485 signals :

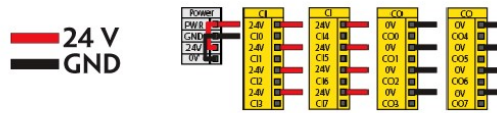
- minimum #24 AWG TEW, 300 V or 600 V
- A and B signals must be balanced at 120 Ohms

Single Gripper

Prior to any software installation on Universal Robots, connect the white, green and bare wires to the Robotiq RS-485 signal converter (ACC-ADT-USB-RS485) as shown in the figure below. Also connect the red (24V) and black (0V) wires in the controller according to that same figure.



POWER SUPPLY WIRING
(ACCORDING TO UR CONTROLLER MODEL)



COMMUNICATION WIRING

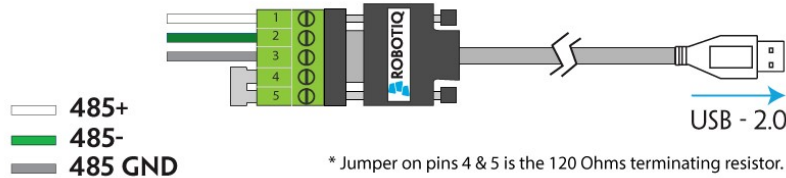


Fig. 3-10: 2-Finger Adaptive Robot Gripper wiring to robot controller.

Multiple Grippers

It is possible to connect and control up to four grippers on the same UR robot. Only one RS485 to USB converter (ACC-ADT-USB-RS485) must be used. Use M12 splitters (ACC-SPLIT-M12-2:1) to connect all the grippers pigtails to one 10m cable (CBL-COM-2065-10-HF) that connects to the RS485 to USB converter.

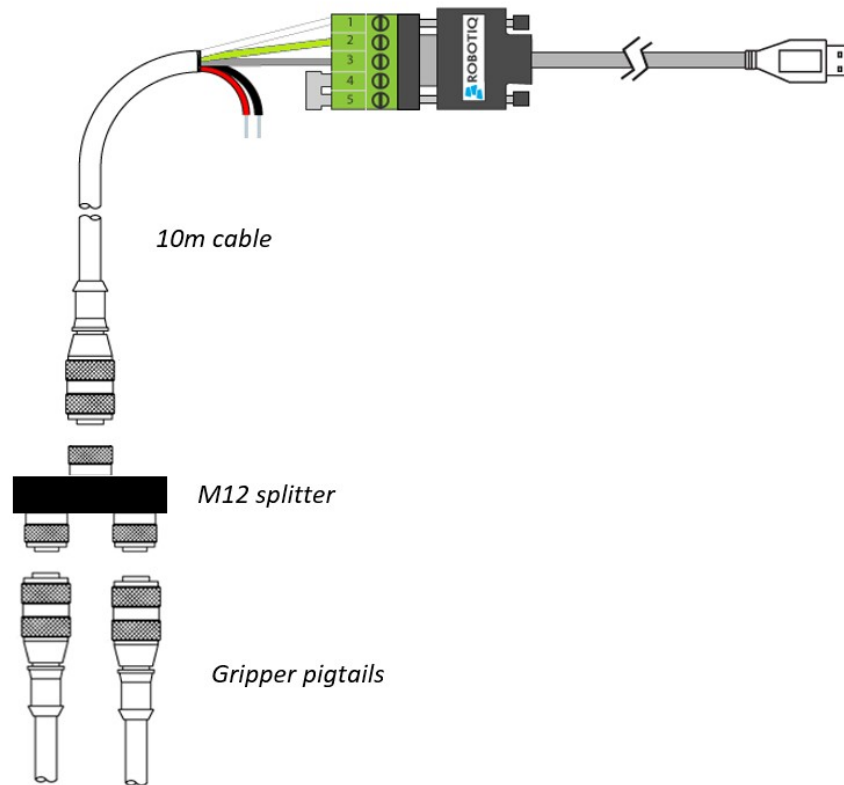


Fig. 3-11: Multiple grippers wiring on CB-Series Universal Robots.

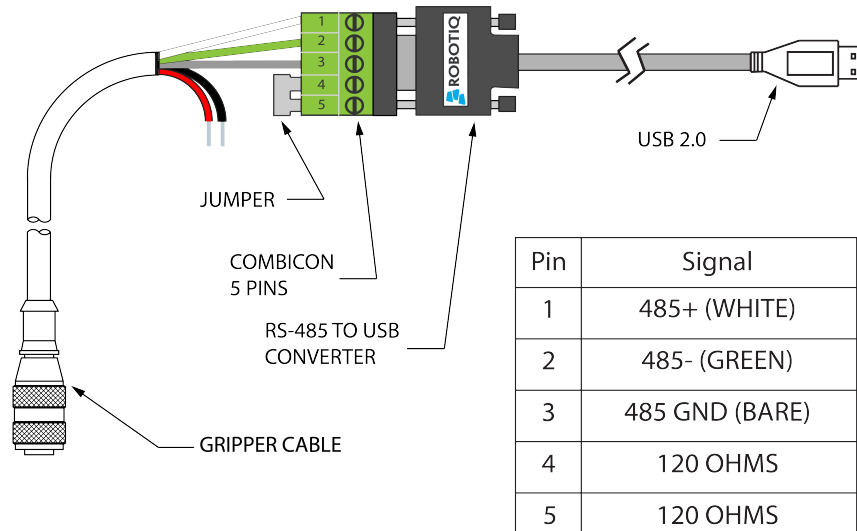
3.6. Testing the Gripper with the Robotiq User Interface (RUI)

Once installed and properly secured, your Robotiq 2-Finger Adaptive Gripper should be tested with the Robotiq User Interface test software using the provided USB converter. To do so:

1. Follow the instructions to install the Robotiq User Interface.
Use the provided RS-485 to USB converter ACC-ADT-USB-RS485 (see the schematic in the figure below) to plug into a PC with the Robotiq User Interface installed.
2. Power up your Gripper with the previously recommended power supply.
3. Execute the R.U.I. software and select "auto-connect" on the connection screen.
4. You are now connected to your Gripper, you can click "activate" to begin using the Gripper.

Info

The Activate command will initiate movement of the Gripper for auto-calibration procedures. Do not interfere with the Gripper. Be sure you have satisfied robot safety measures.



*24 V AND GND ARE NOT SUPPLIED VIA USB

*120 Ohms resistance JUMPER BETWEEN PINS 4 AND 5

Fig. 3-12: RS-485 to USB converter ACC-ADT-USB-RS485 pinout.

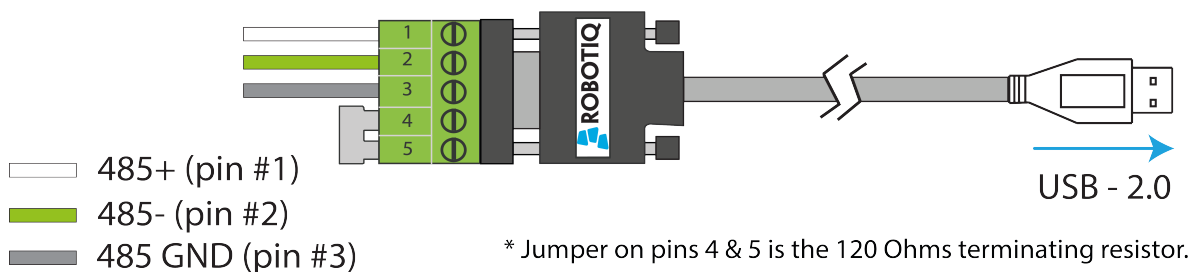


Fig. 3-13: Wiring of the USB to RS-485 converter.



Tip

With the R.U.I. controlling the Gripper, you can go to the "view" menu to see input and output register values to further your understanding on how to command the Gripper. You can also test grasping objects with various speed and force settings. See the **Control** section for details.

3.7. Installation for Universal Robots

The table below shows which Robotiq software to use with your Universal Robots' controller. If you are using a CB3, CB3.1 or e-Series controller, it is recommended to use a Robotiq Grippers URCap Package.

Robotiq Software	Controller CB1	Controller CB2	Controller CB3	Controller CB3.1	e-Series Controller
Driver Package (includes Gripper toolbar)	Incompatible	Compatible	Compatible	Compatible	Incompatible
Robotiq Grippers URCap Package 1.2.1 and earlier versions	Incompatible	Incompatible	Compatible	Compatible	Incompatible
Robotiq Grippers URCap Package 1.3.0	Incompatible	Incompatible	Incompatible	Compatible	Incompatible
Robotiq Grippers URCap Package 1.3.1 and later versions	Incompatible	Incompatible	Incompatible	Compatible	Compatible

Table 3-3: Compatibility between Robotiq software and Universal Robots controller

Refer to the appropriate section depending on your controller version:

- Section 3.8 covers the installation of the Robotiq Grippers URCap Package.
- Section 3.9 covers the software installation when not using a URCap.

Caution

The robot's PolyScope version must be 3.5 and later in order to install the URCap.

Caution

Prior to use over *Universal Robots*, adjust the payload and the center of gravity from the Installation tab (refer to the **Mechanical specifications** section).

Caution

Make sure to refer to the **Installing URCap Package** section to configure the grippers properly before controlling and programming them.



3.8. URCap Package

Robotiq provides you with a Universal Robots URCap package that enables direct serial communication to your UR controller.

Info

To get the URCaps package for your UR controller, visit support.robotiq.com.

Make sure the 2-Finger Adaptive Gripper is properly mounted to the robot arm. Refer to the **Mechanical Installation** section for detailed information on the mechanical installation. Before proceeding with the installation of the URCap package, make sure your Universal Robots controller is compatible with the package (refer to the **Installation for Universal Robots** section).

Info

The robot's PolyScope version must be 3.5 and later in order to install the URCap.

The Gripper's URCap package contains:

- The URCap for the Gripper;
- The Gripper toolbar;
- The Gripper nodes.

Tip

For other robots, where no driver package is available, we recommend the use of the Robotiq Universal Controller which allows fieldbus communication. Available communication protocols with this Universal Controller are:

- Modbus TCP
- EtherNet IP
- EtherCAT
- PROFINET
- DeviceNET
- CANopen

For details on controlling the Gripper, please refer to the **Control** section.

Info

Visit support.robotiq.com for detailed information on how to program using the URCap package (please refer to the **Control** section).

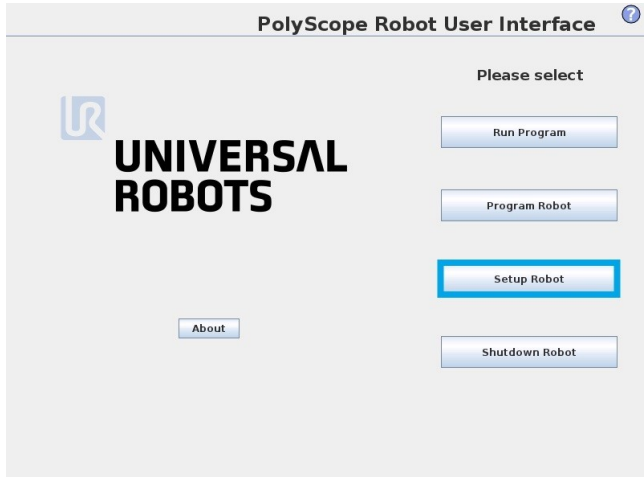


3.8.1. Installing URCap Package

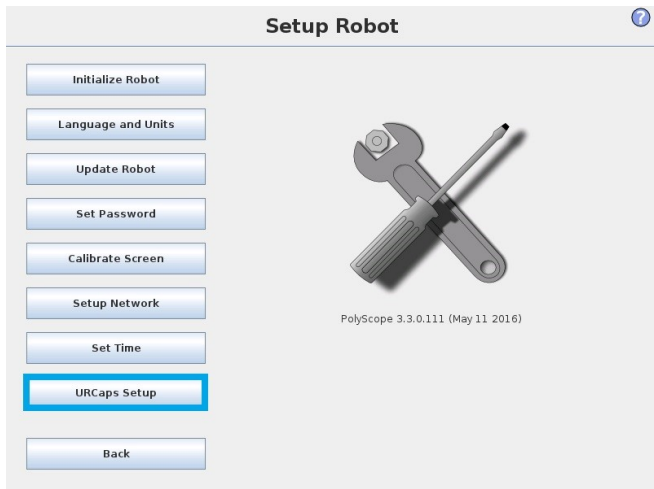
Single Gripper

Make sure the 2-Finger Adaptive Gripper is properly mounted to the robot arm. Refer to the **Mechanical Installation** section for detailed information on the mechanical installation. Before proceeding with the installation of the URCap package, make sure your Universal Robots' controller is compatible with the package (refer to the **Installation for Universal Robots** section).

Follow this procedure to install the 2-Finger Adaptive Gripper URCap package:



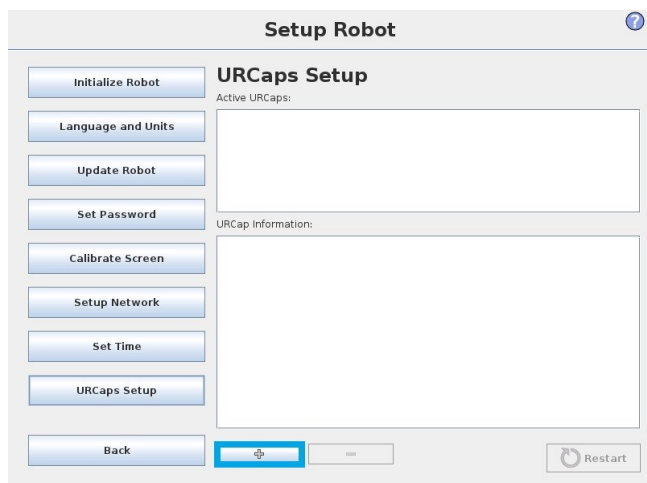
- Make sure that your PolyScope version is up-to-date and that your Universal Robots controller is compatible with the Gripper's URCap package.
- Go to support.robotiq.com and click on the 2-Finger Adaptive Gripper product page.
- Download the UCG-X.X.X and extract on the root of a blank USB stick.
- Insert the USB stick in the UR teach pendant or controller.
- Go to **Setup Robot**.
- Tap **URCaps Setup**.



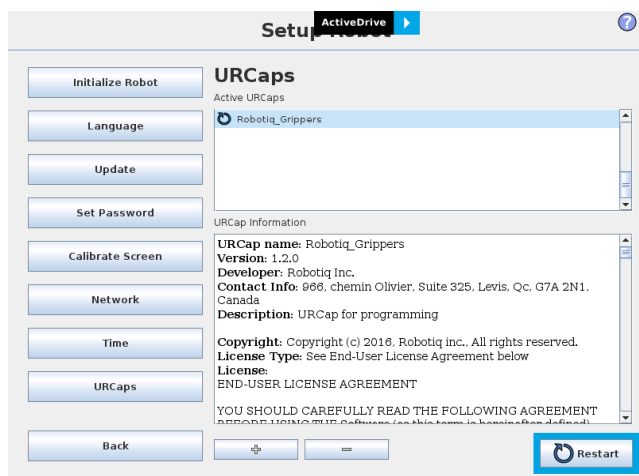
Tip

In PolyScope, go to the Home page and tap the **About** button. A window containing the Universal Robots software version will pop up.

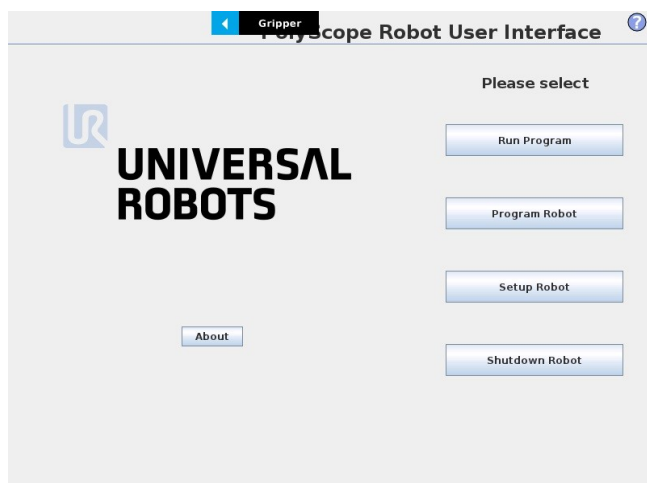




- Tap the **plus** button (+) to add the Gripper's URCap package.
- Open **Robotiq_Grippers-X.X.X.urcap**.



- Restart PolyScope to complete the URCap installation. By doing so, you accept the License Agreement that is detailed in the URCap Information text box (see below for the License Agreement).



- When PolyScope reopens, the Gripper toolbar will appear on the screen.

Multiple Grippers

It is possible to connect and control up to four grippers on the same UR robot.

Only one USB to RS485 converter (ACC- ADT-USB-RS485) must be used. Use M12 splitters (ACC-SPLIT-M12-2:1) to connect all the grippers pigtails to one 10m cable (CBL-COM-2065-10-HF) that connects to the USB to RS485 converter.



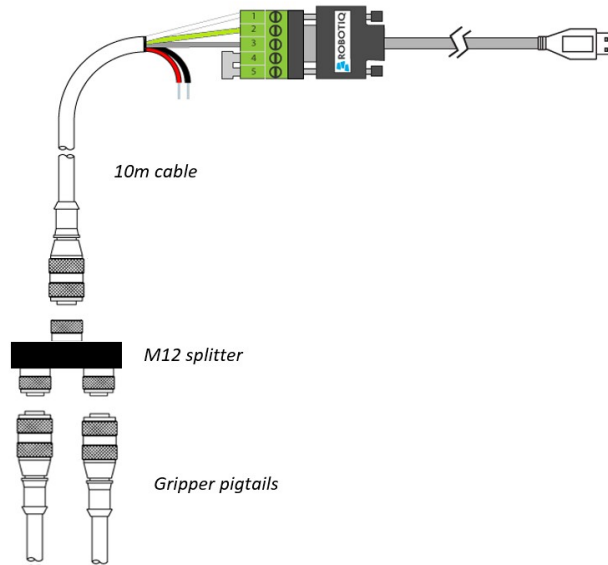
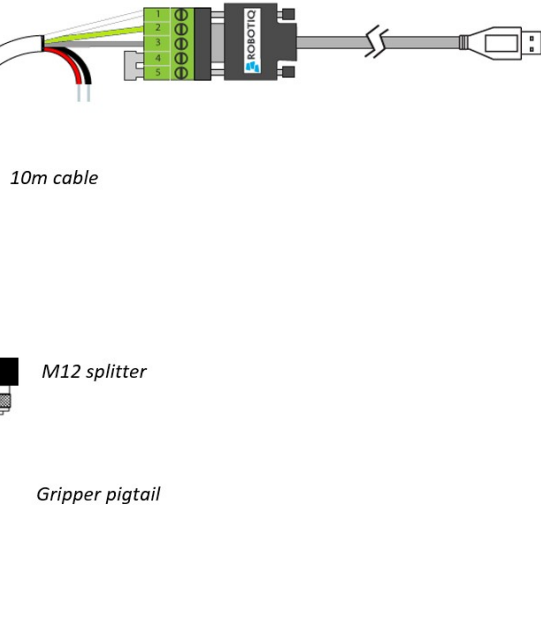
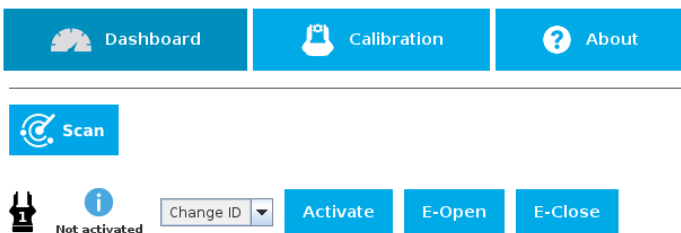


Fig. 3-14: Wiring for multiple grippers.

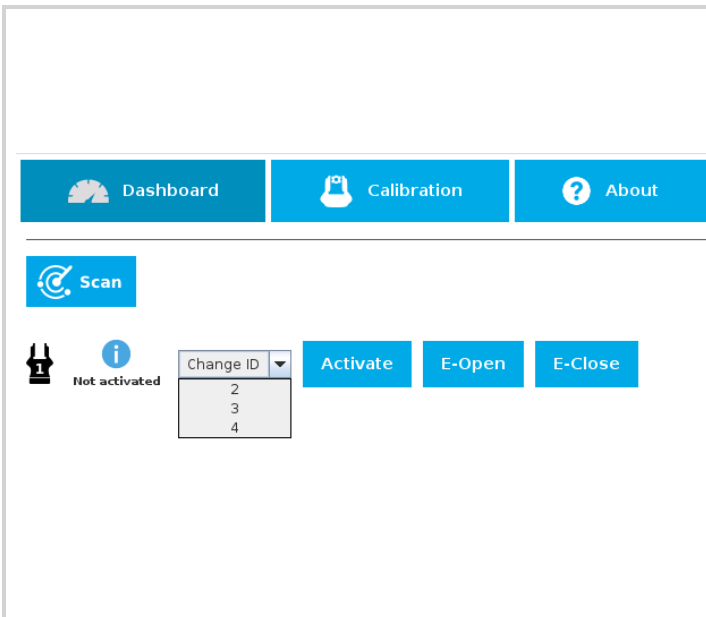
When installing multiple grippers on one UR robot, their gripper ID must be set properly. To do so, perform the following steps for each gripper:



- Connect only one gripper at a time using the M12 splitter.



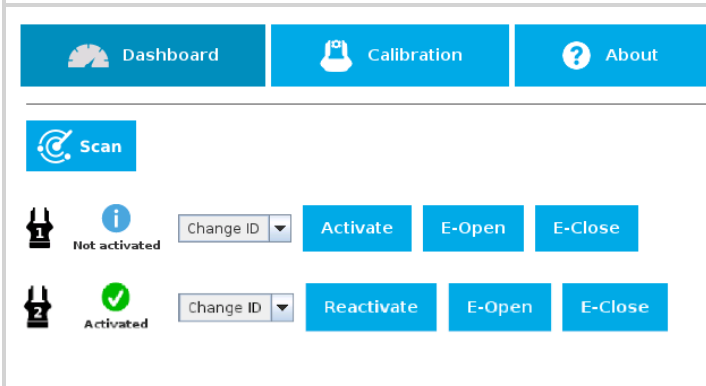
- From the PolyScope home page, go to **Program Robot** and then tap the **Installation** tab.
- Tap on **Gripper**.
- Select the **Dashboard** menu.
- Tap the **Scan** button to detect Grippers.



- Change the Gripper ID to the desired one.
- Make sure it is different from the other Gripper IDs.
- Gripper ID is saved to the Gripper's internal memory. ID will be kept at all time.
- Click on **Activate** to activate the Gripper.
- Perform the same routine for all grippers, if applicable, connecting only one at a time.

Caution

Make sure all grippers have different IDs. With their factory settings, all grippers have Gripper ID set to 1. If you have more than one gripper connected with the same ID, communication issues will arise.

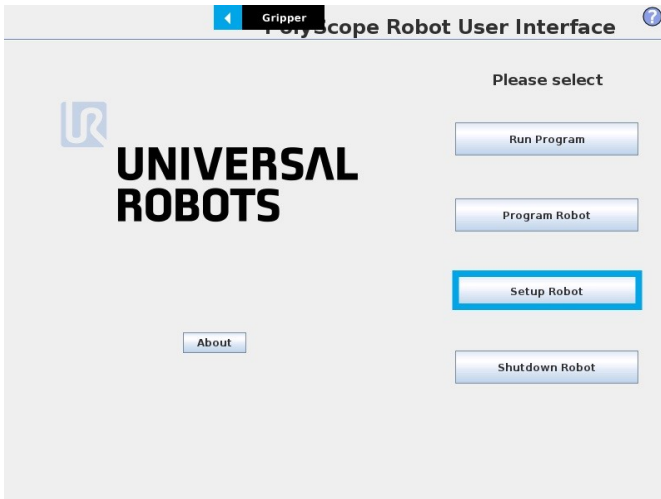


- Once all grippers IDs have been set, connect them all to the M12 splitter.
- Go back to the **Dashboard** tab and confirm that all grippers are recognized and can be controlled.

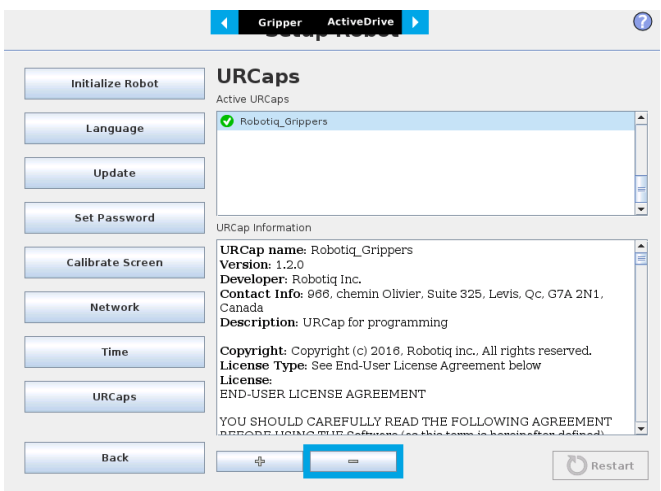
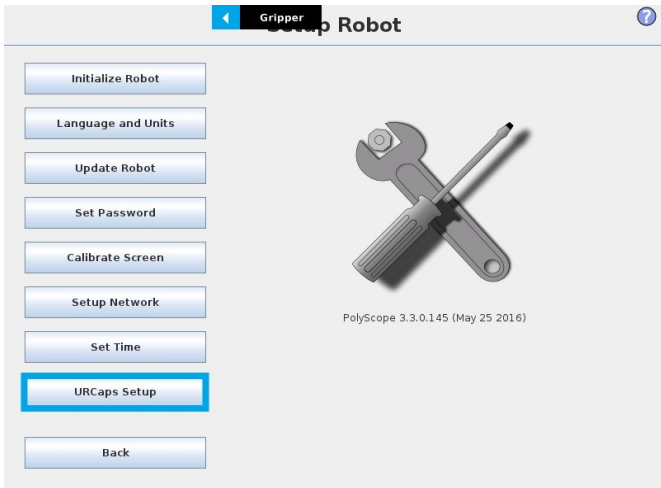


3.8.2. Uninstalling URCap Package

If you wish to uninstall the 2-Finger Gripper URCap, follow this procedure:



- Go to **Setup Robot**.
- Tap **URCaps Setup**.



- In the **Active URCaps** box, tap the URCap to uninstall.
- Tap the **minus (-)** button to uninstall the URCap.
- Restart PolyScope to complete the uninstallation process.



3.8.3. License Agreement

END-USER LICENSE AGREEMENT

YOU SHOULD CAREFULLY READ THE FOLLOWING AGREEMENT BEFORE USING THE Software (as this term is hereinafter defined). Using the Software indicates your acceptance of the agreement. If you do not agree with it, you are not authorized to use the Software.

IMPORTANT-READ CAREFULLY: This End-User License Agreement (the “Agreement”) is a legal agreement between you and the Licensor (as this term is hereinafter defined), the licensor of the Software. This Agreement covers the Software. The Software includes any “on-line” or electronic documentation and all modifications and upgrades related thereto. By installing, or otherwise using the Software, you agree to be bound by the terms of this Agreement. If you do not agree to the terms of this Agreement, the Licensor cannot and does not license the Software to you. In such event, you must not use or install the Software.

1. Definition.

1. “UR” means Universal Robots A/S, a corporation incorporated under the laws of Denmark, having its registered office at Energivej 25, DK-5260 Odense S, which specializes into the conception, advanced manufacturing and sale of robotic products (the “UR’s Business”);
2. “Software” means any of the Licensor’s softwares provided to its customers for the purposes mentioned in Sub-section 1.4 hereof including their modifications and upgrades and their related materials;
3. “Licensor” means Robotiq inc., a corporation incorporated under the laws of Quebec, having its registered office at 325-966 chemin Olivier, Lévis, Québec, Canada, G7A 2N1, which specializes into the conception, advanced manufacturing and sale of robotic products (the “Licensor’s Business”);
4. “End-User” means a customer authorized pursuant to this Agreement to install or use the Software in order to make a specific product from the Licensor’s Products compatible and functional with a specific product of the UR’s Product;
5. “Licensor’s Products” means those products developed by the Licensor in the course of the Licensor’s Business;
6. “UR’s Products” means those products developed by UR in the course of the UR’s Business;
7. “Licensor’s Authorized Representatives” means and includes the Licensor and Licensor’s authorized vendors, resellers, distributors and licensors;
8. “Purchase Agreement” means an agreement between the End-User and the Licensor pursuant to which the End-User purchased one or more of the Licensor’s Products.

2. License. Subject to the terms and conditions hereof, the Licensor grants to the End-User a personal, temporary, non-exclusive, non-assignable and non-transferable and revocable license to use the Software in accordance with the terms and conditions hereof.
3. Software and Documentation. The Licensor may provide, if applicable, all documentation containing the detailed specifications for operation and use of the Software, which Software shall be used in accordance with such documentation. This documentation, if applicable, will be provided, wholly or in part, within (i) this Agreement, (ii) the Licensor’s Web site <http://robotiq.com/> (iii) the Licensor’s Products and the Purchase Agreement therewith, or (iv) any other agreement, document, support, whatsoever decided by the Licensor.

The use of the Software requires the Licensor’s Products, UR’s Products, compatible systems and certain software (which may require some expenses), may require periodical updating and may be affected by such elements. Most equipment will be compatible with the Software. However, the Software may not function on certain types of equipment.

4. Modifications and Upgrades. The Licensor shall be under no obligation to provide any upgrade or modification to the Software. However, the End-User shall be entitled to receive free of charge all modifications and upgrades of the Software provided by the Licensor if, at such time, the End-User is not in default in respect of any of its obligation contained herein. Such modifications and upgrades of the Software shall be installed by the End-User itself by consulting the Licensor’s Website <http://robotiq.com/> where a link to proceed to such installation will be made available thereof. A new version of the Software shall not be covered by



this Section 4 but shall require that a new End-User Software License Agreement be entered into between the Licensor and the End-User.

5. **Fees.** The grant by Licensor to the End-User of the present license shall be free to the extent that the End-User agrees and complies to the term and conditions herein at all time.
6. **Maintenance.** During the term of this Agreement, the Licensor will maintain the Software in an operable condition and will make available any corrections and improvements as are generally incorporated in the Software by the Licensor without additional charge to the End-User. The Licensor may temporarily and without notice suspend or limit access to the Software if necessary or desirable in order to maintain, restore, modify or repair any part of the Software or for any reason related to business. During such works, the Software will not be available but the Licensor undertakes to deploy its best efforts to perform such works at appropriate times and to limit any inconvenience arising therefrom.
7. **Title to Software.** The licensed Software is composed of confidential data and trade secrets and is proprietary to and constitutes trade secret information and intellectual property of the Licensor. Title and ownership rights to the Software, including the intellectual property rights related thereto, shall remain with the Licensor. The End-User agrees to maintain the confidential nature of the Software and related materials provided for the End-User's own internal use under this Agreement. The license granted herein does not include the right to sublicense to others, and may not be assigned to others, in whole or in part, without the prior written consent of the Licensor. The End-User may not or allow others to modify or prepare derivative works, copy (except for normal backups for recovery purposes), reproduce, republish, reverse engineer, upload, post, transmit, or distribute, in any manner, the Software.
8. **Restricted Use.** The Software shall be used solely and exclusively by the End-User and its employees for the purpose mentioned in Sub-section 1.4 hereof. Any other use of the Software, including resell derivative modifications or extensions, is expressly prohibited.
9. **Exclusion of Warranty on Software.** The End-User expressly acknowledges and agrees that use of the Software is at the End-User sole risk. The Software is provided "AS IS" and without warranty of any kind. THE LICENSOR AND THE LICENSOR'S AUTHORIZED REPRESENTATIVES DO NOT WARRANT THAT Software WILL BE FREE OF ERRORS AND YOU ACKNOWLEDGE THAT THE EXISTENCE OF ANY SUCH ERRORS DOES NOT CONSTITUTE A BREACH OF THIS AGREEMENT. TO THE EXTENT PERMITTED BY LAW LICENSOR AND LICENSOR'S AUTHORIZED REPRESENTATIVES EXPRESSLY DISCLAIM ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE UNLESS OTHERWISE STATED HEREIN. LICENSOR AND LICENSOR'S AUTHORIZED REPRESENTATIVES DO NOT WARRANT THAT THE FUNCTIONS CONTAINED IN THE Software WILL MEET THE END-USER REQUIREMENTS OR THAT THE OPERATION OF THE Software WILL BE CORRECT. FURTHERMORE, LICENSOR AND LICENSOR'S AUTHORIZED REPRESENTATIVES DO NOT WARRANT OR MAKE ANY REPRESENTATIONS REGARDING THE USE OR THE RESULTS OF THE USE OF THE Software IN TERMS OF ITS CORRECTNESS, ACCURACY, RELIABILITY, OR OTHERWISE. NO ORAL OR WRITTEN INFORMATION OR ADVICE GIVEN BY LICENSOR AND LICENSOR'S AUTHORIZED REPRESENTATIVE SHALL CREATE A WARRANTY OR IN ANY WAY INCREASE THE SCOPE OF THIS WARRANTY. SHOULD THE Software PROVE DEFECTIVE IN YOUR TECHNOLOGY ENVIRONMENT, YOU ASSUME THE ENTIRE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION TO YOUR OWN TECHNOLOGY ENVIRONMENT.
10. **Limitation of liability.** TO THE MAXIMUM EXTENT PERMITTED BY LAW, LICENSOR AND LICENSOR'S AUTHORIZED REPRESENTATIVES SHALL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR BREACH OF ANY EXPRESS OR IMPLIED WARRANTY, BREACH OF CONTRACT, NEGLIGENCE, STRICT LIABILITY OR ANY OTHER LEGAL THEORY RELATED TO THE Software. SUCH DAMAGES INCLUDE, BUT ARE NOT LIMITED TO, LOSS OF PROFITS, LOSS OF REVENUE, LOSS OF DATA, LOSS OF USE OF THE PRODUCT OR ANY ASSOCIATED EQUIPMENT, DOWN TIME AND USER'S TIME, EVEN IF THE LICENSOR HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. IN ANY CASE, LICENSOR ENTIRE LIABILITY UNDER ANY PROVISION OF THIS AGREEMENT SHALL BE LIMITED TO THE AMOUNT ACTUALLY PAID IN RESPECT OF THE LICENSOR'S PRODUCTS PURCHASED BY THE END-USER PURSUANT TO A PURCHASE AGREEMENT.
11. **Training, Maintenance and Support** There is no entitlement to training, maintenance and support under this license unless otherwise specified in the Purchase Agreement or any other written agreement between the End-User and the Licensor. The End-User may provide the Licensor with details regarding any bug, defect or failure in the Software promptly and with no delay from such event; the End-User shall comply with the Licensor's request for information regarding bugs, defects or failures and furnish him with information, screenshots and try to reproduce such bugs, defects or failures upon Licensor's demand.



12. Expiration and Termination. The Licensor may terminate this Agreement for default by the End-User. This Agreement will also be automatically terminated upon the election of such by the Licensor or the official launch of the Software, whichever event comes first. Upon termination of this Agreement for any reason, the End-User shall promptly uninstall the Software on any UR's Products and Licensor's Products, computer, or server on which it has been installed, deliver to the Licensor all CDs, DVDs, magnetic tapes, cards, and other tangible items and materials embodying the Software, and return to the Licensor all copies thereof or destroy such copies and warrant in writing that all copies thereof have been destroyed. In the event of termination of this Agreement, all obligations of the parties under this Agreement due for performance on the date of termination shall survive the termination, and the party terminating shall not be liable to the other party for any damages arising out of the termination.
13. Miscellaneous.
 1. This Agreement constitutes the entire understanding and agreement between the Licensor and the End-User and replaces any prior agreement relating to the same subject matter.
 2. This Agreement shall be governed and construed in accordance with the laws of the province of Quebec and the federal laws of Canada applicable therein. Any legal action or proceeding between the Licensor and the End-User for any purpose concerning this Agreement or the parties' obligations hereunder shall be brought exclusively in a court of competent jurisdiction sitting in the judicial district of Trois-Rivières, Quebec.
 3. The Licensor's failure to insist upon or enforce strict performance of any provision of this Agreement shall not be construed as a waiver of any provision or right. Neither the course of conduct between the parties nor trade practice shall act to modify any provision of this Agreement.
 4. The Licensor may assign its rights and duties under this Agreement to any party at any time without notice to the End-User. The End-User may not assign this Agreement without the prior written consent of the Licensor.
 5. If any part of this Agreement is null, illegal or non-enforceable, this Agreement shall be interpreted as if this part was never part of this Agreement.
 6. The provisions of this Agreement are for the benefit of the Licensor and its officers, directors, employees, agents, licensors and suppliers. Each of these individuals or entities shall have the right to assert and enforce those provisions directly against the End-User on its own behalf. This Agreement is also for the benefit of, and binds, the End-User and its heirs, successors, legal representatives and permitted assigns.
 7. Any rights not expressly granted herein are reserved.
 8. The parties confirm that they have agreed that this Agreement and all related documents be drafted in English only. Les parties aux présentes confirment qu'elles ont accepté que la présente convention et tous les documents y afférents soient rédigés en anglais seulement.



3.9. UR Package without URCaps

Robotiq provides you with a Universal Robots driver package that enables direct serial communication (via USB) to your UR controller.

Info

To get the driver package for your UR controller, visit the [Documentation Archives](#).

If your Universal Robots' controller is not compatible with the URCaps package (refer to the **Installation for Universal Robots** section for compatibility), you can install the driver package. This package allows programming of the Gripper with scripts in a PolyScope program. It includes program templates and examples to help you get started with your own custom program. It also contains the Gripper toolbar for jogging and controlling the Gripper.

Info

The URCap package also contains a driver package that is different from this section's. If you have already installed the URCaps, you do not need to install the driver package. This section applies to Robotiq's software excluding the URCap package. If you wish to use the URCaps package, refer to the **Control** section.

Tip

Make sure your PolyScope version is up-to-date and that your controller is compatible with the driver package for UR (refer to the **Installation for Universal Robots** section for controller compatibility). To view your PolyScope version, go to the PolyScope home page and tap the **About** button. A window containing the Universal Robots software version will pop up.

Tip

For other robots, where no driver package is available, we recommend the use of the Robotiq Universal Controller which allows fieldbus communication. Available communication protocols with this Universal Controller are :

- Modbus TCP
- EtherNet IP
- EtherCAT
- PROFINET
- DeviceNET
- CANopen

For details on controlling the Gripper, refer to the **Control** section.

3.9.1. Installation

To install the driver package, follow this procedure:



- Download the Robotiq 2-Finger Gripper software driver package (DCU-X.X.X) from the Documentation Archives.
- Extract the content of the .zip file onto a blank USB flash drive.
- Plug the flash drive into the robot controller or teach pendant.
- Installation is automatic. The pendant screen will show installation status. Do not unplug the flash drive until the operation is completed.
- When a green "USB" text is shown, unplug the USB drive.

Testing the Gripper:

- When the installation is completed, the Gripper toolbar button will appear on the teach pendant's screen after a short delay.
- Use the toolbar to jog and test the Gripper. Refer to the **Control over Universal Robots without URCaps** section to get detailed information on how to use the Gripper toolbar.

Removing the package

- Locate the uninstall.sh file provided in the driver package.
- Copy the file on a blank USB stick.
- Rename the file to urmagic_uninstall.sh.
- Plug the USB stick into the UR controller or teach pendant.
- Uninstallation is automatic.



4. Control

Info

Unless specified, all values in this section are hexadecimal values.

4.1. Overview

The Robotiq 2-Finger Adaptive Gripper is controlled directly via Modbus RTU using a RS 485 signal. It can also be controlled via an optional Robotiq Universal Controller using an industrial protocol. The programming of the Gripper can be done with the Teach Pendant of the robot or by offline programming. Communication method used to control the 2-Finger Gripper does not change the control logic or the registers setup described in the following subsections.

Tip

Robotiq suggests using the Robotiq User Interface test software to explore the various features of the Gripper, like object detection and force control.

Since the Robotiq 2-Finger has its own embedded controller, high-level commands, such as "Go to requested position" are used to control it.

Info

The operator can:

- Control force, speed and position of the Gripper fingers.
- Finger movement is always synchronized, movement is initiated via a single "Go to requested position" command.
- Parallel or encompassing grasp is done automatically.
- A built in object detection feature is available, the user can be notified after an object is picked once the "Go to" command has been initiated. The feature also works for lost or dropped objects, and the user can be alerted if an object is dropped after being detected.
- Engage directional (open or close) auto-release for emergencies.

Control using registers

The Gripper has an internal memory that is shared with the robot controller. One part of the memory is for the robot output; **gripper functionalities**. The other part of the memory is for the robot input; **gripper status**. Two types of actions can then be done by the robot controller :

1. Write in the **robot output** registers to activate **functionalities**;
2. Read in the **robot input** registers to get the **status** of the Gripper.

The **Gripper Register Mapping** section will map the different registers used to control the Gripper or to read its status while the **Robot Output Registers & Functionalities** section will detail the output (write) register functions, and the **Robot Input Registers & Status** section will detail the input (read) register status. The figure below is a representation of the memory and the control logic of the Gripper.



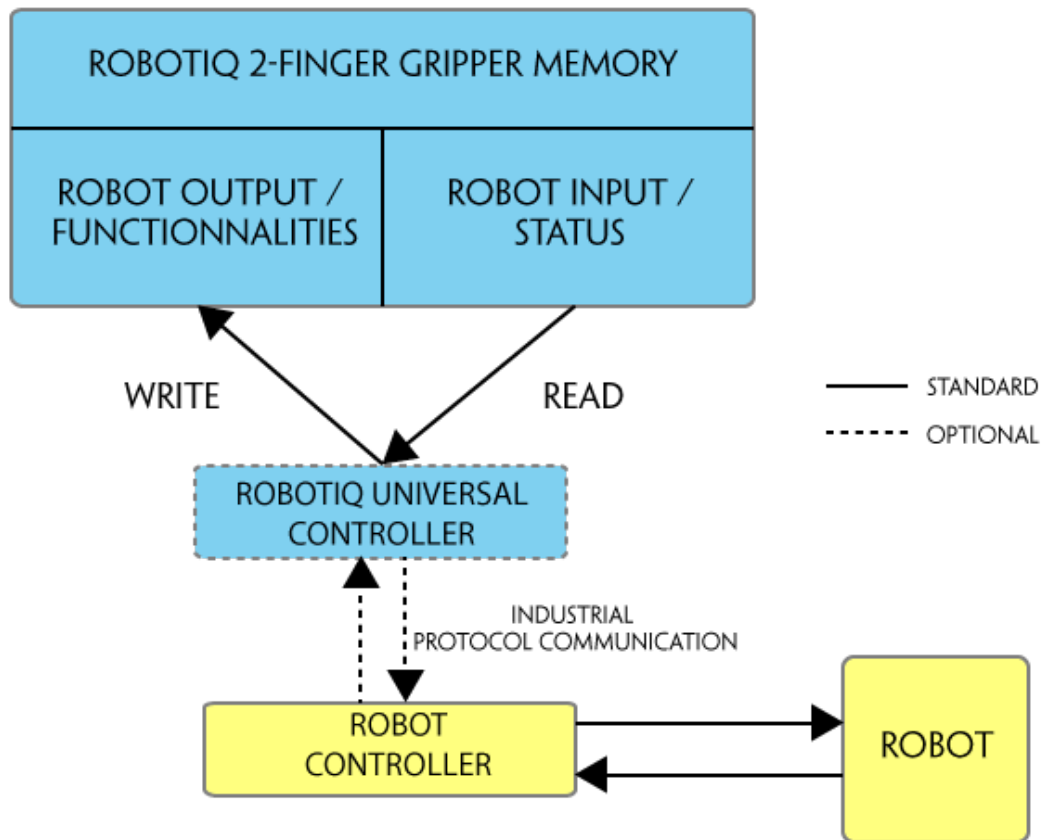


Fig. 4-1: 2-Finger control logic overview

4.2. Gripper Register Mapping

Register mapping

Caution

Byte **numeration starts at zero** and not at 1 for the functionalities and status registers.

Register	Robot Output / Functionalities	Robot Input / Status
Byte 0	ACTION REQUEST	GRIPPER STATUS
Byte 1	RESERVED	RESERVED
Byte 2	RESERVED	FAULT STATUS
Byte 3	POSITION REQUEST	POS REQUEST ECHO
Byte 4	SPEED	POSITION
Byte 5	FORCE	CURRENT
Byte 6 to 15	RESERVED	RESERVED

Table 4-1: Registers of the 2-Finger Gripper.



4.3. Robot Output Registers & Functionalities

Register: ACTION REQUEST

Address: Byte 0

Bits	7	6	5	4	3	2	1	0
Symbols	Reserved		rARD	rATR	rGTO	Reserved		rACT

rACT: First action to be made prior to any other actions, **rACT** bit will activate the Gripper. Clear **rACT** to reset the Gripper and clear fault status.

- 0x0 - Deactivate Gripper.
- 0x1 - Activate Gripper (must stay on after activation routine is completed).

Warning

When setting **rACT** to one, the Gripper will begin movement to complete its auto-calibration feature.

Info

Power loss will set **rACT**; **rACT** bit must then be cleared, then set to allow operation of the Gripper.

Caution

rACT bit must stay on afterwards for any other action to be performed.

rGTO: The "Go To" action moves the Gripper fingers to the requested position using the configuration defined by the other registers, **rGTO** will engage motion while byte 3, 4 and 5 will determine aimed position, force and speed. The only motions performed without the **rGTO** bit are activation and automatic release routines.

- 0x0 - Stop.
- 0x1 - Go to requested position.

rATR: Automatic Release routine action slowly opens the Gripper fingers until all motion axes reach their mechanical limits. After all motion is completed, the Gripper sends a fault signal and needs to be reactivated before any other motion is performed. The **rATR** bit overrides all other commands excluding the activation bit (**rACT**).

- 0x0 - Normal.
- 0x1 - Emergency auto-release.

Caution

The automatic release is meant to disengage the Gripper after an emergency stop of the robot.

The automatic release is not intended to be used under normal operating conditions.

Automatic release will require **rACT** to be cleared (**rACT** == 0) then set (**rACT** == 1).



rARD: Auto-release direction. When auto-releasing, **rARD** commands the direction of the movement. The **rARD** bit should be set prior to or at the same time as the **rATR** bit, as the motion direction is set when the auto-release is initiated.

- 0x0 - Closing auto-release
- 0x1 - Opening auto-release

Register: **GRIPPER OPTIONS**

Address: **Byte 1**

Bits	7	6	5	4	3	2	1	0
Symbol	Reserved							

Register: **GRIPPER OPTIONS 2**

Address: **Byte 2**

Bits	7	6	5	4	3	2	1	0
Symbol	Reserved							

Register: **POSITION REQUEST**

Address: **Byte 3**

Bits	7	6	5	4	3	2	1	0
Symbol	rPR							

This register is used to set the target position for the Gripper's fingers. The positions 0x00 and 0xFF correspond respectively to the fully opened and fully closed mechanical stops. For detailed finger trajectory, please refer to the **Specifications** section.

- 0x00 - Open position, with 85 mm or 140 mm opening respectively
- 0xFF - Closed
- Opening / count: 0.4 mm (for 85 mm stroke) and 0.65 mm (for 140 mm stroke)

Info

The activation feature of the Robotiq Adaptive Gripper will allow the Gripper to adjust to any fingertips. No matter what is the size and shape of the fingertips used, 0 will always be fully opened and 255 fully closed, with a quasi-linear relationship between 0 and 255.

Register: **SPEED**



Address: **Byte 4**

Bits	7	6	5	4	3	2	1	0
Symbol	rSP							

This register is used to set the Gripper closing or opening speed in real time, however, **setting a speed will not initiate a motion.**

- 0x00 - Minimum speed
- 0xFF - Maximum speed

Register: **FORCE**Address: **Byte 5**

Bits	7	6	5	4	3	2	1	0
Symbol	rFR							

The force setting defines the final gripping force for the Gripper. The force will fix the maximum current sent to the motor while in motion. If the current limit is exceeded, the fingers stop and trigger an object detection notification. Please refer to the **Robot Input Registers & Status** section for details on force control.

- 0x00 - Minimum force
- 0xFF - Maximum force

Info

Register bytes 6 to 15 are reserved and should be set to zero.



4.4. Robot Input Registers & Status

Register: **GRIPPER STATUS**

Address: **Byte 0**

Bits	7	6	5	4	3	2	1	0
Symbol s	gOBJ		gSTA		gGTO	Reserved		gACT

gACT: Activation status, echo of the **rACT** bit (activation bit).

- 0x0 - Gripper reset.
- 0x1 - Gripper activation.

gGTO: Action status, echo of the **rGTO** bit (go to bit).

- 0x0 - Stopped (or performing activation / automatic release).
- 0x1 - Go to Position Request.

gSTA: Gripper status, returns the current status & motion of the Gripper fingers.

- 0x00 - Gripper is in reset (or automatic release) state. See Fault Status if Gripper is activated.
- 0x01 - Activation in progress.
- 0x02 - Not used.
- 0x03 - Activation is completed.

gOBJ: Object detection status, is a built-in feature that provides information on possible object pick-up. Ignore if **gGTO** == 0.

- 0x00 - Fingers are in motion towards requested position. No object detected.
- 0x01 - Fingers have stopped due to a contact while opening before requested position. Object detected opening.
- 0x02 - Fingers have stopped due to a contact while closing before requested position. Object detected closing.
- 0x03 - Fingers are at requested position. No object detected or object has been loss / dropped.

Caution

In some circumstances object detection may not detect an object even if it is successfully grasped. For example, picking up a thin object in a fingertip grasp may be successful without object detection occurring. For such reasons, use this feature with caution. In these applications when the "Fingers are at requested position" status of register gOBJ, this is sufficient to proceed to the next step of the routine.

Tip

Checking for the correct position of the fingers (byte 4), **as well as** object detection (byte 0, bit 6 & 7) before proceeding to the next step of a routine is a more reliable method than object detection or finger position alone.

Register: **RESERVED**



Address: **Byte 1**

Bits	7	6	5	4	3	2	1	0
Symbol	Reserved							

Register: **FAULT STATUS**Address: **Byte 2**

Bits	7	6	5	4	3	2	1	0
Symbols	kFLT				gFLT			

gFLT: Fault status returns general error messages that are useful for troubleshooting. Fault LED (red) is present on the Gripper chassis, LED can be blue, red or both and be solid or blinking.

- 0x00 - No fault (LED is blue)
- Priority faults (LED is blue)
 - 0x05 - Action delayed, activation (reactivation) must be completed prior to performing the action.
 - 0x07 - The activation bit must be set prior to action.

Minor faults (LED continuous red)

- 0x08 - Maximum operating temperature exceeded, wait for cool-down.
- 0x09 No communication during at least 1 second.

Major faults (LED blinking red/blue) - Reset is required (rising edge on activation bit **rACT** needed).

- 0x0A - Under minimum operating voltage.
- 0x0B - Automatic release in progress.
- 0x0C - Internal fault; contact support@robotiq.com.
- 0x0D - Activation fault, verify that no interference or other error occurred.
- 0x0E - Overcurrent triggered.
- 0x0F - Automatic release completed.

Info

While booting, status LED will be solid blue / red.

kFLT : See your optional Controller Manual (input registers & status).



Register: **POSITION REQUEST ECHO**

Address: **Byte 3**

Bits	7	6	5	4	3	2	1	0
Symbol	gPR							

gPR: Echo of the requested position for the Gripper, value between 0x00 and 0xFF.

- 0x00 - Full opening.
- 0xFF - Full closing.

Register: **POSITION**

Address: **Byte 4**

Bits	7	6	5	4	3	2	1	0
Symbol	gPO							

gPO: Actual position of the Gripper obtained via the encoders, value between 0x00 and 0xFF.

- 0x00 - Fully opened.
- 0xFF - Fully closed.

Register: **CURRENT**

Address: **Byte 5**

Bits	7	6	5	4	3	2	1	0
Symbol	gCU							

gCU: The current is read instantaneously from the motor drive, value between 0x00 and 0xFF, approximate current equivalent is 10 * value read in mA.

Tip

Built-in features like object detection and force control use the finger's electrical current readings. The user does not need to create these features.



4.5. Picking Features

As stated in previous sections, object picking is done via a simple "Go To" command, **rGTO** bit calls for movement, while **rPR** byte is the aimed position, **rSP** and **rFR** will be the desired speed and force settings respectively. This section describes key features in object picking applications:

- Force control
- Re-grasp
- Object detection

4.5.1. Force control

The 2-Finger Gripper gripping force is controlled via the **rFR** byte (refer to the **Gripper Register Mapping** section). The Gripper behavior will change according to the **rFR** force requested.

- **rFR** = 0 : Very fragile objects or deformable objects mode
 - Lowest force
 - Re-grasp feature is off
- 1 **rFR** 127 : Solid & fragile objects
 - Low torque mode
 - Re-grasp feature is on
- 128 **rFR** 255 : Solid & strong objects
 - High torque mode
 - Re-grasp feature is on

The table below shows the expected applied force according to the payload material hardness, speed setting **rSP** and force setting **rFR**. All tests were done with the 2-Finger Gripper with firmware GC3-1.3.9. Data was obtained with a Load Cell from *Phidget, S Type, model 3138*.



FINGER PAD		PAYLOAD		MEASURED FORCE MIN / MAX (N)	
TYPE	HARDNESS	TYPE	HARDNESS	2-Finger 85	2-Finger 140
Steel 4340	220 HV	Steel 4340	220 HV ³	25 - 220	15 - 120
Aluminium 6061 ¹	95 HV	Aluminium 6061	95 HV	25 - 220	15 - 120
Aluminium 6061 ¹	95 HV	Silicone (TIP-204) ²	60 A Durometer	25 - 220	15 - 120
Aluminium 6061 ¹	95 HV	Silicone rubber	40 A Durometer ⁴	25 - 155	15 - 100
Aluminium 6061 ¹	95 HV	Neoprene rubber	10 A Durometer	25 - 115	15 - 75
Aluminium 6061 ¹	95 HV	Polyurethane rubber	30 OO Durometer	25 - 115	15 - 75

¹ Available with V-Groove fingertip AGC-TIP-205-0085 / AGC-TIP-421-140.

² Available with flat silicone fingertip AGC-TIP-204-085 / AGC-TIP-420-140.

³ HV refers to Vickers hardness test.

⁴ Durometer refers to Shore durometer hardness, scale A or scale OO.



MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 220 HV HARDNESS MATERIAL (STEEL)

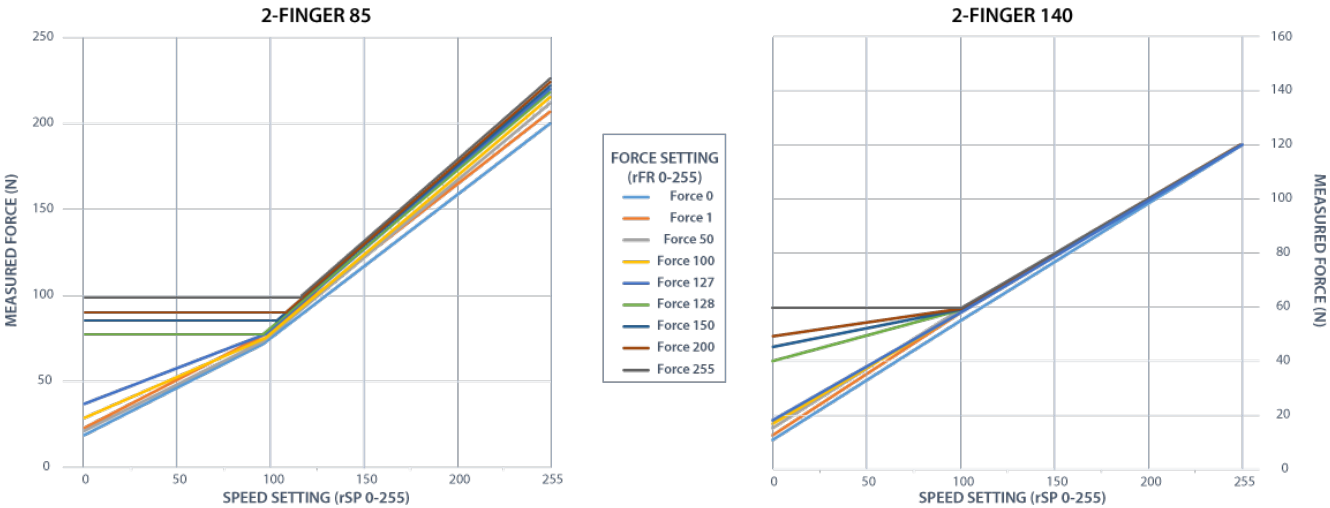


Fig. 4-2: grasp force on hardness 220 HV (4340 annealed carbon steel).

MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 95 HV HARDNESS MATERIAL (ALUMINIUM)

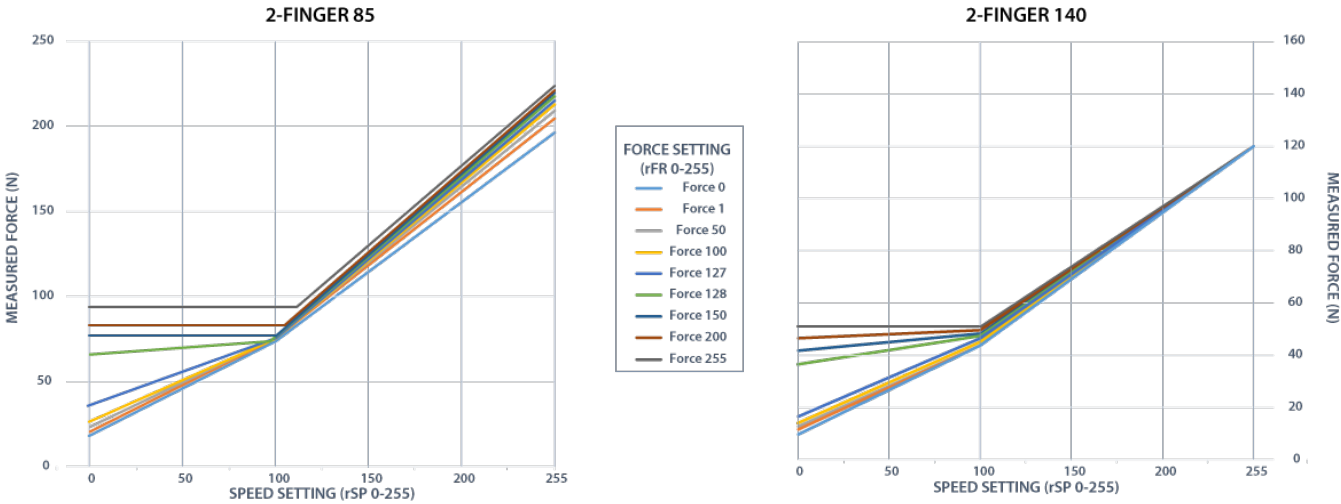


Fig. 4-3: grasp force on hardness 95 HV (6061-T6 aluminium).



MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 60A DUROMETER MATERIAL

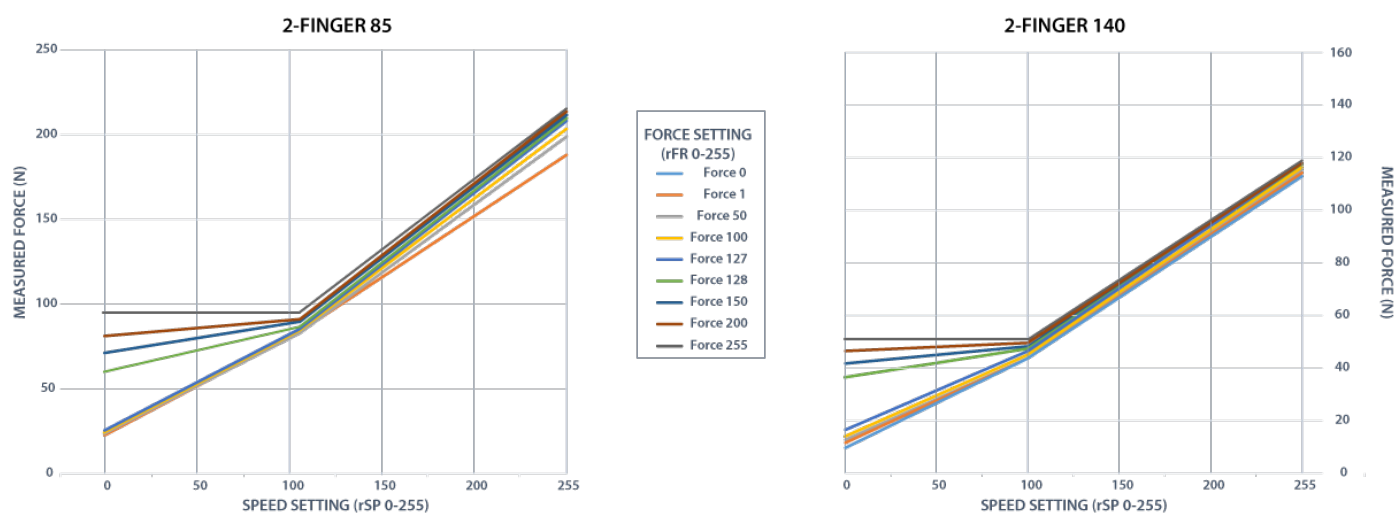


Fig. 4-4: grasp force on hardness 60A (silicone).

MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 40A DUROMETER MATERIAL

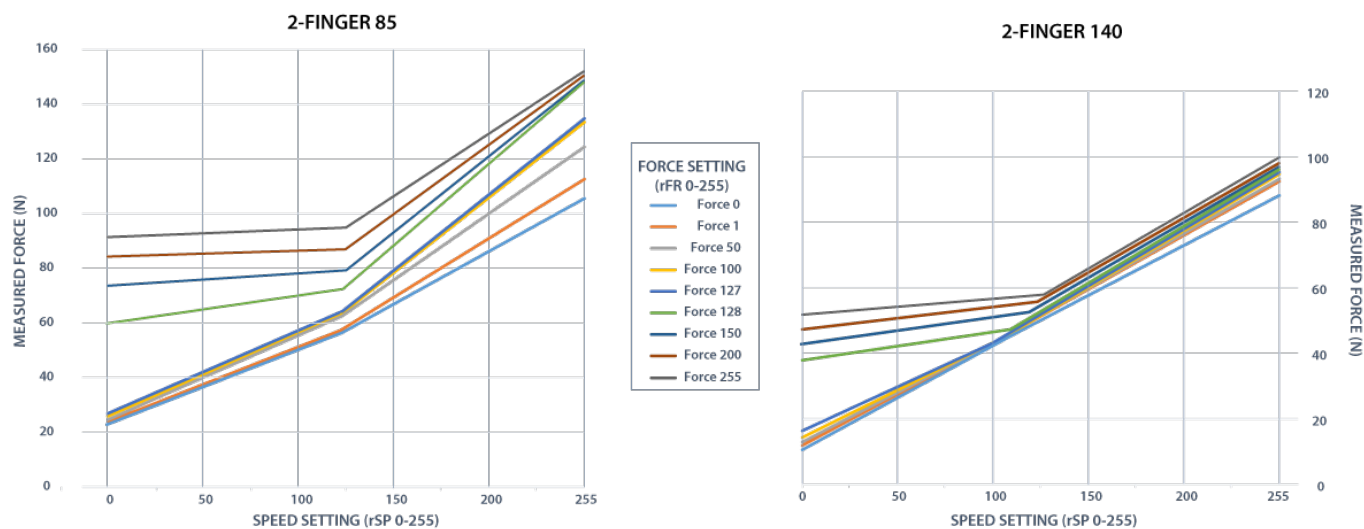


Fig. 4-5: grasp force on hardness 40 A (silicone).



MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 10A DUROMETER MATERIAL

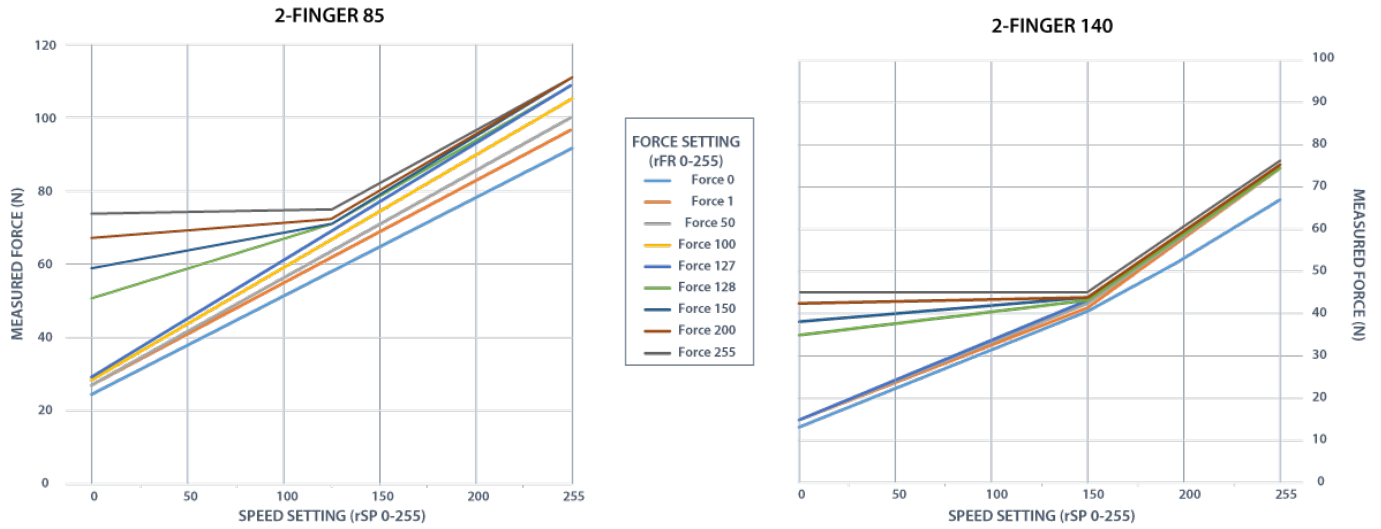


Fig. 4-6: grasp force on hardness 10 A (neoprene).

MEASURED GRIP FORCE ACCORDING TO SPEED & FORCE SETTING
FOR 3000 DUROMETER MATERIAL

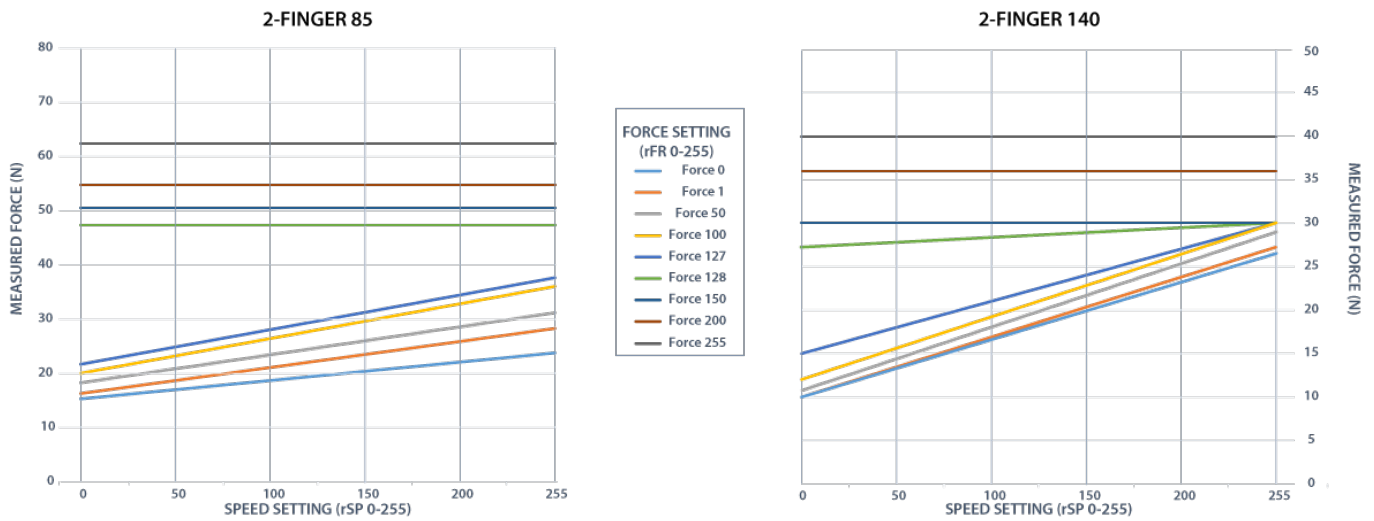


Fig. 4-7: grasp force on hardness 30 OO (polyurethane).



4.5.2. Re-Grasp

Re-grasp feature is a built-in feature meant to prevent object lost due to slipping or inaccurate initial grip. The Re-grasp feature will allow the Gripper to initiate movement when an object is slipping or dropped. When Re-grasping, the Gripper will attempt to close until it reaches the position (**rPR**) request.

- This feature is automatically set according to the force request **rFR**.

Info

Feature is off at force request **rFR** = 0, otherwise it is on.

- Re-grasp will keep the position setting:
 - Finger motion will stop when **rPR** position is reached, even if there is no object.
- Force and speed settings are not used, Re-grasp force and speed will automatically adjust to keep the object from being lost / dropped.

Info

While your initial settings for force and speed are not used for Re-grasp, they will never be exceeded to prevent damaging the object grasped.

Info

The **rOBJ** status is cleared when a motion is detected.



4.5.3. Object detection

When the Gripper grabs an object, **gOBJ** status will allow you to know if object retention was successful. This is a built-in feature for the 2-Finger Grippers meant to be used by the robot controller (or PLC) commanding the overall application. The Object detection feature will change the **gOBJ** status and can be used inside your robot program. As stated in the previous section:

gOBJ: Only valid if **gGTO** = 1.

- 0x00 - Fingers are in motion towards requested position. No object detected.
- 0x01 - Fingers have stopped due to a contact while opening before requested position. Object detected.
- 0x02 - Fingers have stopped due to a contact while closing before requested position. Object detected.
- 0x03 - Fingers are at requested position. No object detected or object has been lost / dropped.

Object detection exemple:

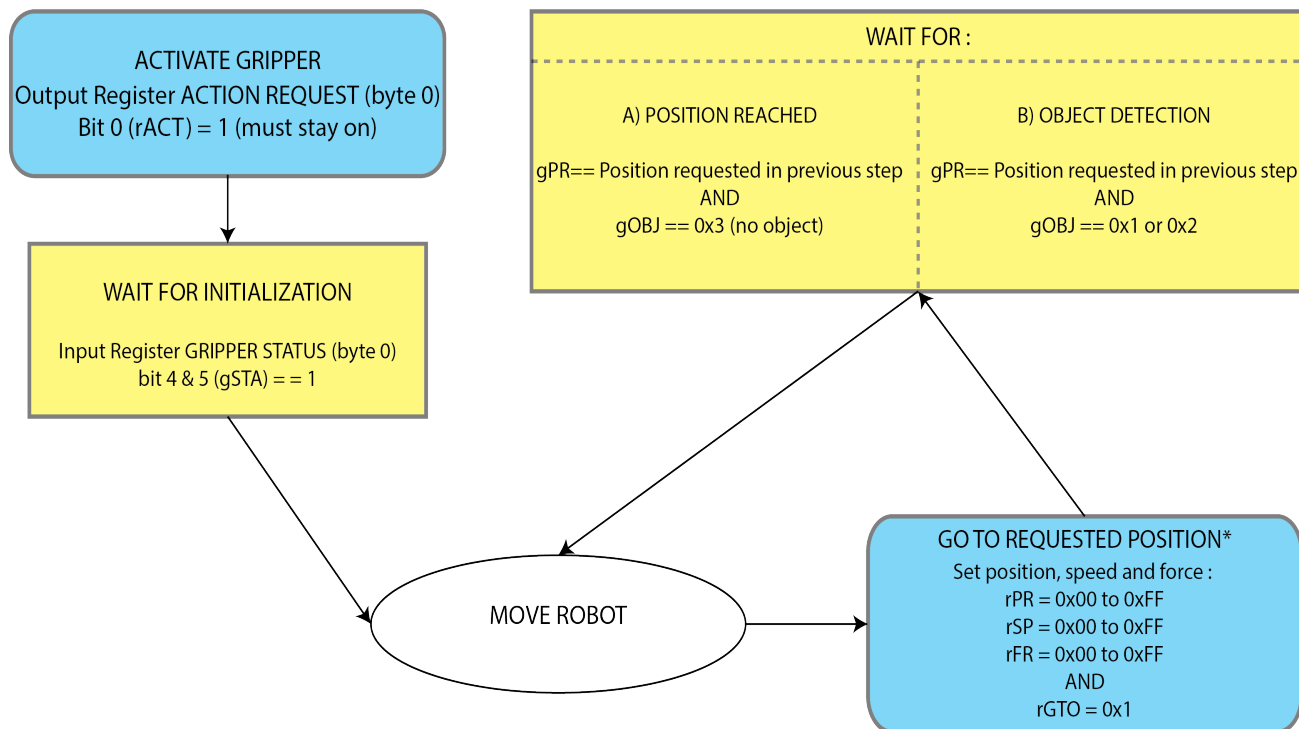
1. Set position, speed and force at maximum (full closing):
 - a. **rPR** == 0xFF, **rSP** == 0xFF, **rFR** == 0xFF,
2. Set "go to requested" will initiate movement :
 - a. **rGTO** == 0x01
3. Then object detection status will be "in motion"
 - a. **gOBJ** = 0x00
4. Until an object is picked, object detection status will then be "stopped due to contact while closing"
 - a. **gOBJ** = 0x02
5. The user can now assume it is holding the payload, and proceed to the next step.

Object lost exemple:

1. From previous exemple, after an object is picked, **gOBJ** = 0x02
2. If **gOBJ** = 0x03 after it was 0x02, user can assume the object as been lost.



4.6. Control Logic



Legend:

Gripper status inquiries



Gripper functionalities



* Go to requested position is used to open / close the Gripper until object is detected or requested position is reached.

Fig. 4-8: Example of the 2-Finger control logic with associated registers.



4.7. Modbus RTU Communication

The Gripper can be controlled by Modbus RTU directly with RS485-RS232 using the ACC-ADT-RS232-RS485, or over USB using the ACC-ADT-USB-RS485. This section is intended to provide guidelines for setting up a Modbus scanner that will adequately communicate with the Gripper.

For a general introduction to Modbus RTU and for details regarding the CRC algorithm, the reader is invited to read the Modbus over serial line specification and implementation guide available at: http://www.modbus.org/docs/Modbus_over_serial_line_V1_02.pdf.

For debugging purposes, the reader is also invited to download one of many free Modbus scanners such as the *CAS Modbus Scanner* from *Chipkin Automation Systems* available at: <http://www.store.chipkin.com/products/tools/cas-modbus-scanner>.

Info

Modbus RTU is a communication protocol based on a Big Endian byte order. Therefore, the 16-bit register addresses are transmitted with the most significant byte first. However, the data port is in the case of Robotiq products based on the Little Endian byte order. As such, the data parts of Modbus RTU messages are sent with the less significant byte first.

Tip

Modbus RTU specification and details can be found at www.modbus.org.



4.7.1. Connection Setup

The following table describes the connection requirements for controlling the Gripper using the Modbus RTU protocol.

PROPRIETY	VALUE
Physical Interface	RS-485 ¹
Baud Rate ²	115,200 bps
Data Bits	8
Stop Bit ²	1
Parity ²	None
Supported Functions	Read Holding Register (FC03) Read Input Registers (FC04) Preset Multiple Register (FC16) Master read & write multiple registers (FC23)
Exception Responses	Not supported
Slave ID ²	0x0009 (9)
Robot Output / Gripper Input First Register	0x03E8 (1000)
Robot Input / Gripper Output First Register	0x07D0 (2000)

¹ Various converters are available in the Spare parts section.

² These parameters can be adjusted using the Robotiq User Interface.

Each register (word - 16 bits) of the Modbus RTU protocol is composed of **2** bytes (8 bits) from the Gripper. The first Gripper output Modbus register(0x07D0) is composed from the first **2** Robotiq Gripper bytes (byte 0 and byte 1).

Info

200 Hz is the usual speed when commanding / reading from the Robotiq Gripper. It is therefore recommended to send commands with a minimum delay of 5 ms between them.

Info

Maximum baud rate of **ACC-ADT-USB-RS485** is 115200 bps.

120 Ohms termination resistor is already present on the converter.



4.7.2. Read holding registers (FC03)

Function code 03 (FC03) is used for reading the status of the Gripper (robot input). Examples of such data are Gripper status, object status, finger position, etc.

Example of FC03 Read function:

This message asks for register 0x07D0 (2000) and register 0x07D1 (2001) which contains Gripper Status, Object Detection, Fault Status and Position Request Echo.

Request is: 09 03 07 D0 00 02 C5 CE

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0002	Number of registers requested (2)
C5CE	Cyclic Redundancy Check (CRC)

Response is: 09 03 04 E0 00 00 00 44 33

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
E000	Content of register 07D0
0000	Content of register 07D1
4433	Cyclic Redundancy Check (CRC)



4.7.3. Read input registers (FC04)

Function code 04 (FC04) is used for requesting the status of the Gripper's analog input register. Examples of such data are Gripper status, object status, finger position, etc.

Example of FC04 read function:

This message asks for register 0x07D0 (2000) and register 0x07D1 (2001) which contains Gripper Status, Object Detection, Fault Status and Position Request Echo.

Request is: 09 04 07 D0 00 02 C5 CE

Bits	Description
09	SlaveID
04	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0002	Number of registers requested (2)
700E	Cyclic Redundancy Check (CRC)

Response is: 09 04 04 E0 00 00 00 44 33

Bits	Description
09	SlaveID
04	Function Code 04 (Read Holding Registers)
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
E000	Content of register 07D0
0000	Content of register 07D1
4584	Cyclic Redundancy Check (CRC)



4.7.4. Preset multiple registers (FC16)

Function code 16 (FC16) is used to activate functionalities of the Gripper (robot output). Examples of such data are action request, speed, force, etc.

Example of setting multiple registers FC16:

This message requests to set position request, speed and force of the Gripper by setting register 0x03E9 (1002) and 0x03EA.

Request is: 09 10 03 E9 00 02 04 60 E6 3C C8 EC 7C

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E9	Address of the first register
0002	Number of registers written to
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
60E6	Value written to register 0x03E9
3CC8	Value written to register 0x03EA
EC7C	Cyclic Redundancy Check (CRC)

Response is: 09 10 03 E9 00 02 91 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E9	Address of the first register
0002	Number of written registers
9130	Cyclic Redundancy Check (CRC)



4.7.5. Master read & write multiple registers FC23

Function code 23 (FC23) is used for reading the status of the Gripper (robot input) and activating functionalities of the Gripper (robot output) simultaneously. Examples of such data are Gripper status, object status, finger position, etc. Action requests are speed, force, etc.

Example of reading and writing multiple registers FC23:

This message reads registers 0x07D0 (2000) and 0x07D1 (2001), which contains Gripper Status, Object Detection, Fault Status and Position Request Echo. It also sets the position request, speed and force of the Gripper by writing to registers 0x03E9 (1001) and 0x03EA (1002).

Request is: 09 17 07 D0 00 02 03 E9 00 02 04 00 E6 3C C8 2D 0C

Bits	Description
09	SlaveID
17	Function Code 23 (read and write multiple registers)
07D0	Address of the first requested register, read
0002	Number of registers requested (2), read
03E9	Address of the first register written to
0002	Number of registers written to (3)
04	Number of data bytes to follow (2 registers X 2 bytes/registers = 4 bytes)
00E6	Value written to register 0x03E9
3CC8	Value written to register 0x03EA
2D0C	Cyclic Redundancy Check (CRC)

Response is: 09 17 04 01 00 09 E6 F6 C1

Bits	Description
09	SlaveID
17	Function Code 23 (read and write multiple registers)
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
1000	Content of register 07D0



Bits	Description
09E6	Content of register 07D1
F6C1	Cyclic Redundancy Check (CRC)

Note that the content of the response might change depending on the Gripper's status.

4.7.6. Modbus RTU example

This section depicts the example given in the **Control Logic** section when programmed using the Modbus RTU protocol. The example is typical of a pick and place application. After activating the Gripper, the robot is moved to a pick-up location to grasp an object. It moves again to a second location to release the grasped object.

Step 1: Activation Request (clear and set rACT)

Request is (clear rAct): 09 10 03 E8 00 03 06 00 00 00 00 00 73 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers written to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0000	Value to write to register 0x03E9 (ACTION REQUEST = 0x01 and GRIPPER OPTIONS = 0x00): rACT = 1 for "Activate Gripper"
0000	Value written to register 0x03EA
0000	Value written to register 0x03EB
7330	Cyclic Redundancy Check (CRC)

Response is: 09 10 03 E8 00 03 01 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)



Request is (set rAct): 09 10 03 E8 00 03 06 01 00 00 00 00 00 72 E1

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers written to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0100	Value to write to register 0x03E9 (ACTION REQUEST = 0x01 and GRIPPER OPTIONS = 0x00): rACT = 1 for "Activate Gripper"
0000	Value written to register 0x03EA
0000	Value written to register 0x03EB
72E1	Cyclic Redundancy Check (CRC)

Response is: 09 10 03 E8 00 03 01 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)



Step 2: Read Gripper status until the activation is completed

Request is: 09 03 07 D0 00 01 85 CF

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0001	Number of registers requested (1)
85CF	Cyclic Redundancy Check (CRC)

Response (if the activation IS NOT completed): 09 03 02 11 00 55 D5

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
02	Number of data bytes to follow (1 register x 2 bytes/register = 2 bytes)
1100	Content of register 07D0 (GRIPPER STATUS = 0x11, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gSTA = 1 for "Activation in progress"
55D5	Cyclic Redundancy Check (CRC)

Response (if the activation IS completed): 09 03 02 31 00 4C 15

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
02	Number of data bytes to follow (1 register x 2 bytes/register = 2 bytes)
3100	Content of register 07D0 (GRIPPER STATUS = 0x31, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gSTA = 3 for "Activation is completed"
4C15	Cyclic Redundancy Check (CRC)



Step 3: Move the robot to the pick-up location

Step 4: Close the Gripper at full speed and full force

Request is: 09 10 03 E8 00 03 06 09 00 00 FF FF FF 42 29

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers written to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0900	Value written to register 0x03E8 (ACTION REQUEST = 0x09 and GRIPPER OPTIONS = 0x00): rACT = 1 for "Activate Gripper", rGTO = 1 for "Go to Requested Position"
00FF	Value written to register 0x03E9 (GRIPPER OPTIONS 2 = 0x00 and POSITION REQUEST = 0xFF): rPR = 255/255 for full closing of the Gripper
FFFF	Value written to register 0x03EA (SPEED = 0xFF and FORCE = 0xFF): full speed and full force
4229	Cyclic Redundancy Check (CRC)

Response is: 09 10 03 E8 00 03 01 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)



Step 5: Read Gripper status until the grasp is completed

Request is: 09 03 07 D0 00 03 04 0E

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0003	Number of registers requested (3)
040E	Cyclic Redundancy Check (CRC)

Example of response if the grasp is **not completed**: 09 03 06 39 00 00 FF 0E 0A F7 8B

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
3900	Content of register 07D0 (GRIPPER STATUS = 0x39, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gGTO = 1 for "Go to Position Request" and gOBJ = 0 for "Fingers are in motion"
00FF	Content of register 07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0xFF): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
0E0A	Content of register 07D2 (POSITION = 0x0E, FINGER CURRENT = 0x0A): the position is 14/255 and the motor current is 100mA (these values will change during motion)
F78B	Cyclic Redundancy Check (CRC)



Example of response if the grasp is **completed**: 09 03 06 B9 00 00 FF BD 00 1D 7C

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
B900	Content of register 07D0 (GRIPPER STATUS = 0xB9, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gGTO = 1 for "Go to Position Request" and gOBJ = 2 for "Fingers have stopped due to a contact while closing"
00FF	Content of register 07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0xFF): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
BD00	Content of register 07D2 (POSITION = 0xBD, FINGER CURRENT = 0x00): the position is 189/255 (can be used to validate the size of the seized object)
1D7C	Cyclic Redundancy Check (CRC)



Step 6: Move the robot to the release location**Step 7: Open the Gripper at full speed and full force**

Request is: 09 10 03 E8 00 03 06 09 00 00 00 FF FF 72 19

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers written to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0900	Value written to register 0x03E8 (ACTION REQUEST = 0x09 and GRIPPER OPTIONS = 0x00): rACT = 1 for "Activate Gripper", rGTO = 1 for "Go to Requested Position"
0000	Value written to register 0x03E9 (GRIPPER OPTIONS 2 = 0x00 and POSITION REQUEST = 0x00): rPR = 0/255 for full opening of the Gripper (partial opening would also be possible)
FFFF	Value written to register 0x03EA (SPEED = 0xFF and FORCE = 0xFF): full speed and full force
7219	Cyclic Redundancy Check (CRC)

Response is: 09 10 03 E8 00 03 01 30

Bits	Description
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)



Step 8: Read Gripper status until the opening is completed

Request is: 09 03 07 D0 00 03 04 0E

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0003	Number of registers requested (3)
040E	Cyclic Redundancy Check (CRC)

Example of response if the opening **is not completed**: 09 03 06 39 00 00 00 BB 10 30 E0

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
3900	Content of register 07D0 (GRIPPER STATUS = 0x39, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gGTO = 1 for "Go to Position Request" and gOBJ = 0 for "Fingers are in motion"
0000	Content of register 07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0x00): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
BB10	Content of register 07D2 (POSITION = 0xBB, FINGER CURRENT = 0x10): the position is 187/255 and the motor current is 160mA (these values will change during motion)
30E0	Cyclic Redundancy Check (CRC)



Example of response if the opening **is completed**: 09 03 06 F9 00 00 00 0D 00 56 4C

Bits	Description
09	SlaveID
03	Function Code 03 (Read Holding Registers)
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
F900	Content of register 07D0 (GRIPPER STATUS = 0xF9, RESERVED = 0x00): gACT = 1 for "Gripper Activation", gGTO = 1 for "Go to Position Request" and gOBJ = 3 for "Fingers are at requested position"
0000	Content of register 07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0x00): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
0D00	Content of register 07D2 (POSITION = 0x0D, FINGER CURRENT = 0x00): the position is 13/255 (the fingers have reached their software limit)
564C	Cyclic Redundancy Check (CRC)

Step 9: Loop back to step 3 if other objects have to be grasped.



4.8. Control over Universal Robots with URCap

The URCap package contains many features to program and control the Gripper. The package provides:

- Gripper toolbar: The Gripper toolbar is automatically installed with the URCap package. It allows jogging and testing of the Gripper. It is a great tool to try grasps with the Gripper while programming.
- Gripper node: The URCap package adds a Gripper node that is used to add a Gripper command. A node can make the Gripper move to a specific opening, grasp an object and modify the speed and force applied by the Gripper.

Caution

Starting from Gripper URCap version 1.0.2, the gripper activation script command (**rq_activate()** or **rq_activate_and_wait()**) must be preceded by the reset command (**rq_reset()**). This will force the gripper's activation routine. Thus, if the gripper was already activated, it will activate again.



4.8.1. Gripper Dashboard

Overview

Single Gripper

Open or create a robot program

- 1 Tap the **Installation** tab
- 2 Select **Gripper** in the left pane
- 3 Tap the **Dashboard** button
- 4 If no Gripper is found, tap the **Scan** button to detect any Gripper that might be connected to the robot
- 5 Tap the **Activate** button

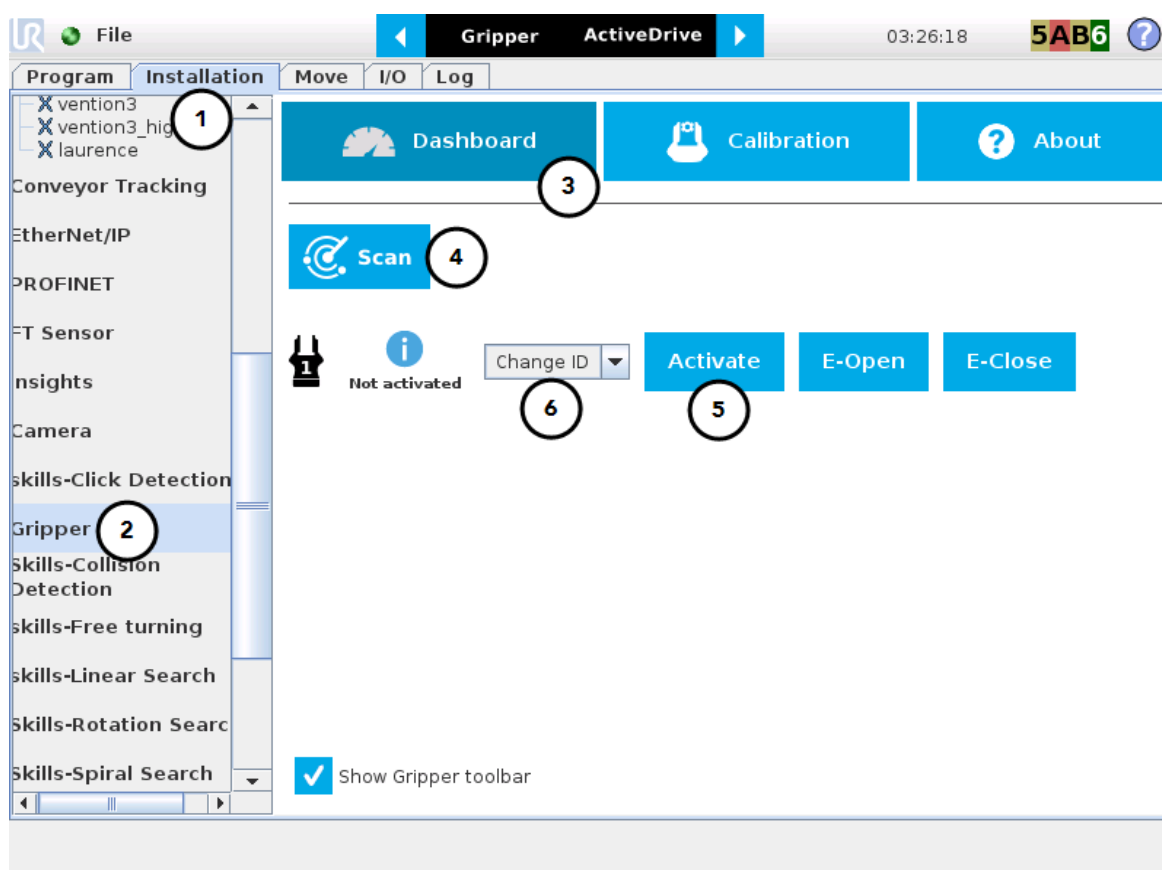


Fig. 4-9: Gripper Dashboard (steps highlighted)

- 6 If required, change the ID of the Gripper; ID numbers range from 1 to 4

Info

The default Gripper ID allocated to the first device connected is 1.



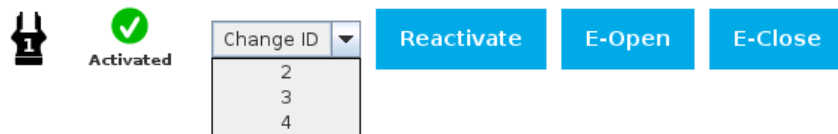


Fig. 4-10: Gripper ID modification in the Gripper Dashboard

Info

The **E-Open** and **E-Close** buttons allow the user to open and close the Gripper even if the latter is not activated. These emergency commands use very low speed and force settings.

Multiple Grippers

Info

The URCap allows the user to connect and control up to **4** Grippers at the same time.

Caution

If you wish to program using multiple grippers at the same time, connect them to the robot **one after the other**; each Gripper connection requiring its own **scan** and **activation** before going on to the next.

For each Gripper connected, follow steps 3 to 6 set out in the **Control** section.

Features

Icon	Functionality Name	Description
	Dashboard	After selecting Gripper in the left pane of the Installation tab, tap this button to access the Gripper Dashboard.
 	Activate/Reactivate	The Activate and Reactivate buttons enables the corresponding Gripper.
 Not activated  Activated	Not activated/Activated	Gripper status: One or the other of these icons is displayed next to the corresponding Gripper icon.
	Gripper	Label indicating the Gripper's ID, ranging from 1 to 4.
	Scan	Command allowing for a detection of connected Grippers. After tapping this button, the latter will display in the Gripper Dashboard.
	Change ID	Tap the arrow next to the textbox to expand a list of available Gripper IDs. Selecting an available number will allocate it to the corresponding Gripper.
	E-Open	Tapping this button sends an emergency open command to the corresponding Gripper, whether it is activated or not.
	E-Close	Tapping this button sends an emergency close command to the corresponding Gripper, whether it is activated or not.



4.8.2. Gripper Calibration menu and wizard

Depending on the fingers/fingertips used, the operator will calibrate the Gripper and measure the stroke by defining the fully open and fully closed positions.

Make sure the Gripper is activated.

- 1 Go to the **Installation** tab
- 2 Select **Gripper** in the left pane

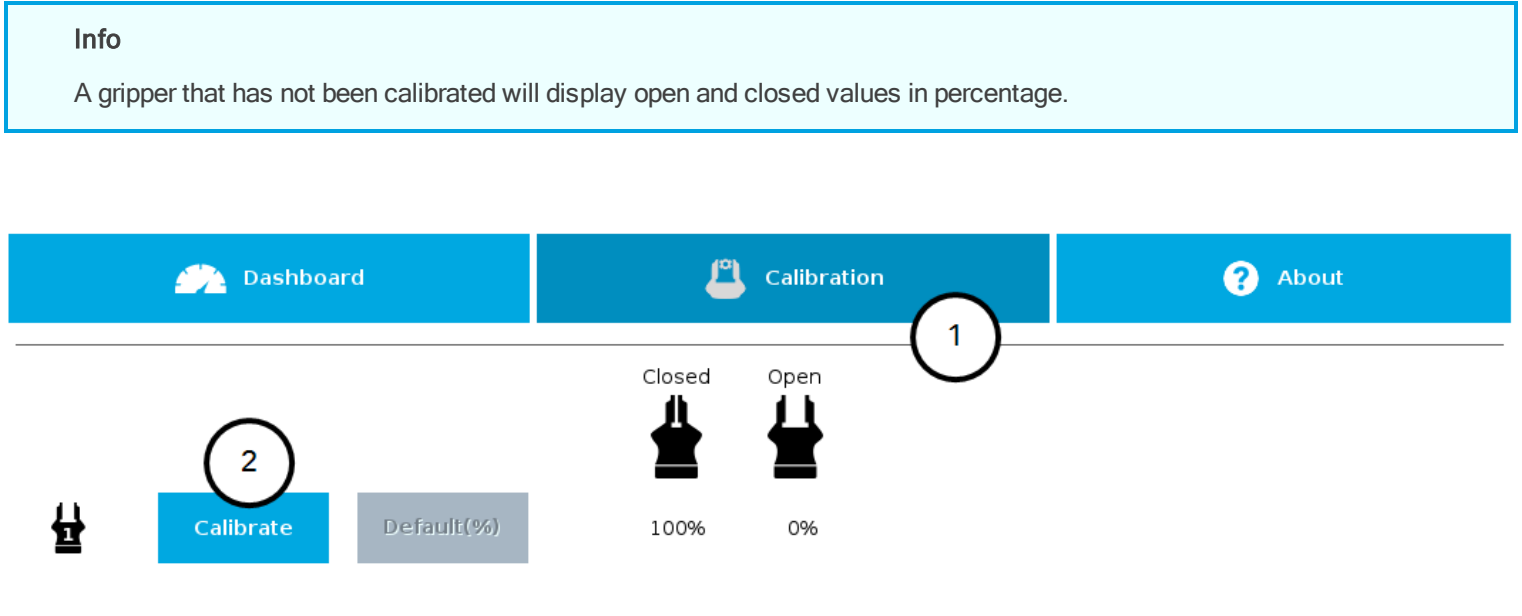


Fig. 4-11: Sequence toward calibration, single Gripper

- 1 Tap the Calibration tab
- 2 Tap the Calibrate button corresponding to the Gripper you want to calibrate

Tapping the **Calibrate** button in the Gripper Calibration menu launches the **Calibration wizard**.

Follow the instructions on the screen of the teach pendant.

Tip

Depending on the PolyScope settings defined by the user, the units of measurement can either display according to the metric or imperial system.

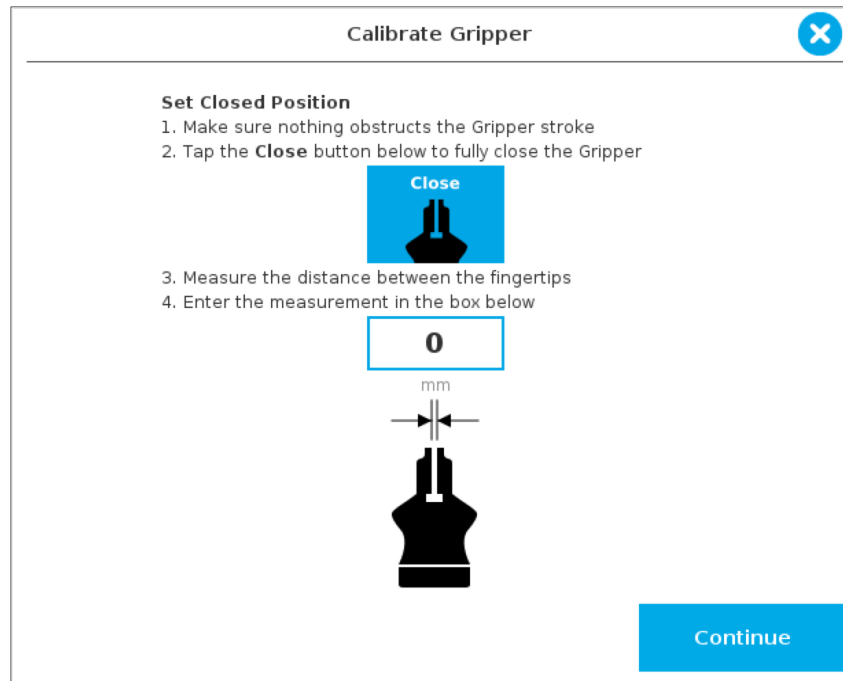


Fig. 4-12: First step of the calibration wizard

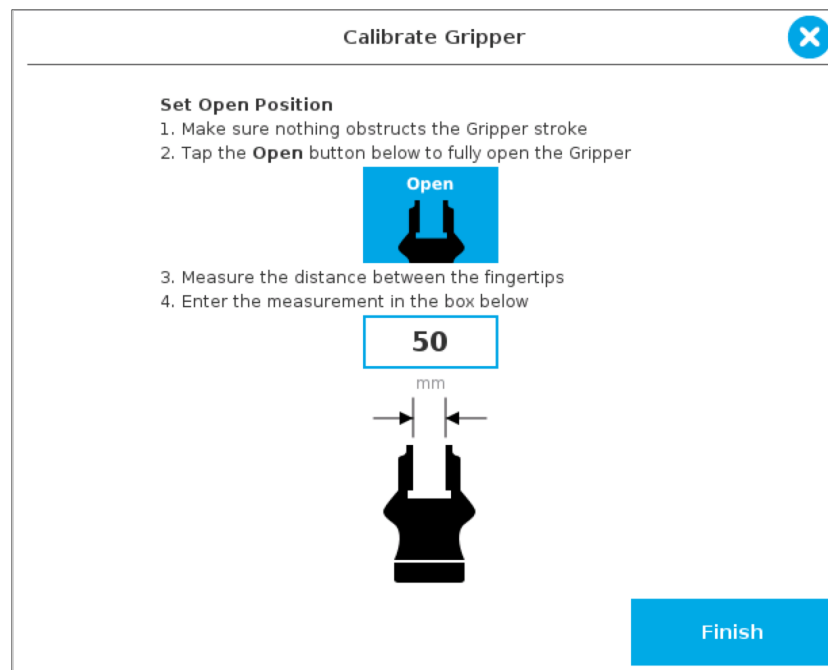


Fig. 4-13: Second step of the calibration wizard

Following their respective calibration, the Grippers will display their actual stroke, from the fully closed to the fully open position.

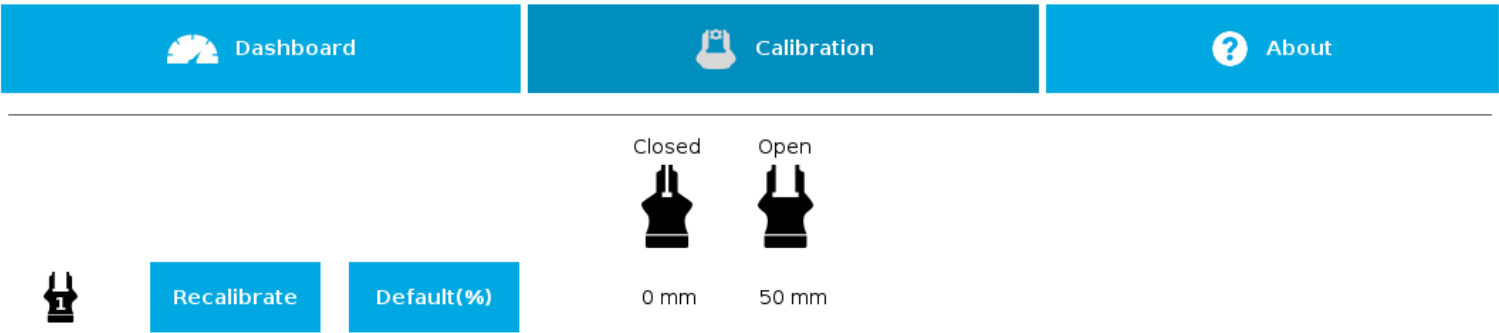
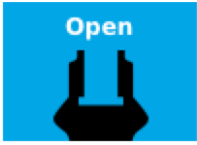


Fig. 4-14: Single Gripper, calibrated

Features

Icon	Functionality Name	Description
 Calibration	Calibration	After selecting the Gripper in the left pane of the Installation tab, tap this button to access the Gripper Calibration menu.
 	Calibrate/Recalibrate	The Calibrate and Recalibrate buttons launch the Gripper Calibration wizard.
	Default (%)	Following a calibration, tap this button to reset the calibration and return to the default percentage values. The button is greyed out if no calibration has been performed for the corresponding Gripper.
	Gripper	Label indicating the Gripper's ID , ranging from 1 to 4.
	Fully closed dimension table header	Visual aid above the fully closed dimensions in the Calibration menu.
	Fully open dimension table header	Visual aid above the fully open dimensions in the Calibration menu

Icon	Functionality Name	Description
	Close	Tap this button to close the Gripper in the Calibration wizard.
	Open	Tap this button to open the Gripper in the Calibration wizard.
	Fully closed	Visual aid for entering the fully closed dimension in the Calibration wizard.
	Fully open	Visual aid for entering the fully open dimension in the Calibration wizard.
	Continue	Tap this button at the end of the first step of the Calibration wizard in order to go to the next step.
	Finish	Tap this button at the the end of the Calibration wizard to complete the calibration process for the corresponding Gripper.



4.8.3. Gripper Toolbar

The Gripper Toolbar is generally used to test and jog the Gripper, meaning it has no effect on the programming of the robot.

Info

The Gripper URCap package contains the Gripper toolbar. Therefore, the toolbar is automatically installed with the URCap package.

Info

The Gripper toolbar can be installed and used without the URCap package (please refer to the **Control over Universal Robots without URCaps** section for information on how to install the toolbar and the driver package without the URCap).

Overview

Prior to using the Gripper Toolbar, the user needs to go to the Gripper Dashboard and tick the Show Gripper Toolbar. Please refer to the **Control** section for visual support.

Right after the installation of the Gripper URCap, a collapsed Gripper toolbar will display at the top of the screen; please refer to the figure below.

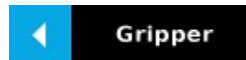


Fig. 4-15: Gripper toolbar (collapsed)

An expanded Gripper Toolbar prior to the activation of the Gripper looks like the figure below:

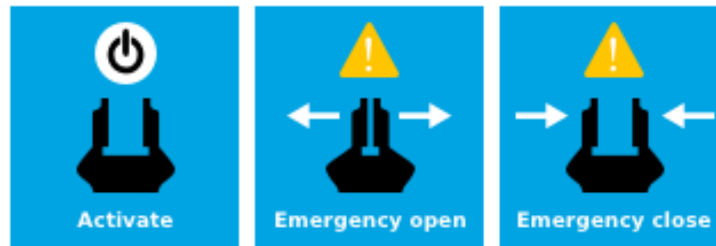


Fig. 4-16: Gripper toolbar before activation

Info

The **Emergency open** and **Emergency close** buttons allow the user to open and close the Gripper even if the latter is not activated.

The emergency commands use very low speed and force settings.

- Tap the **Activate** button to enable the Gripper

Single Gripper

Caution

If the Gripper has not been calibrated at this point, the Open and Close values will remain in percentage. Please refer to the **Control** section for instructions.

Once the Gripper is activated, the toolbar can be expanded and used to test and jog the Gripper

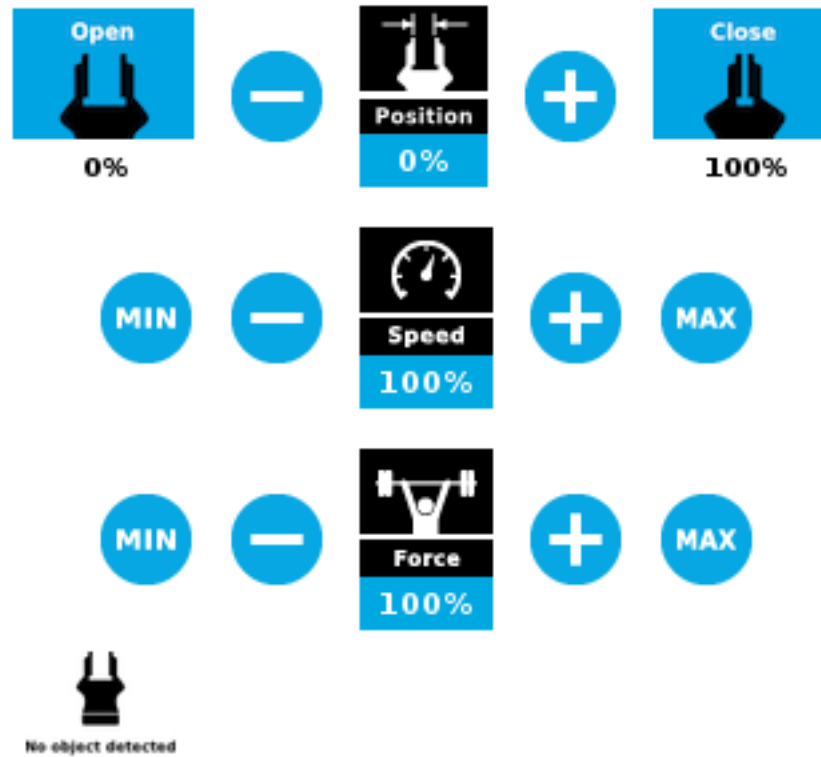


Fig. 4-17: Gripper toolbar after activation



Multiple Grippers

If multiple Grippers are used, check one or several Grippers and test their settings and features.

Tip

The first Gripper selected is the "master", meaning that its settings will be copied over to the others when they are selected. In order to modify each Gripper's settings individually, select a Gripper, edit it, uncheck its box and carry on to the next Gripper.

Info

For consistency, selecting multiple Grippers at the same time displays the open and close values in percentage since their stroke might be different.

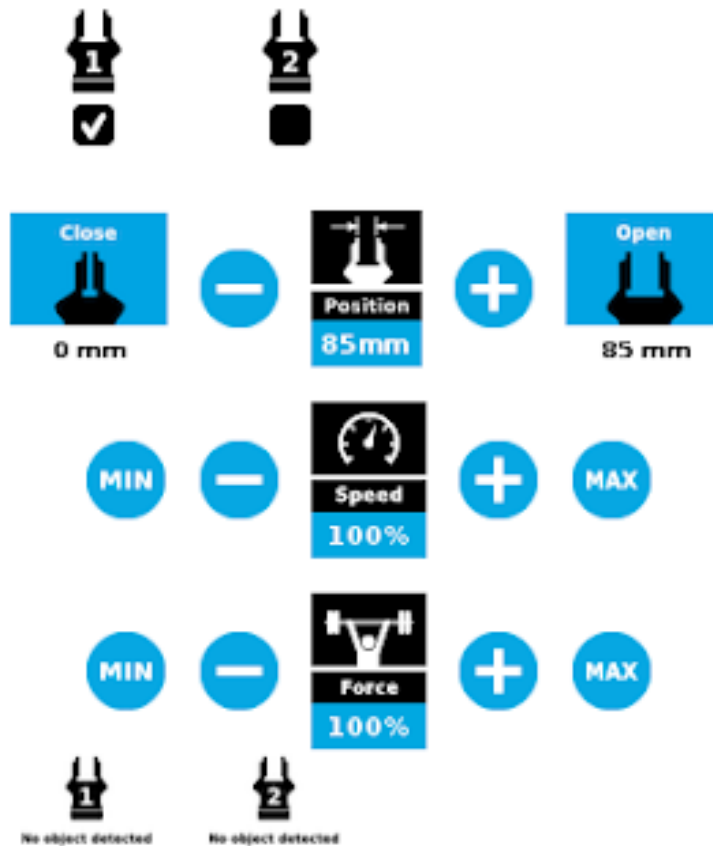
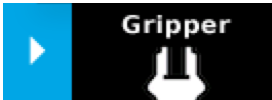


Fig. 4-18: Gripper toolbar (expanded, multiple Grippers)

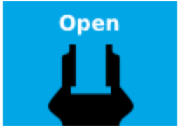
Features

Gripper activation

Icon	Functionality Name	Description
 	Gripper toolbar	Tap to toggle between the expanded and collapsed Gripper toolbar. When greyed out, the functionality is not available.
 	UR+ Menu (Gripper toolbar)	Tap to toggle between the expanded (blue) and collapsed (black) Gripper toolbar.
	Activate	Tap to activate the Gripper. The Gripper will fully open and close to set the zero of the position value.
	Emergency open	Slowly moves the Gripper to its fully closed position.
	Emergency close	Slowly moves the Gripper to its fully open position.



Gripper operation window

Primary Icon	Functionality Name	Description
 	Gripper toolbar	Tap to toggle between expanded and collapsed Gripper toolbar. When greyed out, the functionality is not available.
 	UR+ Menu (Gripper toolbar)	Tap to toggle between the expanded (blue) and collapsed (black) Gripper toolbar.
	Open	Tap to fully open the Gripper.
	Close	Tap to fully close the Gripper.
	Requested position	Shows the actual position of the Gripper according to the system of units defined in PolyScope (imperial vs. metric). If the Gripper is not calibrated, the values will display in percentage.
	Requested speed	Shows the actual speed set of the Gripper. 0% = minimum speed 100% = maximum speed
	Requested force	Shows the actual force set of the Gripper: 0% : minimum force, regrasp feature disabled 1% : minimum force, with regrasp feature enabled 100% : maximum force, with regrasp feature enabled
	Plus	Tap to increase the corresponding parameter.

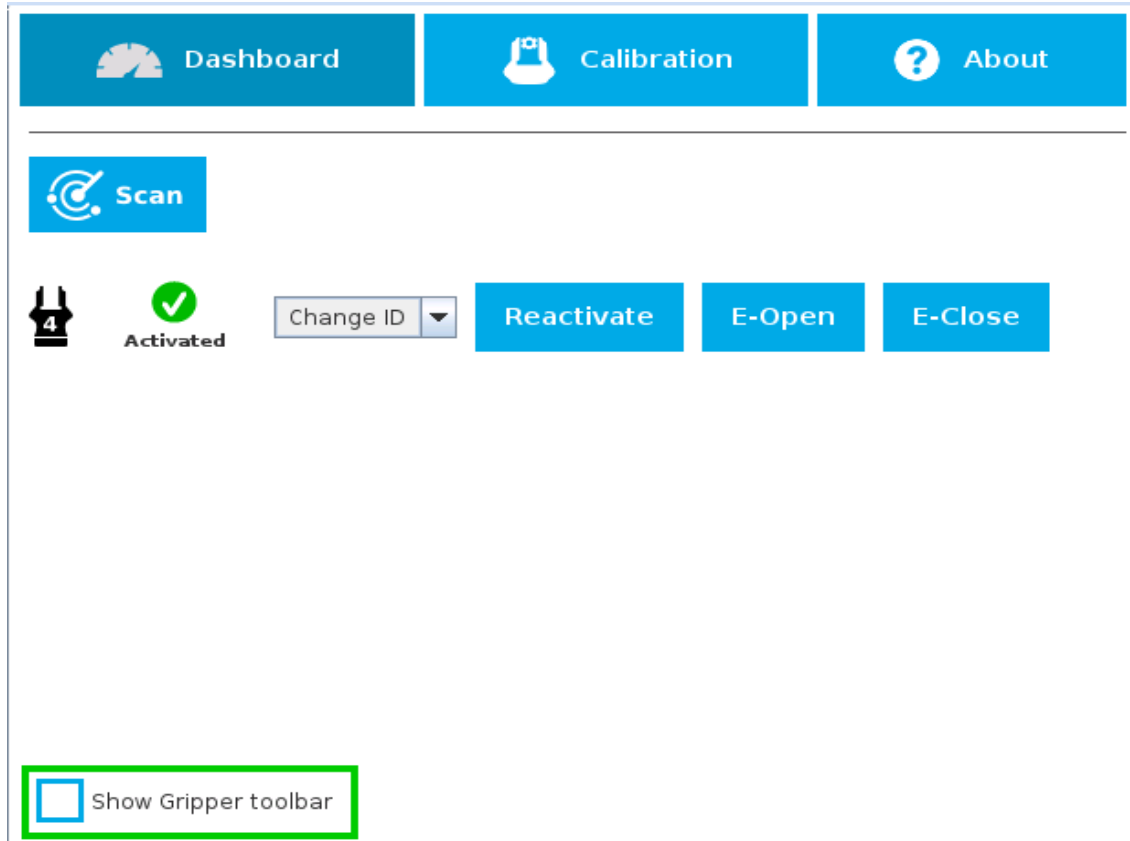
Primary Icon	Functionality Name	Description
	Minus	Tap to decrease the corresponding parameter.
 No object detected	No object detected	Icon shown when no object is detected during a grasp.
 Object detected	Object detected	The icon shows a green checkmark when an object is detected during a grasp.



Show/hide Gripper Toolbar

It is possible to toggle between a shown and a hidden Gripper Toolbar.

- 1 Create or load a robot program
- 2 Tap the **Installation** tab
- 3 Select **Gripper** in the left pane
- 4 Tap the **Dashboard** submenu
- 5 Check/Uncheck the box in the lower left corner



4.8.4. Gripper Node

To add and edit a Gripper node inside your robot program, follow the steps below:

- 1 Tap **Program Robot**.
- 2 Open an empty program or load one.
- 3 Go the **Structure** tab and tap the **URCaps** submenu.
- 4 Tap the **Gripper** button.
- 5 Go to the **Command** tab to edit the **Gripper** node inserted.
- 6 Tap the **Edit** action button to edit the node's parameters.



Info

The Gripper needs to be activated before it can receive a command. To do so, use the `rq_activate_and_wait()` function script at the beginning of the program or activate with the Gripper toolbar. Starting from Gripper URCap version 1.0.2, the gripper activation script command (`rq_activate()` or `rq_activate_and_wait()`) must be preceded by the reset command (`rq_reset()`). This will force the gripper's activation routine. Thus, if the gripper was already activated, it will activate again.

Command window

The Command window shows the requested action parameters for the Gripper node. Depending on the position, speed and force parameters, warning messages may display.

To edit the Gripper node, tap **Edit action** (please refer to the **Edit action screen** section for more details).

Gripper

Gripper 1



✓ Complete gripper motion before next action

⚠ Gripper action undefined. Please edit action.

When the node is not defined, the Command window shows warning messages. Tap **Edit action** to modify the Gripper's action.

Caution

Tap the **Go to position** button prior to editing the action. Whereas the command window displays the command that the Gripper should execute in this particular node, the Edit action screen displays the current Gripper position.

Gripper

Gripper 1



⚠ 50mm



55%



55%



✓ Complete gripper motion before next action

⚠ Warning: if you are picking an object, use close (0mm) or open (50mm) position.

When the node requests the Gripper to close/open, the Command window warns that a partial opening/closing should not be performed for picking an object.

- For positioning and partial movement of the hand, moving to a specific position (between the fully open and fully close positions) is suitable.
- To pick an object, use the close (or open) actions along with the object detection feature (please refer to the **Edit action screen** section).

Info

When the action is set for multiple Grippers, the header lists the selected Grippers (e.g. Grippers 1,2,3,4)



Gripper

Gripper 1



Position

0mm

Close



Speed

55%



Force

55%



Go to position



Edit action

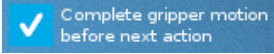
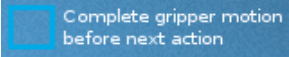


Complete gripper motion
before next action

When the Gripper action is set to **close**, a green "Close" icon appears.

When the Gripper action is set to **open**, a green "Open" icon appears.

Features

Icon	Functionality Name	Description
	Requested position	Shows the position requested for the current Gripper node. Read only.
	Requested speed	Shows the speed requested for the selected Gripper node. Read only.
	Requested force	Shows the force requested for the selected Gripper node. Read only.
	Go to position	Sends all node parameters (position, force, speed) to the Gripper. Note: This button is disabled (grey) when the Gripper is already at the node position or when the Gripper is not powered/activated.
	Edit action	Opens the Edit action screen, where you can edit the Gripper's action parameters.
	Complete motion	Selected by default. Selected: The Gripper completes its motion before executing the next action on the PolyScope program tree. This is slower but safer.
	Do not complete motion	Selected by default. Unselected: As soon as the Gripper starts moving, PolyScope will execute the next action on the program tree. This allows you to move the Gripper and robot at the same time.
	Warning message section	
	Gripper not activated	"Warning, the Gripper is not powered or not activated. The Gripper will not move." Make sure that the Gripper is installed correctly and powered. You need to activate it using the Gripper toolbar before running Gripper commands.
	Gripper action undefined	"Gripper action undefined! Please edit action." When creating a new node, all action parameters are undefined. Tap the edit action button to define parameters.



Icon	Functionality Name	Description
	Object picking warning	“Warning, if you are picking an object, use close (100%) or open (0%) position.” It is highly recommended to only use 0% or 100% position when trying to pick an object.
	Node name	Node name is automatically set to: • Gripper Open, when position = 0% or fully open value as defined in the Calibration wizard • Gripper Close, when position = 100% or fully closed value as defined in the Calibration wizard. • Gripper Move x%/mm/in, when position is between the fully open and fully closed values.
	Open request	When position requested is 0%, or the fully opened value defined in the Calibration wizard.
	Close request	When position request is 100%.

Edit action screen

Tap **Edit action** from the Command window and use the Edit action screen to edit the Gripper's parameters and adjust its position, speed and force. In normal usage conditions, the Gripper will move as soon as you change the position setting. The Gripper will not move if it is not properly powered, connected and activated.

Caution

The Gripper needs to be activated before it can receive a command. To do so, use the `rq_activate_and_wait()` function script at the beginning of the program or activate with the Gripper toolbar. Starting from Gripper URCap version 1.0.2, the gripper activation script command (`rq_activate()` or `rq_activate_and_wait()`) must be preceded by the reset command (`rq_reset()`). This will force the gripper's activation routine. Thus, if the gripper was already activated, it will activate again.

Single Gripper

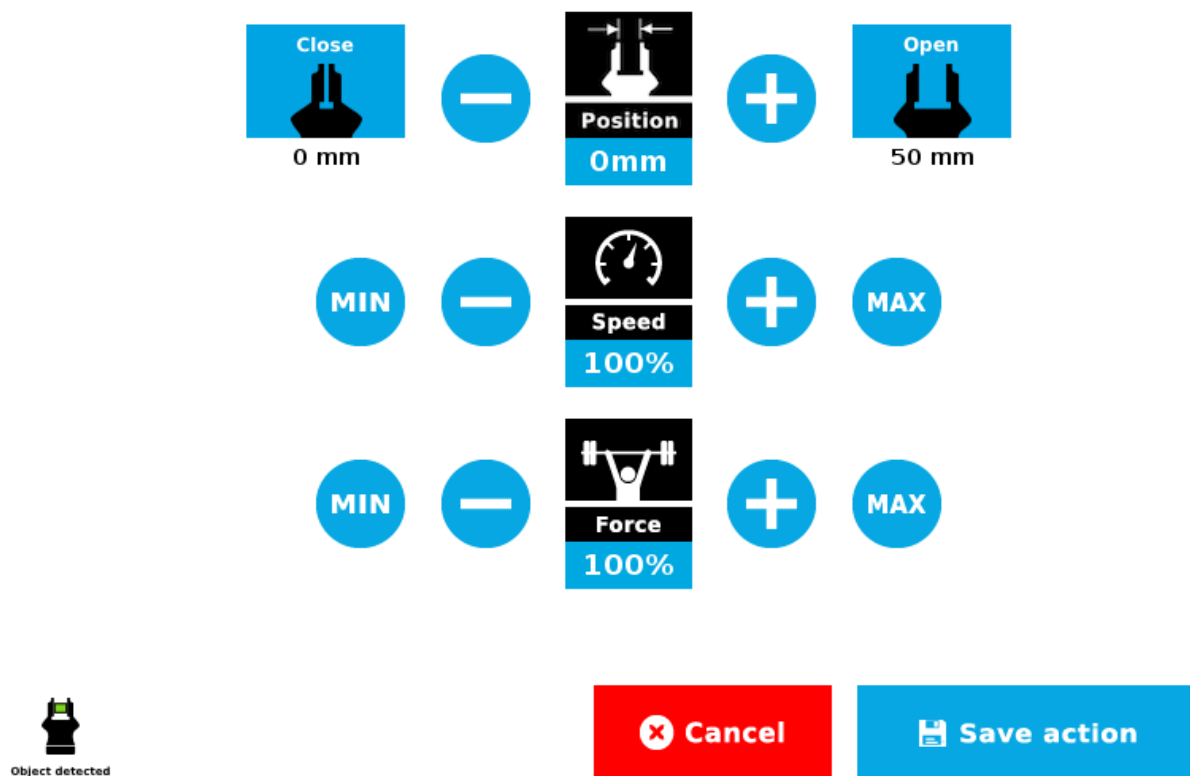
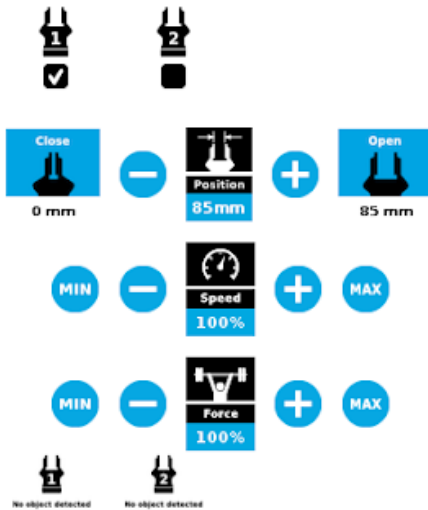


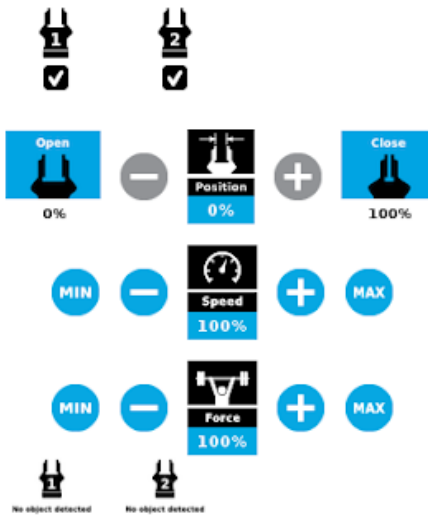
Fig. 4-19: Edit action screen



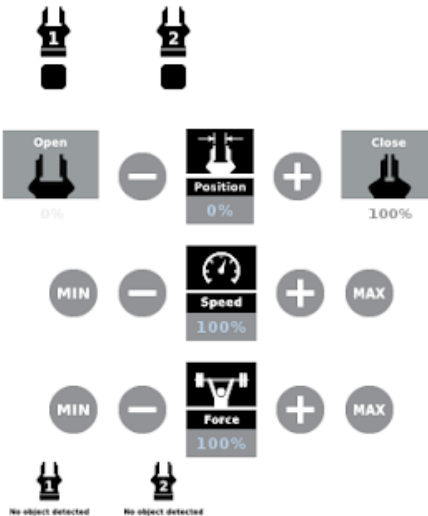
Multiple Grippers



- When selecting one gripper, all buttons are available. The action will apply to the selected gripper.
- When jogging the position, the selected gripper will move automatically.
- It is possible to jog the position (%).



- When selecting two or more grippers, it is not possible to jog the position.
- To jog the position, select only one gripper, jog the position and then select all grippers. The second gripper will move to the requested position.



- When no grippers are selected, the action cannot be edited.

Features

Primary Icon	Functionality Name	Description
	Open	Tap to fully open the Gripper. Use this when you want to grasp an object (internal grip), for example.
	Close	Tap to fully close the Gripper. Use this when you want to grasp an object (external grip), for example.
	Requested position	Shows the position request for the selected Gripper node.
	Requested speed	Shows the speed requested for the selected Gripper node. Use the Plus and Minus icons to modify.
	Requested force	Shows the force requested for the selected Gripper node. Use the Plus and Minus icons to modify.
	Plus	Tap to increase the corresponding parameter.
	Minus	Tap to decrease the corresponding parameter.
	Cancel	Cancels modifications and brings back the Gripper node command screen.
	Save action	Saves settings into the selected Gripper node and brings back the Gripper node command screen. Note: This functionality is not available (grey) when the Gripper has the same action parameters as the selected Gripper node.
	No object detected	Icon shown when no object is detected during a grasp.
	Object detected	Icon shows a green checkmark when an object is detected during a grasp.



Warning

If the Gripper is powered and initialized, it will move as soon as a change to the parameter is made (position, speed, force).

Tip

Change speed and force when grasping objects. In all other cases, it is recommended to use maximum speed and force.

Grip Check node

In order to add and edit a Grip Check node:

Grip Check

Gripper ID 1

Check if object is...

☒ **Detected** ☐ **Not detected**

and...

☐ **Validate grip with dimension**

Fig. 4-20: Grip Check node interface

Grip Check

Gripper ID 1

Check if object is...

☒ Detected ☐ Not detected

and...

☒ Validate grip with dimension

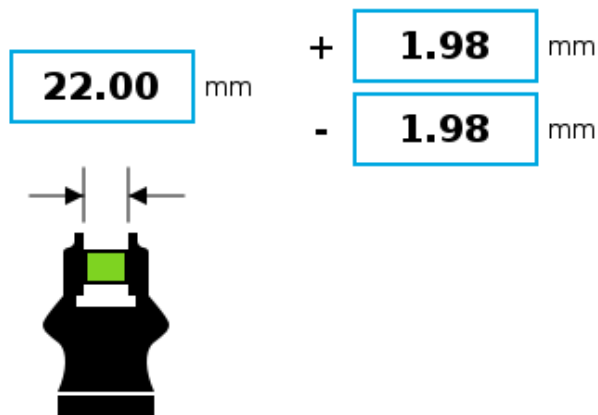


Fig. 4-21: Grip Check node interface with dimension validation

Given that the Grip Check node is an If statement, tapping the actual node in the Command tab brings up the option to add an **Else** instruction to the program tree.

About

In this window, the user can access information such as the Gripper's hardware, software and firmware versions, the number of cycles completed by the device and more.

- 1 Create or load a robot program
- 2 Tap the **Installation** tab
- 3 Select **Gripper** in the left pane
- 4 Tap the **About** submenu



Dashboard

Calibration

About

URCap version: UCG-1.2.1
 PolyScope version: 3.5.4.10845
 Robot serial number: 2016352451

Gripper ID

2-Finger Gripper

Serial number: abc-1684348960
 Firmware version: GC3-1.4.1
 Number of cycles: 73873



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If the Gripper firmware version is obsolete, a warning message will display and an **Upgrade Firmware** button will prompt the user to upgrade the firmware to the latest available version.

Gripper

Dashboard

Calibration

About

URCap version: UCG-1.5.1
 PolyScope version: 3.7.1.40229
 Robot serial number: 2016353035

Gripper ID

2-Finger Gripper

Serial number: N/A
 Number of cycles: 269675
 Firmware version: **GC3-1.6.0**

Robotiq gripper's firmware version is obsolete
 Upgrade firmware

The following grippers have obsolete firmware: Gripper 1, Gripper 2

Visit support.robotiq.com to get the latest instruction manual and further information on how to use your product.

Error messages overview

If a program is running without the Gripper being activated, the program will stop at the execution of a Gripper node and the following message will be displayed in a popup. Tap Stop Program and activate the gripper with the toolbar to continue or add an activation command in the program (see Advanced Gripper Functions).





Caution

Starting from Gripper URCap version 1.0.2, the gripper activation script command (**rq_activate()** or **rq_activate_and_wait()**) must be preceded by the reset command (**rq_reset()**). This will force the gripper's activation routine. Thus, if the gripper was already activated, it will activate again.



Gripper Program Template

With the URCap package installation, a program template will automatically be installed in the program folder of PolyScope. It contains comments and pop up windows that explain Gripper nodes functions and advanced functions.

Load and open the gripper_tutorial.urp program located in the root program folder. Make sure the Gripper is well connected to the UR controller. Run the program. You will see pop up windows with comments on the Gripper nodes and advanced functions.

Tip

Use the gripper_tutorial.urp to know how to program the Gripper using nodes and advanced functions. You can copy the tutorial and start a new program based on it.



Advanced Gripper Functions

The use of Gripper URCaps nodes allows to move the Gripper and modify its speed and force. However, some advanced functions are only available through UR Script commands.

Info

The Gripper needs to be activated before it can receive a command. To do so, use the **rq_activate_and_wait()** function script at the beginning of the program or activate with the Gripper toolbar.

Single Gripper

Name	Description
rq_activate()	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution continues before the end of activation.
rq_activate_and_wait()	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution waits for the activation.
rq_auto_release_close_and_wait()	Slowly moves the Gripper to its maximum closed position. The Gripper must be activated after this command. Meant for emergency procedures.
rq_auto_release_open_and_wait()	Slowly moves the Gripper to its maximum opened position. The Gripper must be activated after this command. Meant for emergency procedures.
rq_close()	Moves the Gripper its fully closed position.
rq_close_and_wait()	Moves the Gripper to its fully closed position and waits until the motion is completed to execute the next command.
rq_current_pos()	Returns the current position of the fingers, from 0 to 255. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
rq_current_pos_norm()	Returns the current position of the fingers, normalized from 0% to 100%. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
rq_current_pos_mm()	Returns the current position of the fingers in millimeters, considering that the PolyScope configuration has been set to use the metric system as measurement system. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
rq_current_pos_inches()	Returns the current position of the fingers in inches, considering that the PolyScope configuration has been set to use the imperial system as measurement system. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
rq_get_gripper_max_cur(gripperId)	Returns the maximum supply current value of the appropriate Gripper in mA.
rq_go_to()	Sets the Go To bit to 1.
rq_is_gripper_activated()	Sets global variable rq_gripper_act to 1 if the Gripper is activated and returns True. Otherwise sets the variable to 0 and returns False.



Name	Description
<code>rq_gripper_led_off()</code>	Forces the LED to OFF status.
<code>rq_gripper_led_on()</code>	Allows the firmware to turn ON the LEDs when they have been forced to OFF status.
<code>rq_is_motion_complete()</code>	Returns True if the motion of the Gripper is complete and False otherwise. Can be useful to synchronize a program after a Gripper node is run without the option "Complete Gripper motion before next action".
<code>rq_is_object_detected()</code>	Returns True if the Gripper motion has stopped due to an object. Returns False if Gripper motion was not impeded by an object. Useful for verifying that an object has been picked correctly before going to the next step. Note: this function is not reliable for very small objects. Refer to the user manual for more details.
<code>rq_move_and_wait("position")</code>	Moves the Gripper to the position defined by the argument and waits until the motion is completed.
<code>rq_move("position")</code>	Moves the Gripper to the position defined by the argument.
<code>rq_move_and_wait_norm("position")</code>	Moves the Gripper to the position defined by the argument, normalized from 0% to 100%. Waits until the motion is completed.
<code>rq_move_mm()</code>	Moves the Gripper to the position defined by the argument.
<code>rq_move_and_wait_mm()</code>	Moves the Gripper to the position defined by the argument.
<code>rq_move_inches()</code>	Moves the Gripper to the position defined by the argument.
<code>rq_move_and_wait_inches()</code>	Moves the Gripper to the position defined by the argument.
<code>rq_move_norm("position")</code>	Moves the Gripper to the position defined by the argument, normalized from 0% to 100%.
<code>rq_open()</code>	Moves the Gripper to its fully opened position.
<code>rq_open_and_wait()</code>	Moves the Gripper to its fully opened position and waits until the motion is completed.
<code>rq_print_gripper_driver_state()</code>	Prints the driver's state to the UR log window.
<code>rq_print_gripper_driver_version()</code>	Prints the driver version to the UR log window.
<code>rq_print_gripper_fault_code()</code>	Prints the Gripper's fault code in the UR log window.
<code>rq_print_gripper_firmware_version()</code>	Prints the Gripper's firmware version to the UR log window.
<code>rq_print_gripper_num_cycles()</code>	Prints the Gripper's number of cycles in the UR log window.
<code>rq_print_gripper_serial_number</code>	Prints the Gripper's serial number in the UR log window.
<code>rq_reset()</code>	Resets the Gripper's activation status. Use after an autorelease before activating the Gripper.
<code>rq_set_force("force")</code>	Sets the Gripper's force defined by the argument, from 0 to 255.
<code>rq_set_force_norm("force")</code>	Sets the Gripper's force defined by the argument, normalized from 0% to 100%.
<code>rq_set_gripper_max_cur(current_mA, gripperId)</code>	Sets the Gripper's maximum supplycurrent in mA.
<code>rq_set_speed("speed")</code>	Sets the Gripper's speed defined by the argument, from 0 to 255.
<code>rq_set_speed_norm("speed")</code>	Sets the Gripper's speed defined by the argument, normalized from 0% to 100%.



Name	Description
rq_stop()	Stops the Gripper's motion. The Go To bit is set to 0.



Multiple Grippers

Here is an example of how to use the script commands with more than one gripper.

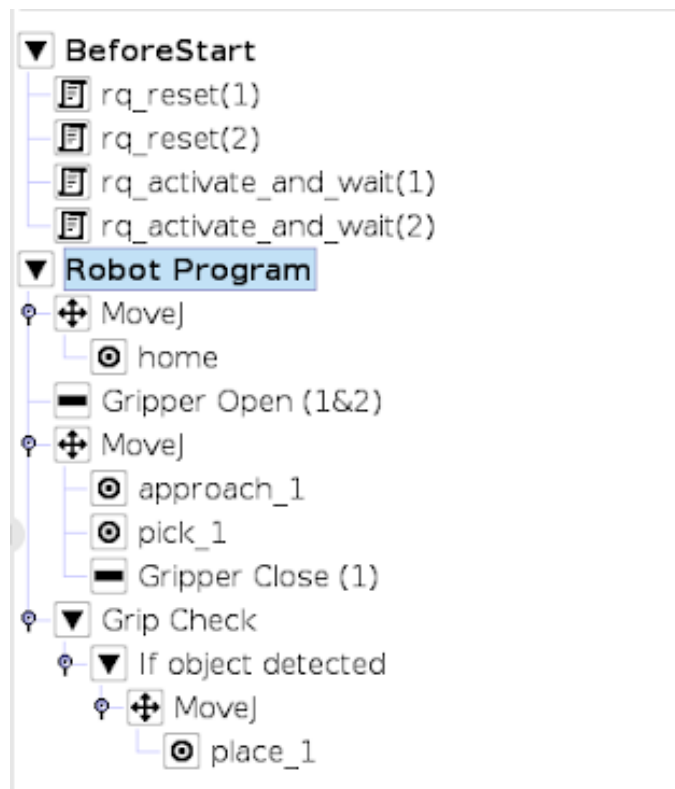


Fig. 4-22: Example of UR script commands with multiple Grippers

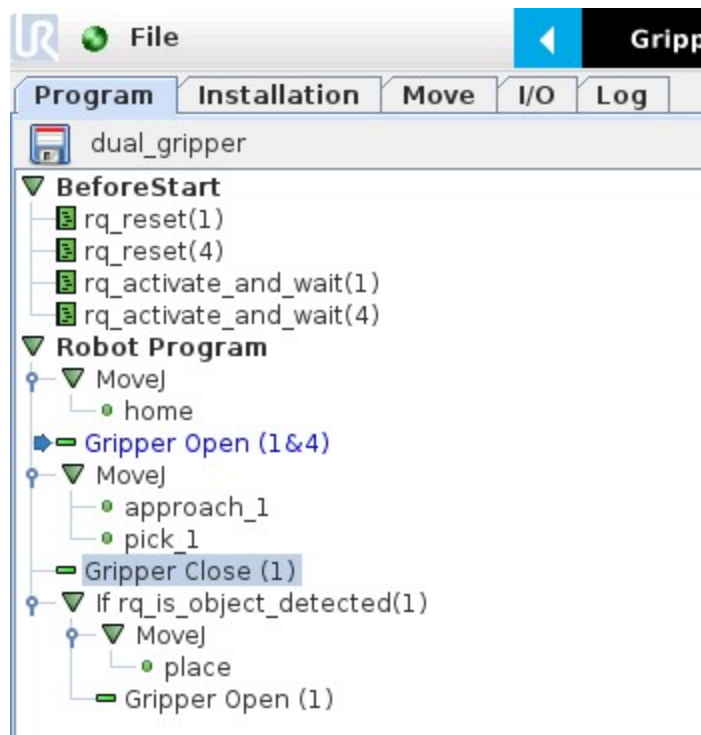


Fig. 4-23: Example of UR script commands with multiple Grippers

The functions listed below can be called using subprograms in PolyScope or with a script. Functions with the suffix **_and_wait** will wait for the Gripper to complete its operation before going to the next step.

Name	Description
<code>rq_activate("gripper ID")</code>	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution continues before the end of activation.
<code>rq_activate_and_wait("gripper ID")</code>	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution waits for the activation.
<code>rq_auto_release_close_and_wait("gripperID")</code>	Slowly moves the Gripper to its maximum closed position. The Gripper must be activated after this command. Meant for emergency procedures.
<code>rq_auto_release_open_and_wait("gripperID")</code>	Slowly moves the Gripper to its maximum opened position. The Gripper must be activated after this command. Meant for emergency procedures.
<code>rq_close("gripper ID")</code>	Moves the Gripper its fully closed position.
<code>rq_close_and_wait("gripper ID")</code>	Moves the Gripper to its fully closed position and waits until the motion is completed to execute the next command.
<code>rq_current_pos("gripper ID")</code>	Returns the current position of the fingers, from 0 to 255. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
<code>rq_current_pos_norm("gripper ID")</code>	Returns the current position of the fingers, normalized from 0% to 100%. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
<code>rq_current_pos_mm("gripper ID")</code>	Returns the current position of the fingers in millimeters, considering that the PolyScope configuration has been set to use the metric system as measurement system. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
<code>rq_current_pos_in("gripper ID")</code>	Returns the current position of the fingers in inches, considering that the PolyScope configuration has been set to use the imperial system as measurement system. Can be used to verify that the pick is successful, by comparing the finger position when the pick was taught with the similar-sized object.
<code>rq_go_to("gripper ID")</code>	Sets the Go To bit to 1.
<code>rq_is_gripper_activated("gripper ID")</code>	Sets global variable rq_gripper_act to 1 if the Gripper is activated and returns True. Otherwise sets the variable to 0 and returns False.
<code>rq_is_motion_complete("gripper ID")</code>	Returns True if the motion of the Gripper is complete and False otherwise. Can be useful to synchronize a program after a Gripper node is run without the option "Complete Gripper motion before next action".
<code>rq_is_object_detected("gripper ID")</code>	Returns True if the Gripper motion has stopped due to an object. Returns False if Gripper motion was not impeded by an object. Useful for verifying that an object has been picked correctly before going to the next step. Note: this function is not reliable for very small objects. Refer to the user manual for more details.
<code>rq_move_and_wait("position", "gripper ID")</code>	Moves the Gripper to the position defined by the argument and waits until the motion is completed.
<code>rq_move("position")</code>	Moves the Gripper to the position defined by the argument.



Name	Description
<code>rq_move_and_wait_norm</code> ("position")	Moves the Gripper to the position defined by the argument, normalized from 0% to 100%. Waits until the motion is completed.
<code>rq_move_norm</code> ("position", "gripper ID")	Moves the Gripper to the position defined by the argument, normalized from 0% to 100%.
<code>rq_open</code> ("gripper ID")	Moves the Gripper to its fully opened position.
<code>rq_open_and_wait</code> ("gripper ID")	Moves the Gripper to its fully opened position and waits until the motion is completed.
<code>rq_print_gripper_driver_state</code> ("gripper ID")	Prints the driver's state to the UR log window.
<code>rq_print_gripper_driver_version</code> ("gripper ID")	Prints the driver version to the UR log window.
<code>rq_print_gripper_fault_code</code> ("gripper ID")	Prints the Gripper's fault code in the UR log window.
<code>rq_print_gripper_firmware_version</code> ("gripper ID")	Prints the Gripper's firmware version to the UR log window.
<code>rq_print_gripper_num_cycles</code> ("gripper ID")	Prints the Gripper's number of cycles in the UR log window.
<code>rq_reset</code> ("gripper ID")	Resets the Gripper's activation status. Use after an autorelease before activating the Gripper.
<code>rq_set_force</code> ("force", "gripper ID")	Sets the Gripper's force defined by the argument, from 0 to 255.
<code>rq_set_force_norm</code> ("force", "gripper ID")	Sets the Gripper's force defined by the argument, normalized from 0% to 100%.
<code>rq_set_speed</code> ("speed", "gripper ID")	Sets the Gripper's speed defined by the argument, from 0 to 255.
<code>rq_set_speed_norm</code> ("speed", "gripper ID")	Sets the Gripper's speed defined by the argument, normalized from 0% to 100%.
<code>rq_stop</code> ("gripper ID")	Stops the Gripper's motion. The Go To bit is set to 0.



4.8.5. Retro-compatibility of URCaps with legacy driver programs

The Gripper URCap can be used within programs that were made with the legacy driver package. To do so, you need to do the following modifications:

1. Open your .urp program in PolyScope and execute it.
 - a. You should see an error message that tells you that some functions are double defined. This is because the URCap embeds the functions directly in the program preamble, so there is no need for a BeforeStart section.
2. Comment all the lines of the BeforeStart section from your older .urp program.
 - a. You will still get an error. It arises because we changed some functions names due to a naming conflict.
3. To solve the problem, you can either suppress those functions or rename them.

Info

Since they are probably not used in a production program, the fastest method would be to suppress them, as shown in the figure below.



Fig. 4-24: Older program's subprograms to suppress or rename to use with URCaps package.

Here is the list of the functions that were renamed:

- rq_print_fault_code -> rq_print_gripper_fault_code
- rq_print_num_cycles -> rq_print_gripper_num_cycles
- rq_print_driver_state -> rq_print_gripper_driver_state
- rq_print_firmware_version -> rq_print_gripper_firmware_version
- rq_print_driver_version -> rq_print_gripper_driver_version



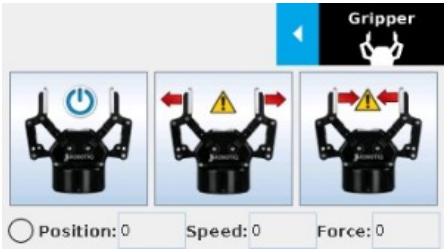
4.9. Control over Universal Robots without URCaps

If your Universal Robots’ controller is not compatible with the URCap package (refer to the **Installation for Universal Robots** section), you can install the driver package. This package allows programming of the Gripper with scripts in a PolyScope program. It includes program templates and examples to help you get started with your own custom program. It also contains the Gripper toolbar for jogging and controlling the Gripper.

4.9.1. Gripper Toolbar

Overview

The Gripper toolbar is automatically installed with the driver package. The Gripper toolbar allows you to jog and test the Gripper. It is a great tool to try grasps with the Gripper while programming.

Toolbar collapsed	Toolbar expanded, Gripper activation window	Toolbar expanded, Gripper operation window :
		
Tap the Gripper button to expand the toolbar.	• When the Gripper is not activated, the toolbar shows this window. • You need to tap the Activate button to be able to jog the Gripper. • Emergency open and close allows you to control the Gripper without activating, this mode will use a very low speed and force setting	Use the buttons of this window to jog and test the Gripper.

Features

Toolbar collapsed

Icon	Functionality Name	Description
 	Gripper toolbar	Tap to toggle between expand and collapse the 2-Finger Adaptive Gripper toolbar. When grey, the functionality is not available.
	Activate	Tap to activate the 2-Finger Adaptive Gripper. The Gripper will fully open and close to set the zero of the position value..

Icon	Functionality Name	Description
	Emergency open	Slowly moves the Gripper to its fully closed position.
	Emergency close	Slowly moves the Gripper to its fully open position.

Toolbar expanded

Icon	Functionality Name	Description
 	Gripper toolbar	Tap to toggle between expand and collapse the 2-Finger Adaptive Gripper toolbar. When grey, the functionality is not available.
	Open	Tap to fully open the Gripper.
	Close	Tap to fully close the Gripper.
Position	Requested position	Shows the actual position of the Gripper: 0% : fully open 100% : fully closed
Speed	Requested speed	Shows the actual speed set of the Gripper: 0% : minimum speed 100% : maximum speed
Force	Requested force	Shows the actual force set of the Gripper: 0% : minimum force, regrasp feature disabled 1% : minimum force, with regrasp feature enabled 100% : maximum force, with regrasp feature enabled
	Increase force	Tap to increase the force request.



Icon	Functionality Name	Description
	Decrease force	Tap to decrease the force request.
	Increase speed	Tap to increase the speed request..
	Decrease speed	Tap to decrease the speed request.
	No object detected	Icon shown when no object is detected during a grasp.
	Object detected	Icon shown when an object is detected during a grasp.

When communication with the Gripper is not established, the expanded toolbar shows the driver version:

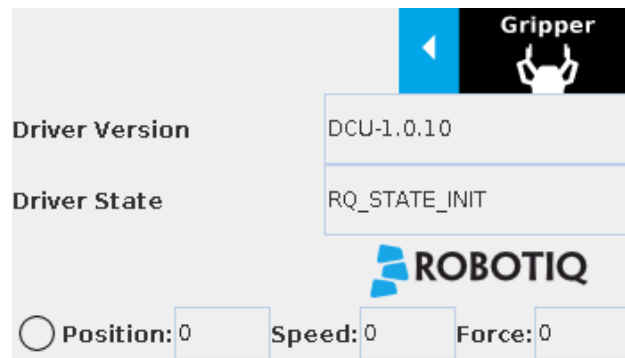


Fig. 4-25: Gripper toolbar expanded with driver version

Info

The driver state "RQ_STATE_INIT" means the driver is attempting to connect to a Robotiq Gripper. When connection is established, the normal toolbar detailed above will appear.

Tip

If you see the following toolbar with communication not established, check if your Gripper is powered first, then check if the RS-485 to USB converter is properly wired.

4.9.2. Demo Scripts

The following section details the demo scripts provided with the driver package.

pick_and_place_demo_with_subprograms.script is a demo script for pick and place applications using provided subprograms. The script uses subprograms included in the package such as **rq_set_force**, **rq_set_speed**, etc.



Info

All provided subprograms are identified with the prefix **rq_**.

The script uses **rq_speed** and **rq_force** as speed and force parameters to be used during the program. They can be modified using values from 0 to 255 (please refer to the **Provided Variables and Functions** section).

The script does the following actions in sequence :

- Assign initial values to the global variables in the **Init Variables** section.
- Initiate the communication with the Gripper in the **BeforeStart** section.
- The **Robot Program** section contains the commands sent to the Gripper:
 - Activate the Gripper with **SubP_rq_activate_and_wait**.

Info

Remember that all **_and_wait** subprograms will wait for the action to be completed before going to the next step.

- Move the robot to a predetermined position

Tip

Run with Universal Robots simulator first or make sure that the UR robot work area is totally cleared before running the script, as it will move the robot.

- Close the Gripper with **rq_close_and_wait**.
- Watch for object detection status:
 - If an object is detected, the script moves the robot and opens the Gripper.
 - If no object is detected, the script prompts a warning

.pick_and_place_demo_async_partial_opening_without_subprograms.script is similar to the previous demo, but without using subprograms. This demo uses asynchronous commands so that the robot and the Gripper will move at the same time (the previous example had the Gripper and the robot move separately).

4.9.3. Custom Programs

You can create your own program that commands the Gripper with the provided templates :

- Open **basic_template.script** in the list of provided templates;
- Push play to test the Gripper. The program will activate the Gripper and then do a loop of closing and opening the Gripper;
- Add your instructions under the robot program section. Program instructions can be added with PolyScope.

Tip

When programming an object pick up, use the **rq_is_object_detected** subprogram and **rq_object_detect** variable to know if an object has been picked. The subprogram sets the **rq_object_detect** variable to 1 if an object is detected, 0 otherwise.



Info

Subprograms with the **_and_wait** will wait for the instruction to be completed before going to the next step. For example, **rq_close_and_wait** will wait for the motion to be completed before continuing to the next step, while **rq_close** will initiate motion and go to the next programmed step.

As shown in the figure below, the **basic_template** program will execute these instructions in a sequence:

- Assign initial values to the global variables with the Init Variables section.
- Initiate communication with the Gripper with the Before Start section.
- The Robot Program section contains the commands sent to the Gripper:
 - Activate the Gripper with **SubP_rq_activate_and_wait**.
 - Close the Gripper with **rq_close_and_wait**.
 - Open the Gripper with **rq_open_and_wait**.

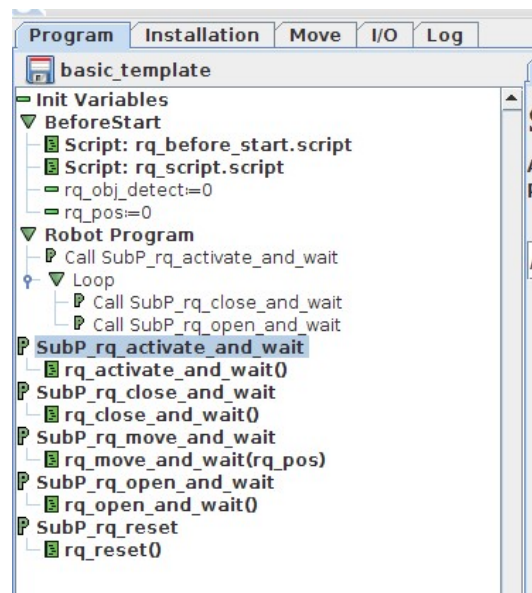


Fig. 4-26: Basic template as shown in UR PolyScope.

When using the **advanced_template**, you have access to all of the subprograms listed in the **Provided Variables and Functions** section. Unused subprograms can be removed from the list. You must use global variables to pass information, not arguments.

4.9.4. Provided Variables and Functions

The file **rq_script.script** contains function definitions and variables that enhance the programming of the Gripper. Here is the list of these variables and functions.

Variables are used when programming the Gripper using subprograms. Since it is not possible to pass arguments to subprograms, the global variables listed below must be used.

Name	Range	Description
rq_force()	[0-255]	Force set point. Change this variable by calling the subprogram SubP_rq_set_force .
rq_gripper_act()	[0-1]	1 if the Gripper is activated. This variable is updated by calling either SubP_rq_is_gripper_activated or SubP_rq_activate_and_wait .



Name	Range	Description
rq_move_complete()	[0-1]	1 if the motion is complete. This variable is updated by calling one of the following: • SubP_rq_move_and_wait; • SubP_rq_open_and_wait; • SubP_rq_close_and_wait; • SubP_rq_is_motion_complete.
rq_object_detect()	[0-1]	1 if an object is detected. This variable is updated by calling one of the following: • SubP_rq_move_and_wait; • SubP_rq_open_and_wait; • SubP_rq_close_and_wait; • SubP_rq_is_object_detected.
rq_pos()	[0-255]	Position set point. This variable is updated by calling SubP_rq_current_pos .
rq_speed()	[0-255]	Speed set point. Change this variable by calling the subprogram SubP_rq_set_speed .

Name	Description
rq_activate()	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution continues before the end of activation.
rq_activate_and_wait()	Sends the Gripper activation command. If it is already activated, nothing happens. Note that the Gripper must be activated to complete any other operation. Program execution waits for the activation.
rq_auto_release_close_and_wait()	Slowly moves the Gripper to its maximum closed position. The Gripper must be activated after this command. Meant for emergency procedures.
rq_auto_release_open_and_wait()	Slowly moves the Gripper to its maximum opened position. The Gripper must be activated after this command. Meant for emergency procedures.
rq_close()	Moves the Gripper its fully closed position.
rq_close_and_wait()	Moves the Gripper to its fully closed position and waits until the motion is completed to execute the next command.
rq_current_pos()	Sets global variable rq_pos to the current Gripper position and returns this value.
rq_is_gripper_activated()	Sets global variable rq_gripper_act to 1 if the Gripper is activated and returns True. Otherwise sets the variable to 0 and returns False.
rq_is_motion_complete()	Sets global variable rq_mov_complete to 1 if the Gripper motion is complete and returns True. Otherwise sets the variable to 0 and returns False.
rq_is_object_detected()	Sets global variable rq_obj_detect to 1 if the Gripper has detected an object and returns True. Otherwise, sets the variable to 0 and returns False.



Name	Description
<code>rq_move_and_wait()</code>	Moves the Gripper to the position defined by the argument and waits until the motion is completed.
<code>rq_move()</code>	Moves the Gripper to the position defined by the argument.
<code>rq_open()</code>	Moves the Gripper to its fully opened position.
<code>rq_open_and_wait()</code>	Moves the Gripper to its fully opened position and waits until the motion is completed.
<code>rq_set_force()</code>	Writes the value of <code>rq_force</code> into the force setting.
<code>rq_set_speed()</code>	Writes the value of <code>rq_speed</code> into the speed setting.



5. User Interface

Visit support.robotiq.com to get the latest installer of the Robotiq User Interface along with appropriate documentation.

See the [instruction manual of the Robotiq User Interface](#) for more details.



6. Specifications

Caution

The following manual uses the metric system, unless specified, **all dimensions are in millimeters**.

The following subsections provide data on the various specifications for the Robotiq 2-Finger 85 and 140 Adaptive Grippers.

- Section 6.1 lists the technical dimensions of the Grippers
 - Dimensions for custom (blank) coupling
 - Dimensions of all available couplings
 - Dimensions for custom fingertip
 - Dimensions of all available fingertips
- Section 6.2 presents the mechanical specifications of the Grippers.
- Section 6.3 gives electrical specifications for the Grippers.



6.1. Technical dimensions

The 2-Finger 85 and 2-Finger 140 share the same basic chassis and thus have the same technical dimensions for everything except the fingers. Figure 6-1 represents the Robotiq 2-Finger 85 Adaptive Robot Gripper's dimensions with axis X, Y, Z and origin referenced for finger motion. Figure 6-3 will show the equivalent with 140 mm fingers (2-Finger 140).

Info

All technical drawings in the present section are shown with silicone flat fingertip option: **AGC-TIP-204-085** (2-Finger 85) or **AGC-TIP-420-140** (2-Finger 140).

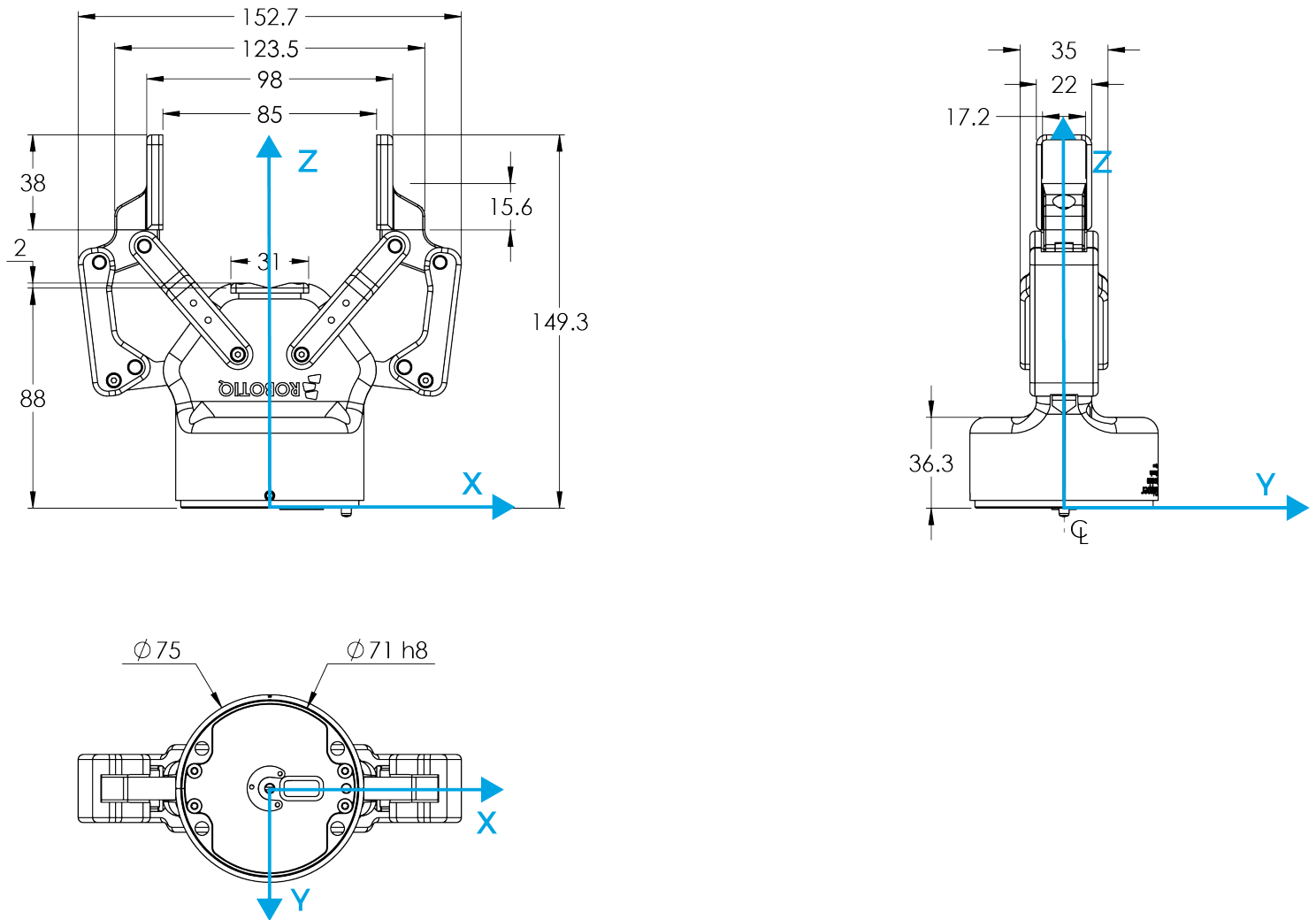


Fig. 6-1: 2F-85 general dimensions (opened).

As mentioned in the figure above, height and width of the fingers vary with opening position. Figure 6-1 represents the 2F-85 Gripper in the opened position (position request = 0), while Figure 6-2 represents the 2F-85 Gripper in the closed position (position request = 255).

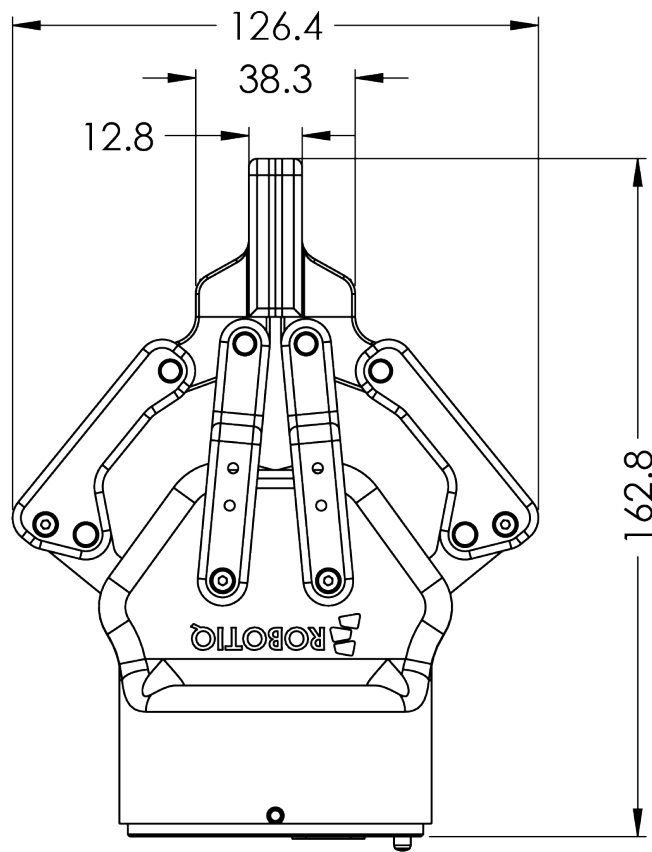


Fig. 6-2: 2F-85 dimensions (closed).



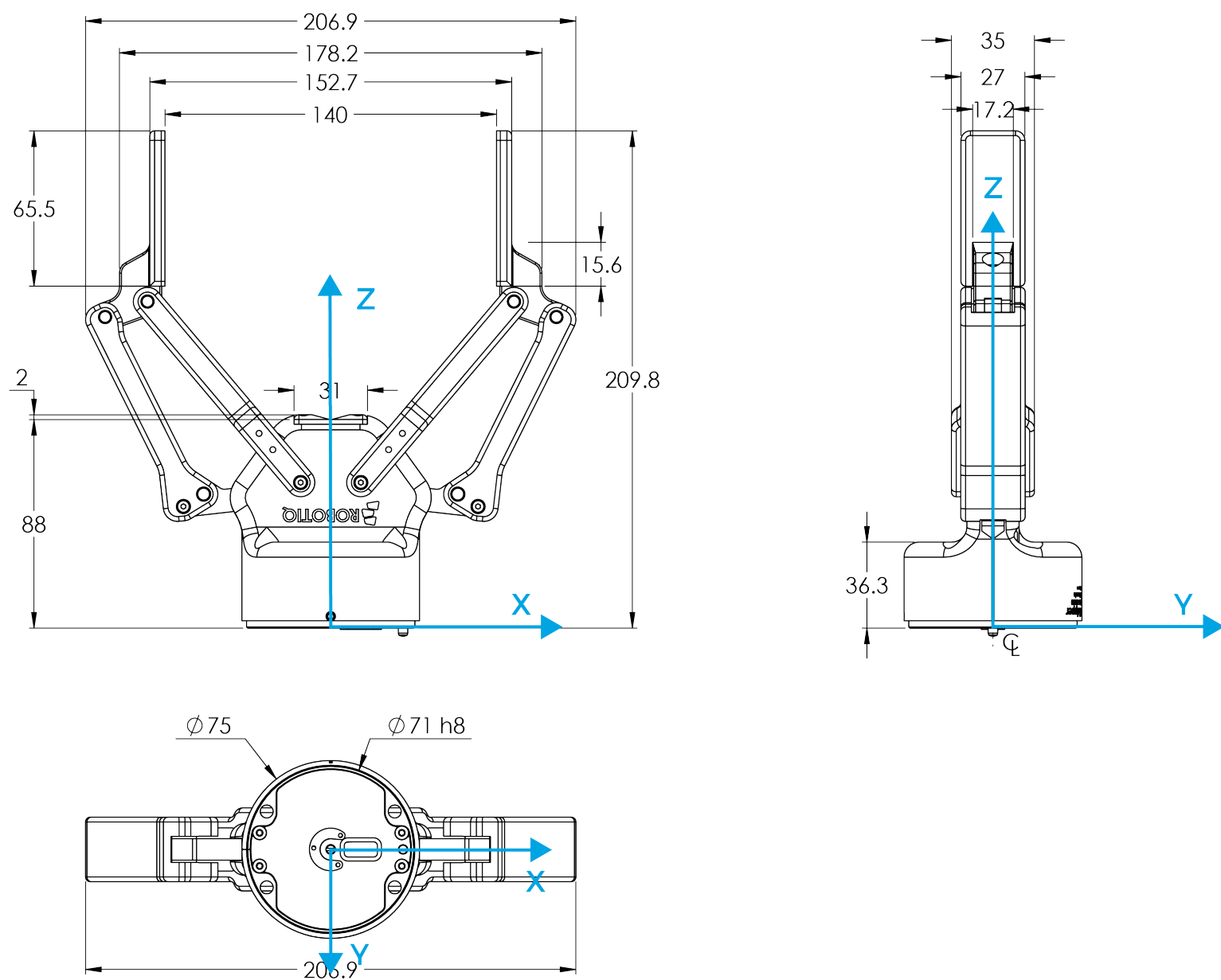


Fig. 6-3: 2F-140 general dimensions (opened).

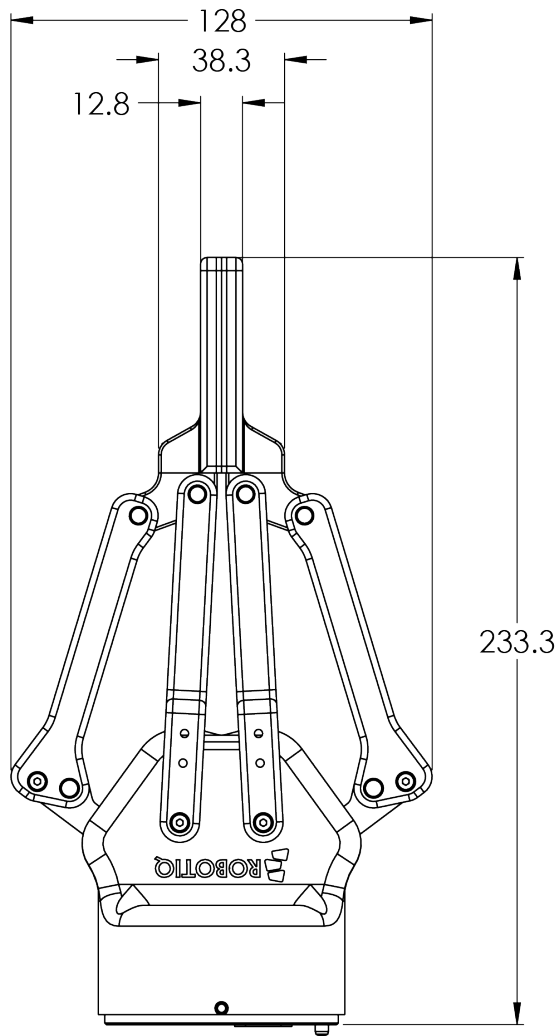


Fig. 6-4: 2F-140 dimensions (closed).

As mentioned in the figure above, the height and width of the fingers vary with opening position. Figure 6-3 represents the 2F-140 Gripper in the opened position (position request = 0), while Figure 6-4 represents the 2F-140 Gripper in the closed position (position request = 255).

6.1.1. Couplings

The Robotiq 2-Finger Adaptive Robot Gripper requires a coupling provided by Robotiq to operate. The coupling is mandatory since it integrates electronics and electrical contacts.

Info

The coupling is common to both the 2F-85 and the 2F-140.

Blank coupling

Below are the dimensions of the blank coupling, **AGC-CPL-BLANK-002** (refer to the **Spare Parts, Kits and Accessories** section), available to create a custom bolt pattern. Blue section can be fully customized (holes can be placed in any part of this section) while the grey section can only be worked to a depth of 3 mm.

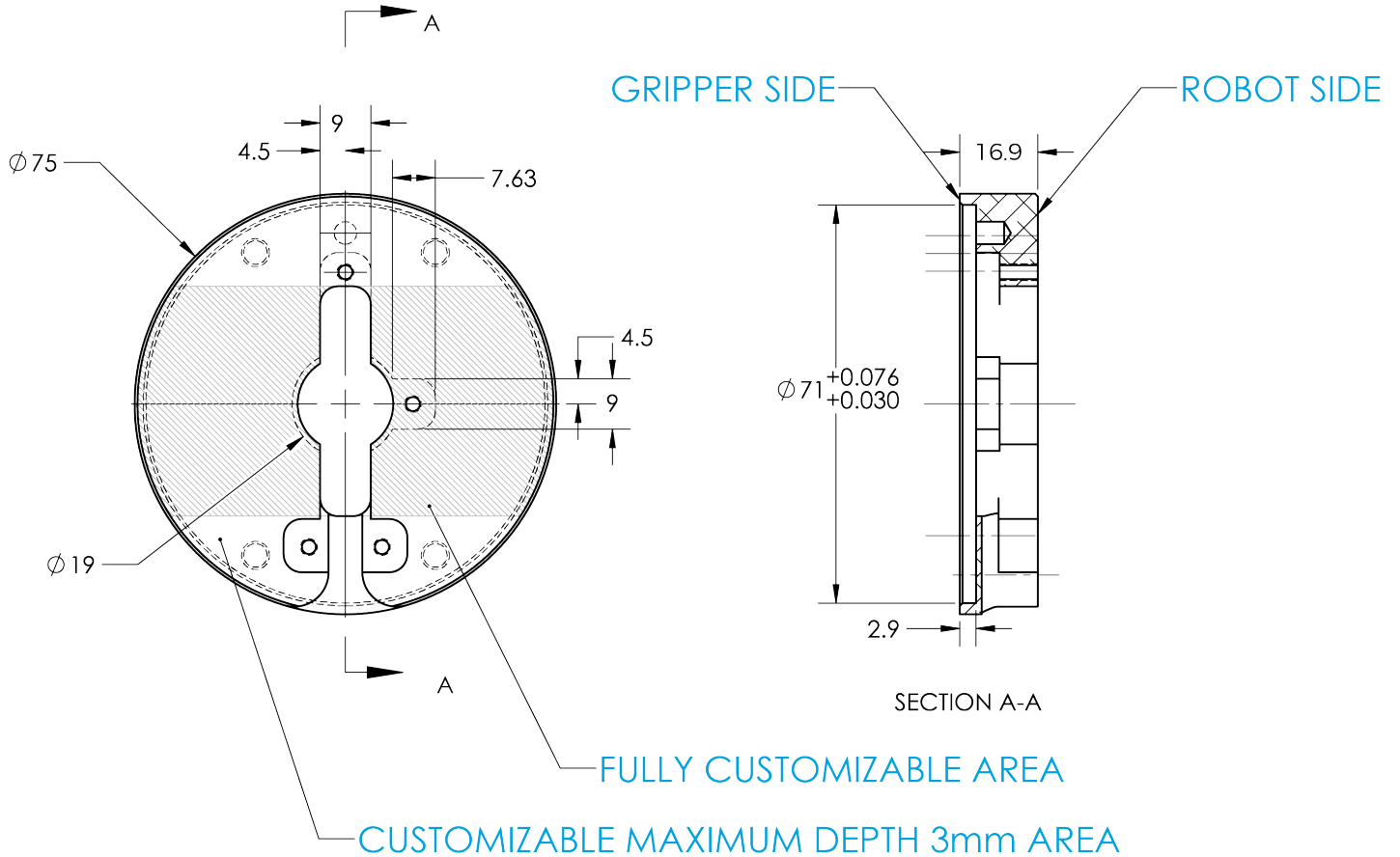


Fig. 6-5: Blank coupling AGC-CPL-BLANK-002 workable area dimensions.

Coupling for ISO 9409-1-50-4-M6

Bolt pattern for coupling **AGC-CPL-062-002** (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 50 mm pitch circle diameter :
 - (4) M6-1.0 low head socket cap screw clearance
 - (1) M6 indexing pin
 - ISO 9409-1 standard 50-4-M6

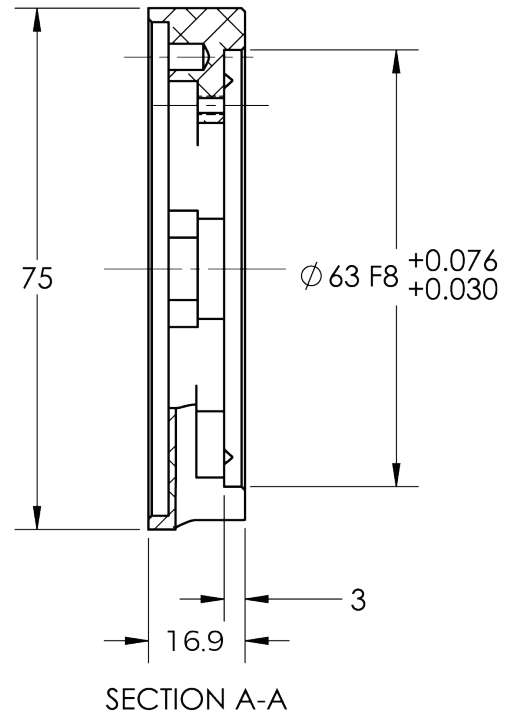
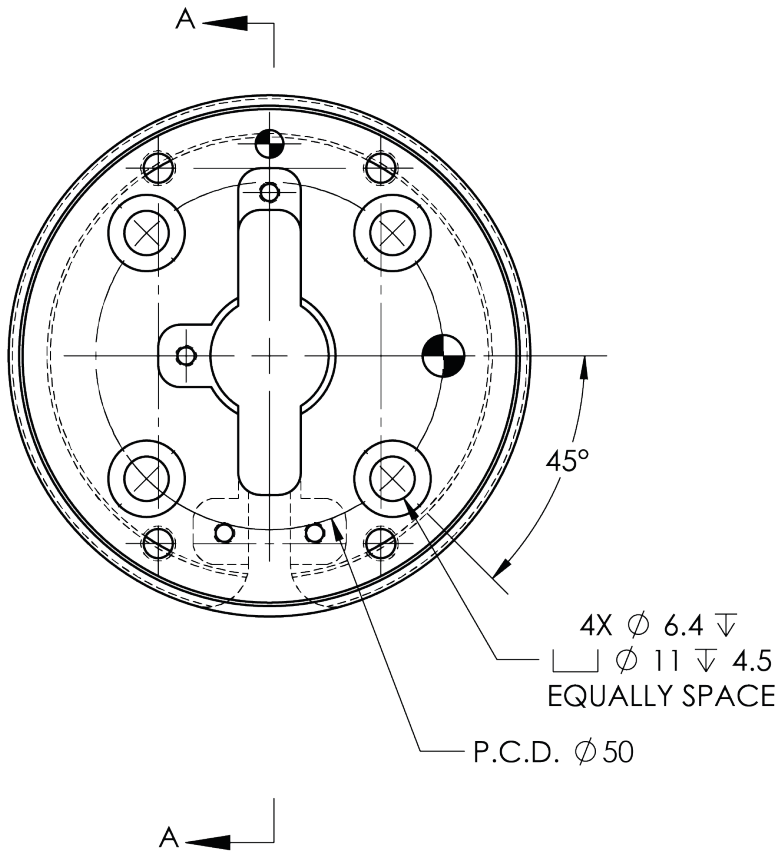


Fig. 6-6: Coupling for ISO 9409-1-50-4-M6.



Coupling for ISO 9409-1-31.5-4-M5

Bolt pattern for coupling **AGC-CPL-063-002** (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 31.5 mm pitch circle diameter :
- (4) M5-0.8 low head socket cap screw clearance
- (1) M5 indexing pin
- ISO 9409-1 standard 31.5-4-M5

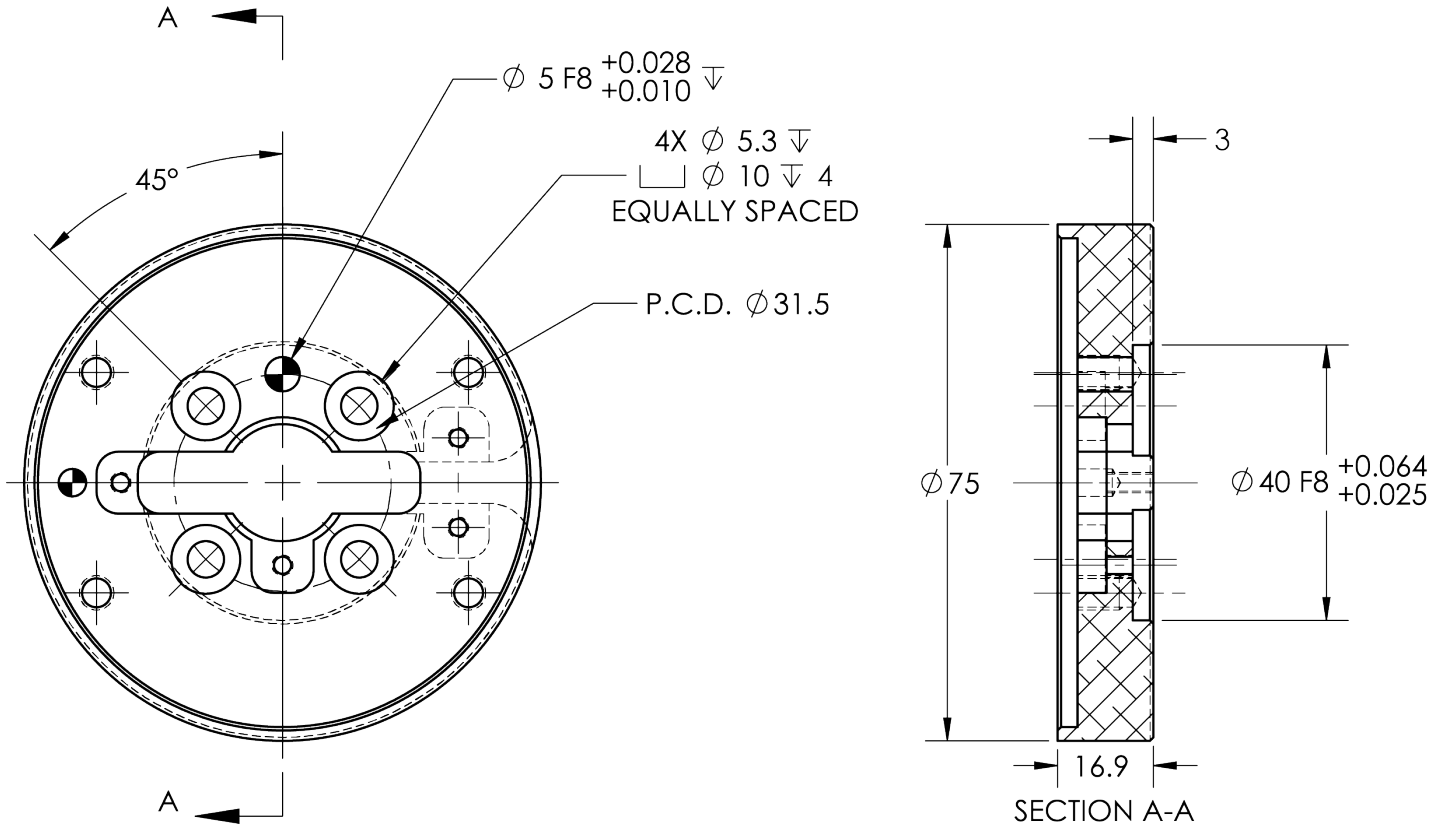


Fig. 6-7: Coupling for ISO 9409-1-31.5-4-M5.

Coupling for ISO 9409-1-40-4-M6

Bolt pattern for coupling AGC-CPL-064-002 (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 40 mm pitch circle diameter :
 - (4) M6-1.0 low head socket cap screw clearance
 - (1) M6 indexing pin
- ISO 9409-1 standard 40-4-M6

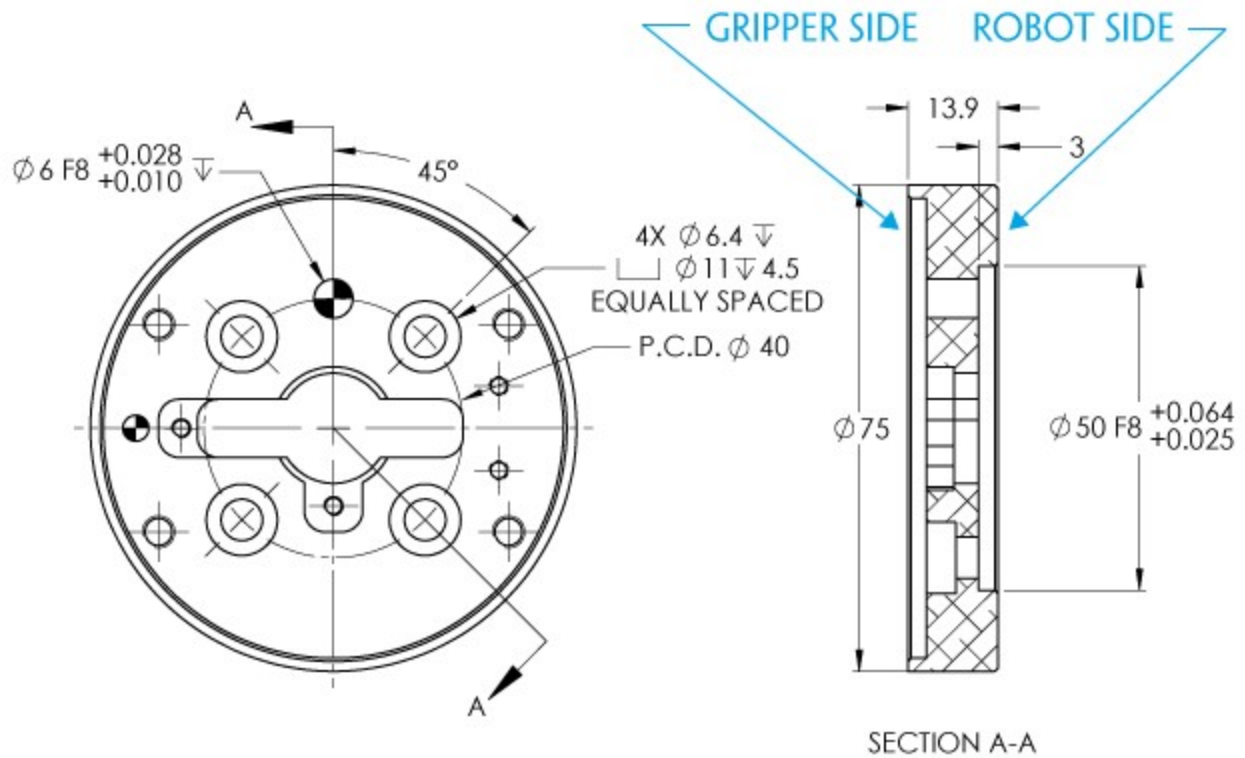


Fig. 6-8: Coupling for ISO 9409-1-40-4-M6.



Coupling for PCD 56 with 8 x M4

Bolt pattern for coupling AGC-CPL-065-002 (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 56 mm pitch circle diameter :
 - (8) M4-0.7 low head socket cap screw clearance
 - (1) M4 indexing pin
 - 62 mm diameter internal insert

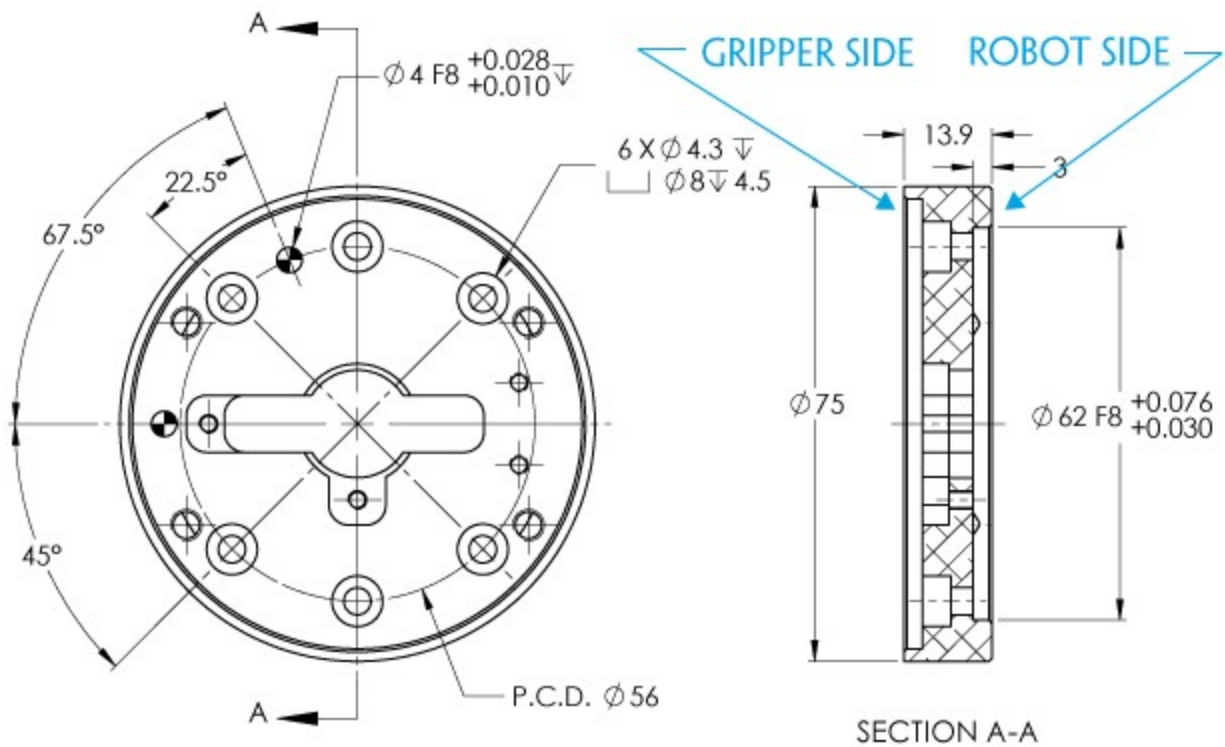


Fig. 6-9: Coupling for PCD 56 mm with 8 x M4 clearance.

Info

Although coupling AGC-CPL-065-002 is compatible with 8 x M4 threads on a 56 mm PCD it uses only 6 of the 8 normally present holes.

Coupling for PCD 56 with 6 x M4

Bolt pattern for coupling AGC-CPL-066-002 (refer to the **Spare Parts, Kits and Accessories** section) is compatible with:

- 56 mm pitch circle diameter:
 - (6) M4-0.7 low head socket cap screw clearance
 - (1) M6 indexing pin
 - 42 mm diameter external insert

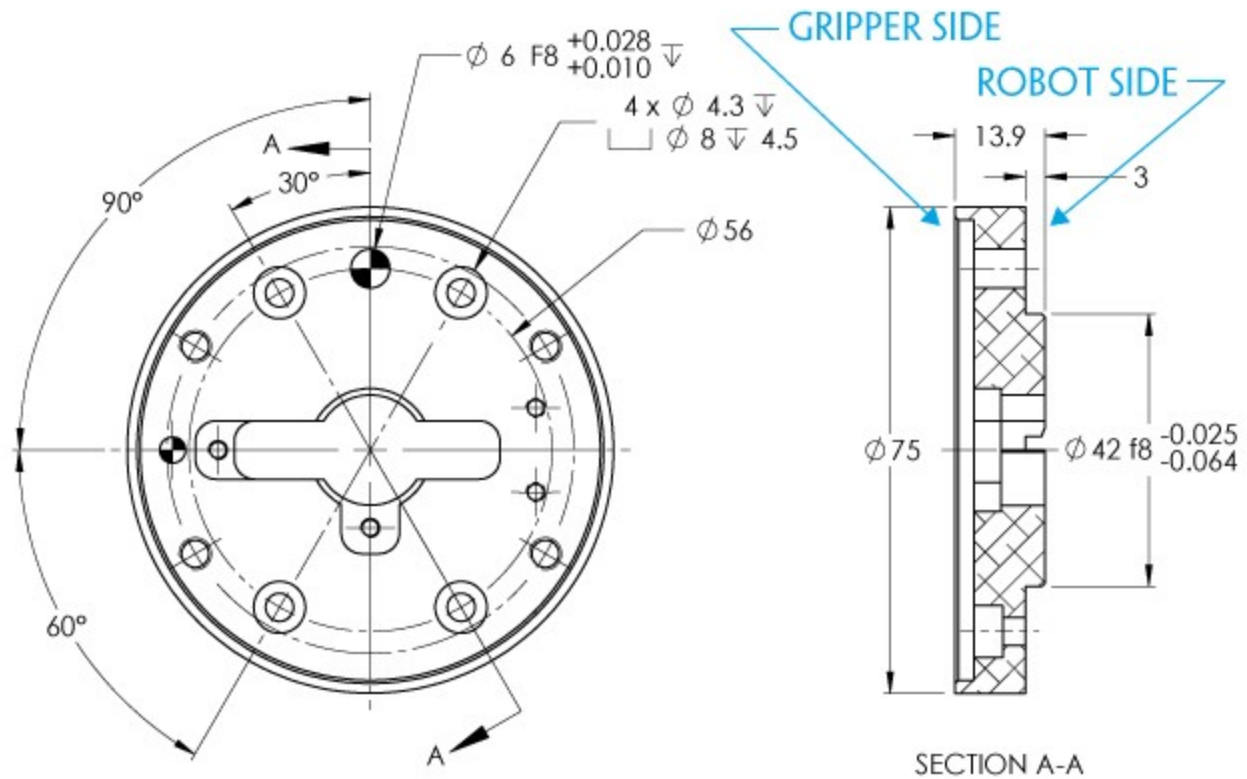


Fig. 6-10: Coupling for PCD 56 mm with 6 x M4 clearance.



Coupling for PCD 60 with 4 x M5

Bolt pattern for coupling **AGC-CPL-067-002** (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 60 mm pitch circle diameter :
 - (4) M5-0.8 low head socket cap screw clearance
 - (1) M5 indexing pin
 - 34 mm diameter external insert

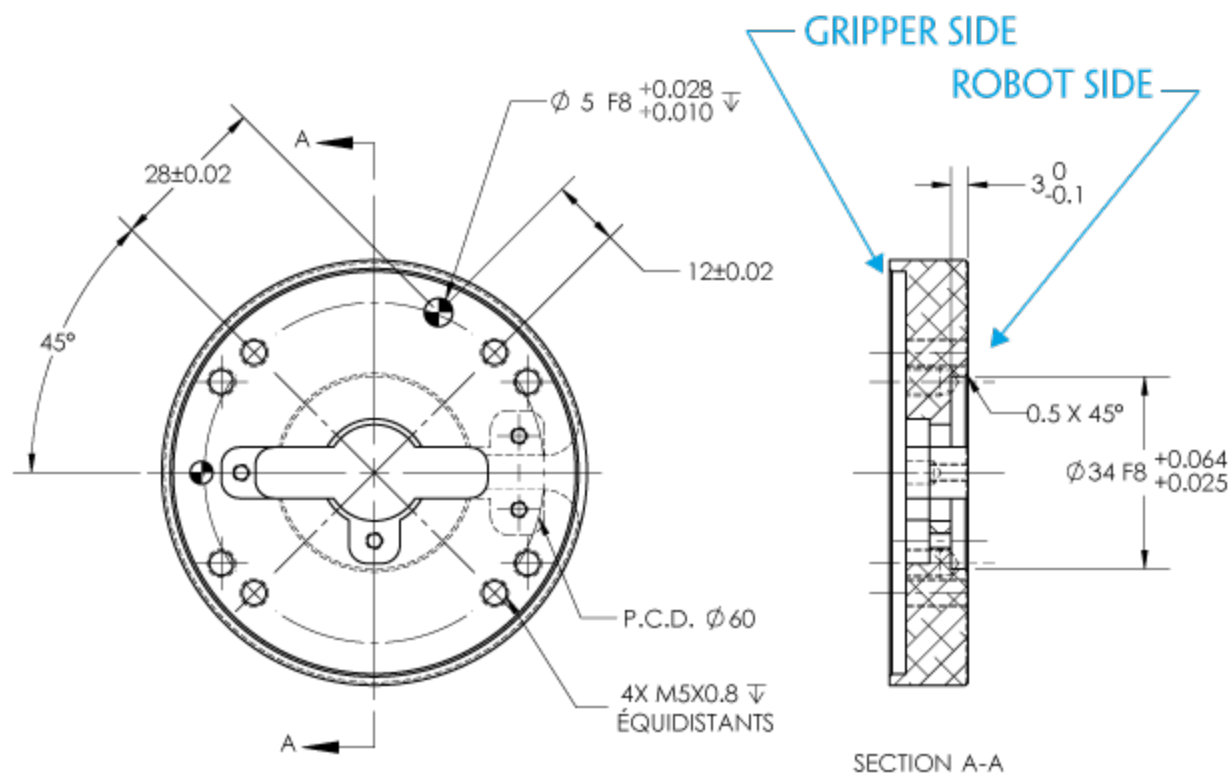


Fig. 6-11: Coupling for PCD 60 mm with 4 x M5 clearance.

Coupling for PCD 63 with 6 x M6

Bolt pattern for coupling AGC-CPL-068-002 (refer to the **Spare Parts, Kits and Accessories** section) is compatible with :

- 63 mm pitch circle diameter :
 - (6) M6-1.0 low head socket cap screw clearance
 - (2) M6 indexing pins

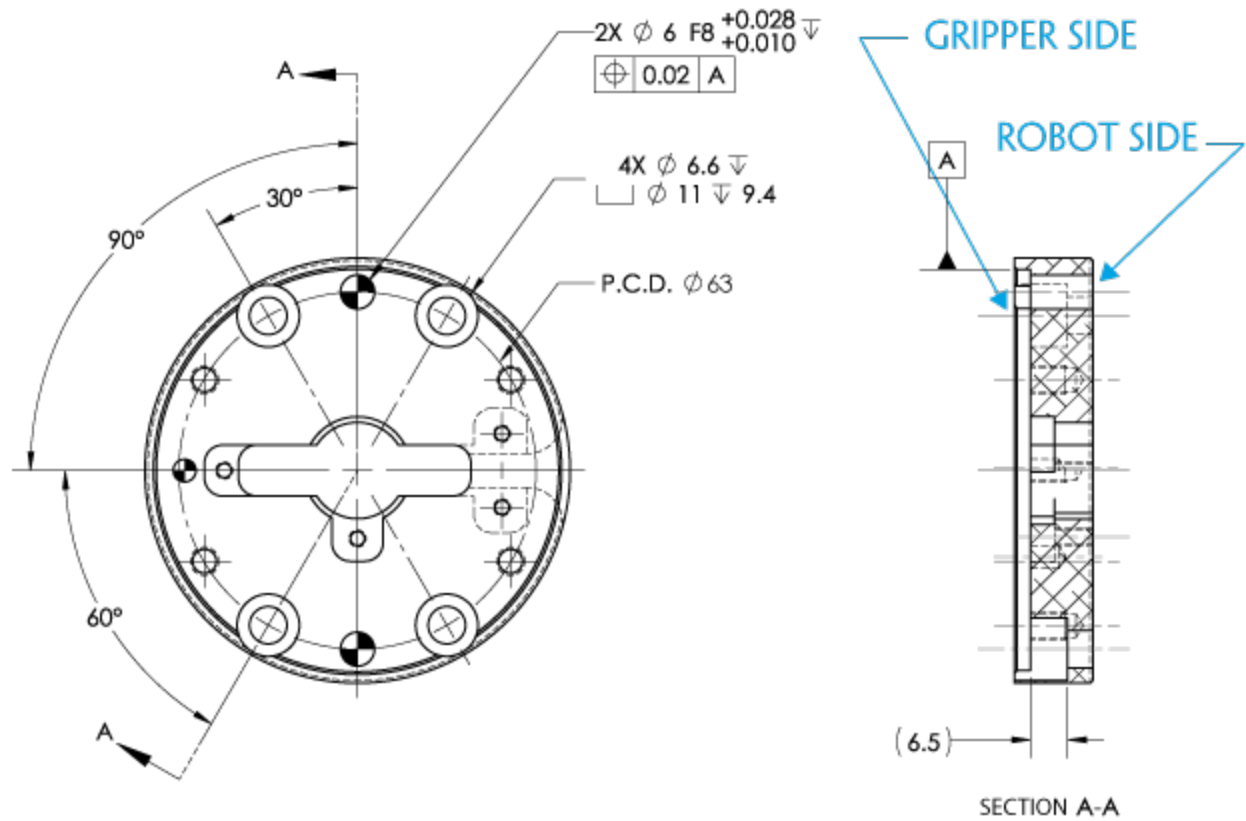


Fig. 6-12: Coupling for PCD 63 mm with 6 x M6 clearance.



6.1.2. Fingertips

The contact grasp points for the Robotiq 2-Finger Adaptive Robot Gripper are its two fingertip pads and palm pad. Many fingertips are available from Robotiq (refer to the **Spare Parts, Kits and Accessories** section). The user can customize their own fingertips from blanks or create them from scratch. The figure below represents the distal phalanx (which acts as the fingertip holder) the permanent, non customizable part of the Gripper finger on which the fingertip must be mounted.

Custom fingertip designs must abide by the following:

- Fingertip must not exceed 100 mm in height from the fingertip's base.
- Fingertip must not exceed 100 mm in width from the fingertip's base (refer to Y axis from figure 6-18).
- Refer to the **Mechanical specifications** section to evaluate the grasp force according to your fingertip design.
- Applied forces to the gripper must not exceed moment and force limits detailed in the **Moment and force limits** section.

Info

Both 2-Finger 85 and 2-Finger 140 use the same fingertips and finger holder.

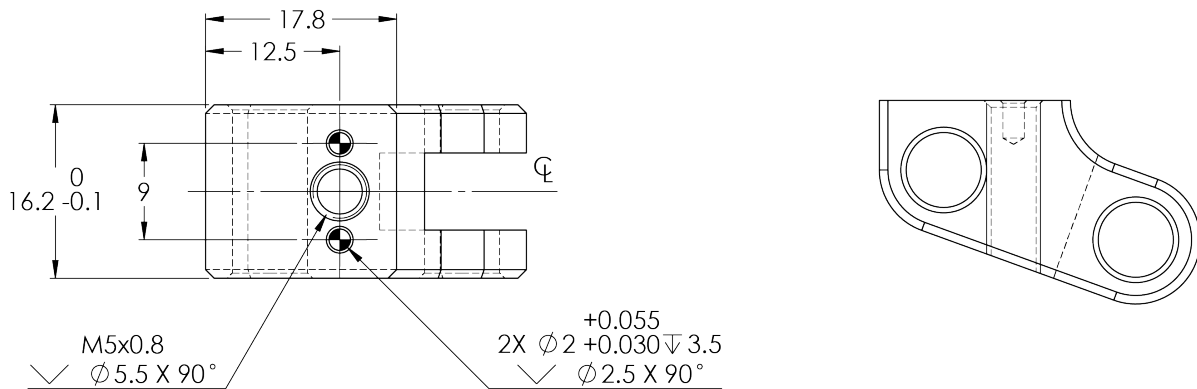


Fig. 6-13: Distal phalanx for holding standard or custom fingertips.

Tip

Custom fingertips will still be subject to the equilibrium line rule for proper actuation of the Gripper, see General Presentation section.

Flat silicone fingertip

The figure below represents a flat silicone fingertip **AGC-TIP-204-085** (2F-85) and **AGC-TIP-420-140** (2F-140); please refer to the **Spare Parts, Kits and Accessories** section. This fingertip has a flat silicone surface with an optimal friction coefficient for picking objects while the other surface will mount onto the fingertip holder shown in the **Fingertips** section.

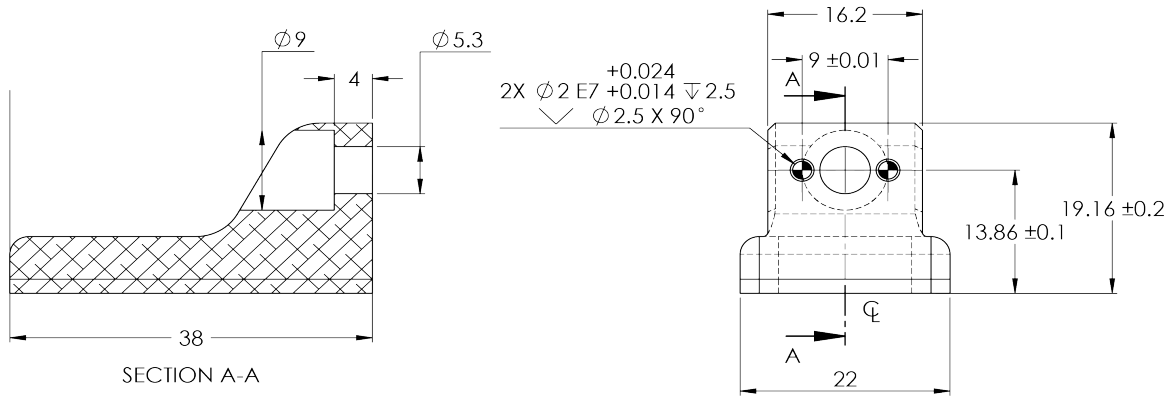


Fig. 6-14: Flat silicone fingertip AGC-TIP-204-085

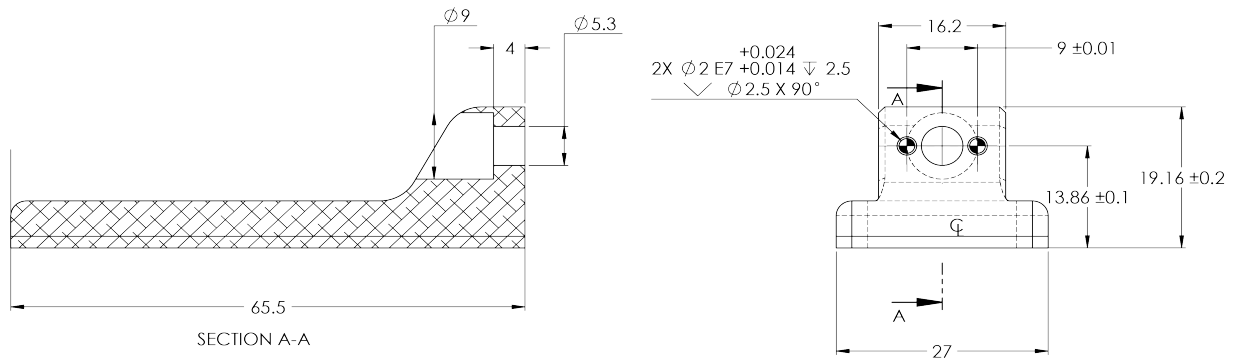


Fig. 6-15: Flat silicone fingertip AGC-TIP-420-140



Grooved fingertip

The figure below represents the available grooved fingertip **AGC-TIP-205-085** (2-Finger 85) and **AGC-TIP-421-140** (2-Finger 140); please refer to the **Spare Parts, Kits and Accessories** section. This fingertip has a grooved surface with an optimal shape for picking cylindrical objects (with its horizontal and vertical grooves) while the other surface will mount onto the fingertip holder shown in the **Fingertips** section.

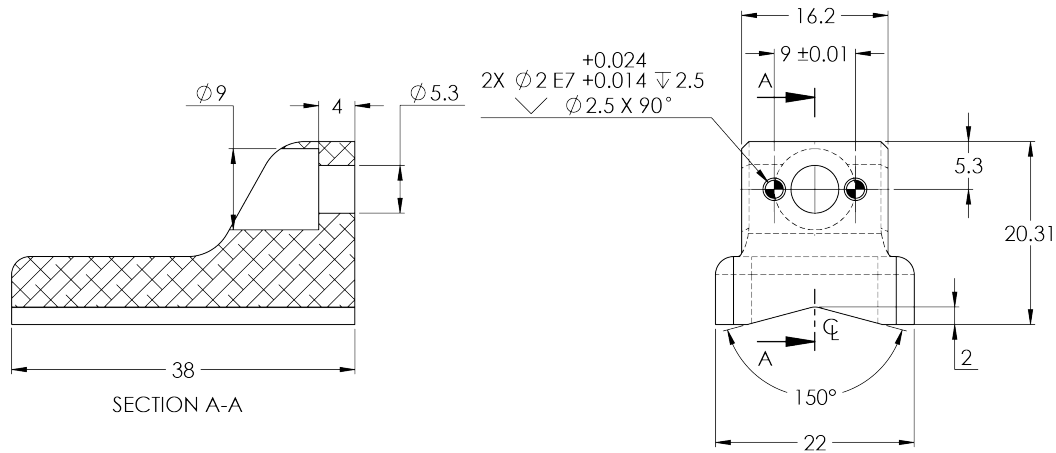


Fig. 6-16: Grooved fingertip AGC-TIP-205-085

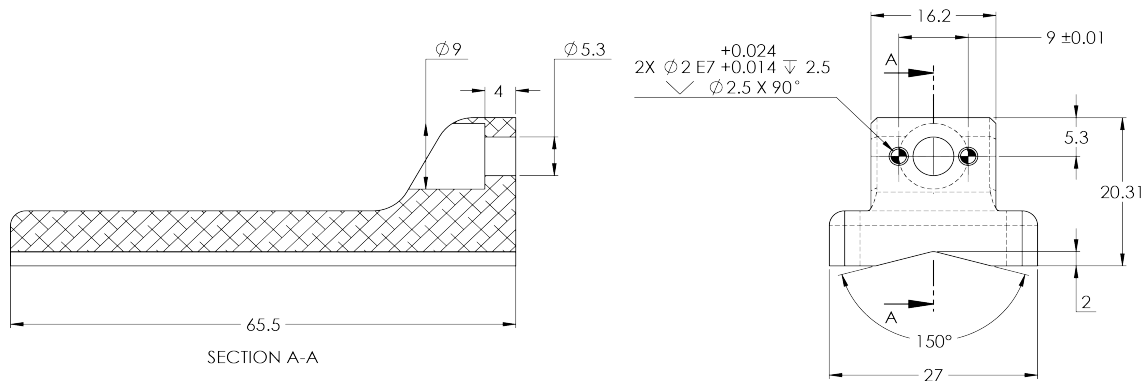


Fig. 6-17: Grooved fingertip AGC-TIP-421-140

6.2. Mechanical specifications

Specification	2-FINGER 85		2-FINGER 140	
	Metric Units	Imperial Units	Metric Units	Imperial Units
Gripper Opening	85 mm	3.35 in	140 mm	5.5 in
Minimum diameter for encompassing	43 mm	1.69 in	90 mm	3.5 in
Maximum height	162.8 mm	6.4 in	232.8 mm	9.15 mm
Maximum width	148.6 mm	5.85 in	202.1 mm	8.0 in
Weight	925 g	2.04 lbs	1,025 g	2.25 lbs
Grasp Force	20 to 235 N	4.49 to 52.83 lbf	10 to 125 N	2.25 to 28.10 lbf
Finger speed	20 to 150 mm/s	0.8 to 5.9 in/s	30 to 250 mm/s	1.2 to 9.8 in/s
Position repeatability ¹	0.05 mm	0.002 in	0.08 mm	0.003 in
Force repeatability	+/- 10%			
Position resolution ²	0.4 mm	0.016 in	0.6 mm	0.022 in
Grasp force resolution	Maximum force calculation below; refer to the Force control section			

Info

All specs are measured with coupling **GRP-CPL-062** and fingertip **AGC-TIP-204-085** (2-Finger 85) or **AGC-TIP-420-140** (2-Finger 140).

¹Repeatability is defined as the positional deviation resulting from the average displacement determined when picking an object with a parallel grasp using standard silicone fingertips. For more details see the blog.robotiq.com article on repeatability.

²Resolution is the increment modified from a 1 bit difference of position/speed/force request (from 0 to 255).



6.2.1. Payload and force

Actuation force model used to calculate recommended payload is described in the figure below, the user must not exceed the force and torque limits:

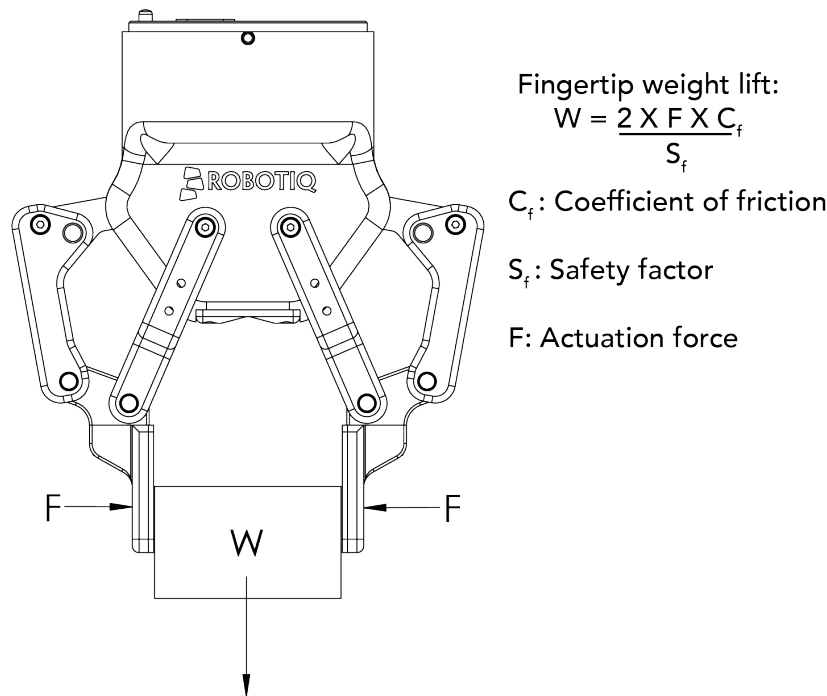


Fig. 6-18: Actuation force on the fingertip of the Adaptive Gripper 2-Finger (see charts below for 2-Finger 85 and 2-Finger 140 force).

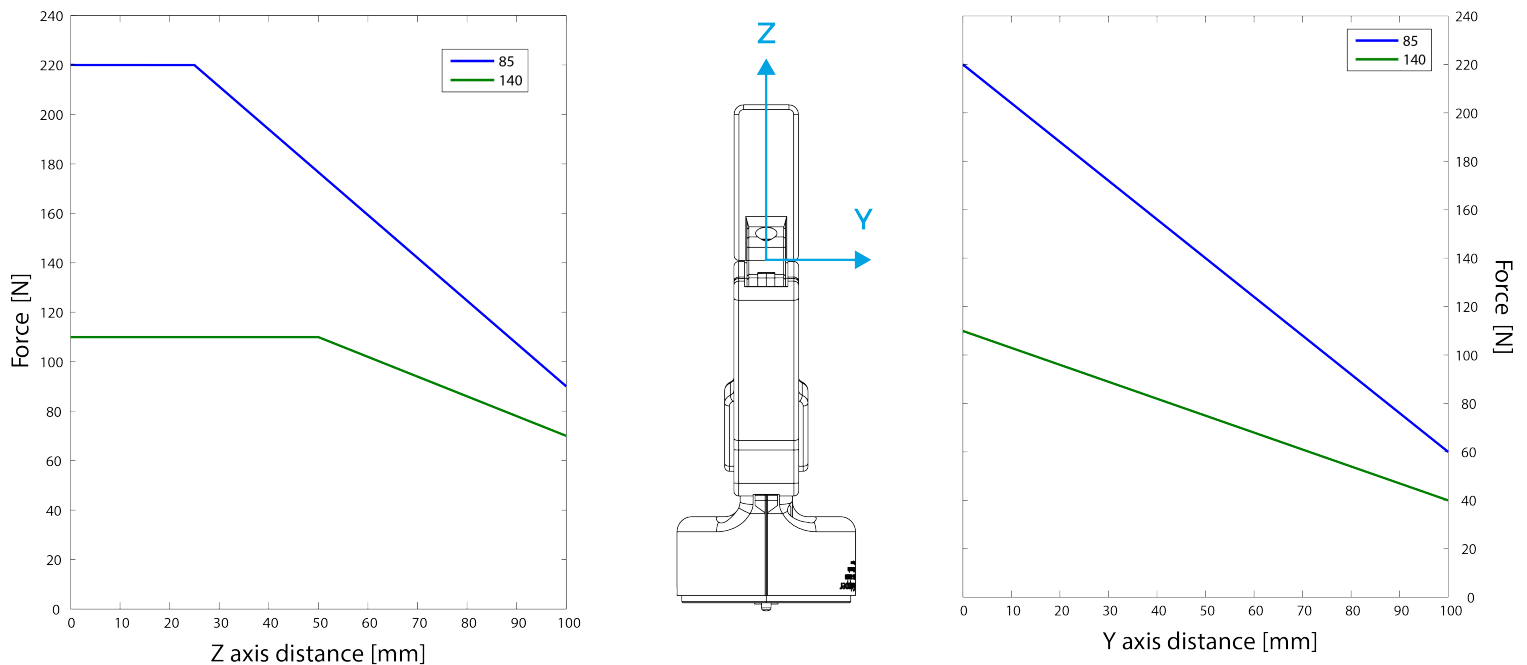


Fig. 6-19: Grasp force in the Y and Z axis for the 2-Finger 85 and 2-Finger 140.

Info

- The "Actuation Force" is the force that can be applied to an object by the motor of the Gripper.
- The Gripper is self-locking.

Info

The user of the Gripper must always ensure that the result of the forces against the finger is always lower than the maximum holding force as seen in figure 6-18.

As defined in figure 6-18, the weight that can be lifted is defined by :

- F is the force that is applied to the load by the Gripper.
- Cf is the friction coefficient between the fingertip and the object load.
- Sf is a safety factor to be determined by the robot integrator.

Info

For example, if the silicone fingertips **AGC-TIP-204** are used to lift a lubricated steel object (machine tending with cutting oils), the friction coefficient would be 0.3 (tested static coefficient of friction).

Maximum weight with a safety factor of 2.4 would be :

$$W = (2 \times 200 \text{ N} \times 0.3) / 2.4 = 50 \text{ N}$$

This calculation means that a 5 kg object will be held by the Gripper when not moving (standing still). When accelerating, the payload will decrease.

For example, if your robot accelerates at 2g then the 5 kg object would weigh 100 N and would therefore be dropped.

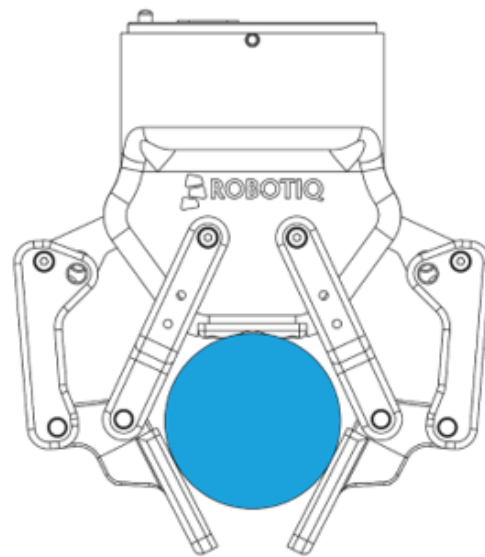
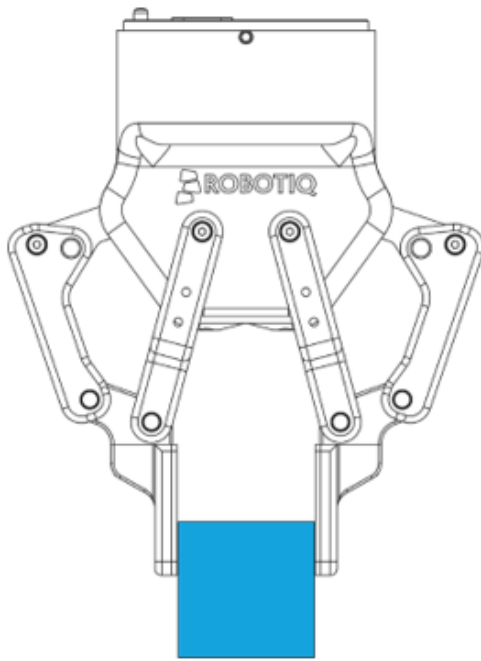
The biggest factor in such calculations will always be the friction coefficient, we recommend testing the coefficient.

Warning

You must consider the **robot acceleration** in your payload calculations.
Robot emergency stops will lead to major deceleration velocities.



Friction grasp and form-fit grasp



Maximum payload by grasp type

Grasp Type	2F-85	2F-140
Friction grasp	5 kg	2.5 kg
Form-fit grasp	5 kg	2.5 kg

6.2.2. Equilibrium Line

Equilibrium line position (explained in the **General Presentation** section) is detailed in the figure below, where:

- θ is the opening angle between the Gripper proximal bar and center line
- d is the distance between the bottom of the finger pads and the equilibrium line as seen on the Z axis in figure 1-5.

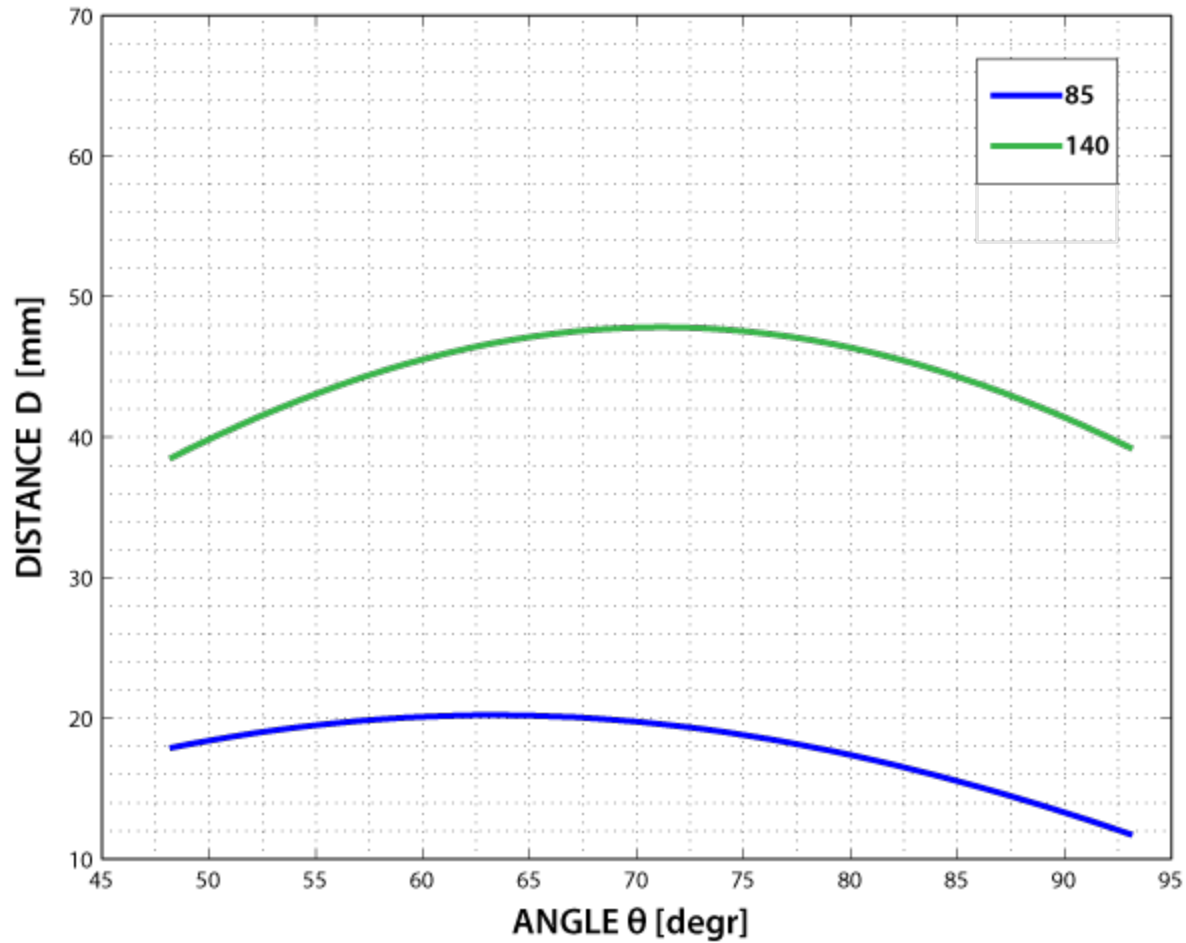


Fig. 6-20: Position of the Gripper equilibrium line according to the opening angle for 2-Finger 85 and 2-Finger 140 options.



6.2.3. Center of mass, tool centere point and moment of inertia

Coupling is included when Gripper is not mounted on the Camera. Dual Gripper adapter plate included where appropriate.

Products	Center of mass (mm)			TCP (mm)						Mass (g)
	X	Y	Z	Gripper 1			Gripper 2			
				X	Y	Z	X	Y	Z	
FT Sensor	-3.0	0.0	17.0	0.0	0.0	37.5	-	-	-	300.0
Camera	-3.0	4.5	9.0	0.0	0.0	23.5	-	-	-	245.0
2F-85	0.0	0.0	58.0	0.0	0.0	174.0	-	-	-	925.0
2F-140	0.0	0.0	73.0	0.0	0.0	244.0	-	-	-	1025.0
Hand-E	0.0	0.0	57.5	0.0	0.0	157.0	-	-	-	1090.0
Dual grippers plate	0.0	0.0	15.5	-	-	-	-	-	-	265.0
FT Sensor + 2F-85	-0.7	0.0	76.3	0.0	0.0	211.5	-	-	-	1225.0
FT Sensor + 2F-140	-0.7	0.0	89.3	0.0	0.0	281.5	-	-	-	1325.0
FT Sensor + Hand-E	-0.6	0.0	78.2	0.0	0.0	194.5	-	-	-	1390.0
Camera + 2F-85	-0.7	1.2	57.1	0.0	0.0	175.5	-	-	-	975.0
Camera + 2F-140	-0.7	1.1	72.4	0.0	0.0	246.5	-	-	-	1075.0
Camera + Hand-E	-0.6	1.0	57.4	0.0	0.0	159.5	-	-	-	1135.0
FT Sensor + Camera + 2F-85	-1.3	0.9	76.3	0.0	0.0	213.0	-	-	-	1275.0
FT Sensor + Camera + 2F-140	-1.2	0.8	89.6	0.0	0.0	284.0	-	-	-	1375.0
FT Sensor + Camera + Hand-E	-1.1	0.8	78.6	0.0	0.0	197.0	-	-	-	1435.0
Hand-E/Hand-E	0.0	0.0	62.0	154.0	0.0	138.0	-154.0	0.0	138.0	2445.0
Hand-E/2F-85	5.9	0.0	61.7	154.0	0.0	138.0	-166.0	0.0	150.0	2280.0
Hand-E/2F-140	-2.4	0.0	66.6	154.0	0.0	138.0	-215.5	0.0	199.5	2380.0
2F-85/2F-85	0.0	0.0	61.4	166.0	0.0	150.0	-166.0	0.0	150.0	2115.0
2F-85/2F-140	-8.7	0.0	66.6	166.0	0.0	150.0	-215.5	0.0	199.5	2215.0
2F-140/2F-140	0.0	0.0	71.4	215.5	0.0	199.5	-215.5	0.0	199.5	2315.0
Camera + Hand-E/Hand-E	-0.3	0.4	78.5	154.0	0.0	161.5	-154.0	0.0	161.5	2690.0
Camera + Hand-E/2F-85	5.0	0.4	77.8	154.0	0.0	161.5	-166.0	0.0	173.5	2525.0
Camera + Hand-E/2F-140	-2.5	0.4	82.5	154.0	0.0	161.5	-215.5	0.0	223.0	2625.0
Camera + 2F-85/2F-85	-0.3	0.5	77.0	166.0	0.0	173.5	-166.0	0.0	173.5	2360.0
Camera + 2F-85/2F-140	-8.1	0.4	82.1	166.0	0.0	173.5	-215.5	0.0	223.0	2460.0
Camera + 2F-140/2F-140	-0.3	0.4	86.7	215.5	0.0	223.0	-215.5	0.0	223.0	2560.0
FT Sensor + Hand-E/Hand-E	-0.3	0.0	90.5	154.0	0.0	175.5	-154.0	0.0	175.5	2745.0
FT Sensor + Hand-E/2F-85	4.9	0.0	89.7	154.0	0.0	175.5	-166.0	0.0	187.5	2580.0
FT Sensor + Hand-E/2F-140	-2.5	0.0	94.3	154.0	0.0	175.5	-215.5	0.0	237.0	2680.0
FT Sensor + 2F-85/2F-85	-0.4	0.0	88.8	166.0	0.0	187.5	-166.0	0.0	187.5	2415.0
FT Sensor + 2F-85/2F-140	-8.0	0.0	93.7	166.0	0.0	187.5	-215.5	0.0	237.0	2515.0
FT Sensor + 2F-140/2F-140	-0.3	0.0	98.4	215.5	0.0	237.0	-215.5	0.0	237.0	2615.0
FT Sensor + Camera + Hand-E/Hand-E	-0.5	0.4	106.1	154.0	0.0	199.0	-154.0	0.0	199.0	2990.0
FT Sensor + Camera + Hand-E/2F-85	4.2	0.4	104.9	154.0	0.0	199.0	-166.0	0.0	211.0	2825.0
FT Sensor + Camera + Hand-E/2F-140	-2.5	0.4	109.4	154.0	0.0	199.0	-215.5	0.0	260.5	2925.0
FT Sensor + Camera + 2F-85/2F-85	-0.6	0.4	103.5	166.0	0.0	211.0	-166.0	0.0	211.0	2660.0
FT Sensor + Camera + 2F-85/2F-140	-7.6	0.4	108.4	166.0	0.0	211.0	-215.5	0.0	260.5	2760.0
FT Sensor + Camera + 2F-140/2F-140	-0.6	0.4	112.9	215.5	0.0	260.5	-215.5	0.0	260.5	2860.0



Info

The angle to calculate the TCP for Grippers mounted on a dual gripper assembly is as follows:

- $R_x = 0$
- $R_y+ / R_y- = 0.7854$
- $R_z = 0$

The moment of inertia are calculated for a configuration where the fingers are fully open. Here is the approximate moment of inertia matrix for the Gripper:

2-FINGER 85 OPTION

$$I = \begin{bmatrix} I_{XX} & I_{XY} & I_{XZ} \\ I_{YX} & I_{YY} & I_{YZ} \\ I_{ZX} & I_{ZY} & I_{ZZ} \end{bmatrix} = \begin{bmatrix} 4180 & 0 & 0 \\ 0 & 5080 & 0 \\ 0 & 0 & 1250 \end{bmatrix} = \begin{bmatrix} 14.3 & 0 & 0 \\ 0 & 17.4 & 0 \\ 0 & 0 & 4.3 \end{bmatrix}$$

$kg * mm^2$ $lb * in^2$

2-FINGER 140 OPTION

$$I = \begin{bmatrix} I_{XX} & I_{XY} & I_{XZ} \\ I_{YX} & I_{YY} & I_{YZ} \\ I_{ZX} & I_{ZY} & I_{ZZ} \end{bmatrix} = \begin{bmatrix} 7400 & 0 & 0 \\ 0 & 9320 & 0 \\ 0 & 0 & 2260 \end{bmatrix} = \begin{bmatrix} 25.3 & 0 & 0 \\ 0 & 31.8 & 0 \\ 0 & 0 & 7.7 \end{bmatrix}$$

$kg * mm^2$ $lb * in^2$

Fig. 6-21: Robotiq 2-Finger inertia matrix.



6.2.4. Moment and force limits

The 2-Finger Adaptive Gripper has maximum moments and force limit. The listed moments and forces are independent to the force applied by the Gripper itself on it's payload. For payload calculation, refer to the **Mechanical specifications** section.

Warning

The following limits must be respected at all time. Calculation of maximum moment and force should include the robot acceleration and a safety factor.

Parameters	Finger Option	
	2-Finger 85	2-Finger 140
F_x, F_y, F_z	50 N	25 N
M_x^*	5 Nm	5 Nm
M_y^*	5 Nm	5 Nm
M_z	3 Nm	3 Nm

* Moments in x and y are calculated from the base of the fingertips as shown in figure 6-19.

Example usage of the listed limit :

- After picking it's normal payload, the robot can use the 2-Finger 85 Gripper to apply up to 50 N of force in any direction. Applying more than 50 N could damage the Gripper or result in payload loss.
- The Gripper can pick a screwdriver and apply 3 Nm of torque to screw (such moment would be in the Z axis).



6.3. Electrical specifications

SPECIFICATION	VALUE
Operating supply voltage	24 V DC $\pm 10\%$
Absolute maximum supply voltage	28 V DC
Quiescent power (minimum power consumption)	< 1 W
Peak current	1 A



7. Maintenance

The Adaptive Gripper requires only external maintenance with limited downtime. Maintenance for both 2-Finger Adaptive Robot Grippers is required after specified usage, measured in time (normal 40h week) or in cycles (requesting an open and closed movement from the Gripper).

Following the maintenance interval will ensure :

- Correct functioning of your Gripper.
- Validity of your warranty.
- Proper lifetime for your Gripper.

Please visit support.robotiq.com for details on the maintenance operation.

Warning

Unless specified, any **repairs** done on the Gripper will be done by Robotiq.

Info

A cycle is defined as a **go to requested position** command that results in grasp force being applied (picking an object while opening or closing or closing the fingers on themselves).



Operation	Daily	Weekly	Semiannually (or 1 M cycles)	Annually (or 2 M cycles)
Gripper Cleaning	Dirty conditions	Normal conditions		
Periodic Inspection			X	
Finger Pad Replacement ¹			X	
Overhaul ²				X

Table 7-1: Maintenance intervals for the 2-Finger Grippers

¹Replace pads before if wear is visible.

²Overhaul is recommended after 2M cycles and is done by Robotiq at the user's expense, please contact Robotiq support.

Caution

Maintenance operations are for the average normal usage of the Gripper, the maintenance intervals must be adjusted according to environmental conditions such as:

- Operating temperature
- Humidity
- Presence of chemical(s)
- Presence of physical objects (debris, scraps, dust, grease etc.)
- Interaction with operated parts (sharp or rough)
- Dynamics of the operation (accelerations)

7.1. Gripper cleaning

Maintenance Interval	Tools You Need	Parts You Need
Weekly or daily in dirty operating conditions	• Flat head precision 2 mm screwdriver • 4 mm hex key • Dry tissue or towel • Medium strength thread locker	None

Caution

The Robotiq 2-Finger Adaptive Robot Gripper is not waterproof or water resistant without additional protection, only clean the Gripper with a dry towel.

Info



Always turn off the robot and Gripper power supply before performing any maintenance operations on the Gripper.

Caution

Maintenance operator must be grounded to prevent electrostatic discharge that could damage the Gripper electronics.

1. Remove the Gripper from its coupling using the 4 mm hex key to unscrew the four (4) M5-0.8 x 25mm socket head cap screws. Note that each screw uses a tooth lock washer, do not lose them.
2. Use the flat head precision screwdriver to open or close the Gripper by accessing the transmission shaft under the Gripper, where the coupling connects. Illustrated in the figure below.
 - a. Open or close to access the palm pad and other parts of the Gripper.
3. Clean the Gripper with a dry towel, remove all debris, dirt and dust from the surface of the Gripper, clean all pads, dry thoroughly.
4. Clean the coupling with a dry towel, pay particular attention to the electrical contact.
5. Visually inspect the Gripper and pay attention to any visible damage.
6. Put the coupling back on and fix it with the four (4) M5-0.8 x 25mm socket head cap screws, use the tooth lock washers, apply medium strength thread locker to the M5 screws.

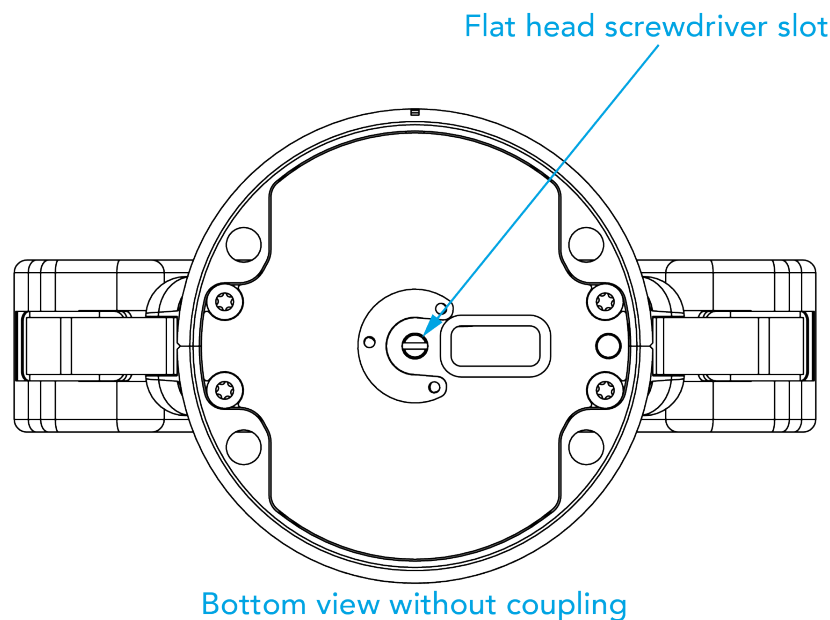


Fig. 7-1: Flat head screwdriver slot for manual opening and closing of the Gripper.



7.2. Periodic inspection

Maintenance Interval	Tools You Need	Parts You Need
Monthly	• Flat head precision 2 mm screwdriver • 4 mm hex key • Dry tissue or towel • Medium strength thread locker	None (unless damage is detected)

Info

Always turn off robot and Gripper power supply before doing maintenance operations on the Gripper.

1. Remove and clean the Gripper following instructions in the **Gripper cleaning** section.
2. Inspect the Gripper :
 - a. Finger movement must be symmetric and fluid:
 - i. Test the return movement by pushing fingers open, the fingers must come back to the initial starting position on its own.
 - ii. Test the general movement of the fingers by actuation using the screwdriver insertion hole in the chassis.
 - b. Finger pad wear must not affect grasping, if wear is visible and affects movement, change fingerpad(s).
 - c. Check for any collision damage, if damage is visible, contact support@robotiq.com.
 - d. Check for any sign of wear on the Gripper chassis, if wear is present and may affect the Gripper, contact support@robotiq.com.
3. Put back in place respecting instructions from the **Gripper cleaning** section.



7.3. Fingertip replacement

Maintenance Interval	Tools You Need	Parts You Need
1 Million cycles or when wear is visible	• Flat head 2 mm precision screwdriver • 4 mm hex key • Dry tissue or towel • low strength thread locker • medium strength thread locker	2 x Robotiq 2-Finger Adaptive Robot Gripper fingertip (silicone pad or V-grooved).

Refer to the **Spare Parts, Kits and Accessories** section section to order Robotiq 2-Finger Adaptive Robot Gripper replacement parts.

Info

Always turn off robot and Gripper power supply before doing maintenance operations on the Gripper.

1. Remove and clean the Gripper following instructions from the **Gripper cleaning** section.
2. Remove the worn fingertip by removing the M5 screws with the 4 mm hex key; make sure not to mislay the 2 mm indexing pins.
3. Clean the fingertip holder and dry thoroughly.
4. Insert the new fingertip by inserting the indexing pin in the fingertip holder.
5. Fix the fingertip using the provided M5-0.5 x 8 mm socket head cap screws, apply low strength threadlocker to the M5 screw threads.
6. Repeat for remaining finger.



7.4. Overhaul

Maintenance Interval	Tools You Need	Parts You Need
2 Million cycles or at warranty expiration	None	None

Gripper overhaul is necessary when the Gripper reaches 2 Million cycles or when warranty expires. Overhaul is done by Robotiq, please contact Robotiq support service.

Gripper overhaul includes, but is not limited to :

- Worn parts changed
 - Power transmission gearing mechanism
 - Plain bearings
 - Ball bearings
- Quality control
 - Specification test (force, speed, position)

Overhaul takes a maximum of 5 business days after reaching Robotiq, shipping is at customer's expense.

Tip

Loan units are available while your Gripper is under maintenance.



8. Spare Parts, Kits and Accessories

Spare parts, kits and accessories list:

Info

The following list is up to date at print time and is subject to change, check online for updates.

Info

Unless specified, screws, dowel pins and other hardware are included only for the Gripper side, never for the robot side.

Item	Description	Ordering Number (2-Finger 85)	Ordering Number (2-Finger 140)
Gripper basic unit	Adaptive Robot Gripper 2-Finger basic unit with fingers (no fingertips, no coupling)	AGC-GRP-2F85	AGC-GRP-2F140
Kit for <i>Universal Robots</i> (CB-Series)	Gripper basic unit, 10 m cable and coupling for Universal Robots with silicone fingertips	AGC-UR-KIT-002	AGC-UR-KIT-140
Kit for <i>Universal Robots</i> (e-Series)	Gripper basic unit, silicone fingertips, and cable-to-wrist coupling for Universal Robots	AGC-ES-UR-KIT-85	AGC-ES-UR-KIT-140
Controller	Optional controller for industrial communications see Robotiq Universal Controller Items	UNI-CTR-XXXX	
Finger Kit	Finger upgrade kit for 85 or 140 mm option. Included: • 2 x Replacement finger • 2 x Hardware kit • 1 x hex key	AGC-FIN-KIT-85-V4	AGC-FIN-KIT-140-V4
Blank coupling	Blank coupling for Adaptive Robot Gripper 2-Finger, with screws for Gripper fixation and 1 m pigtail cable	AGC-CPL-BLANK-002	
ISO 9409-1-50-4-M6 coupling (coupling-to-wrist)	ISO 9409-1-50-4-M6 coupling for 2-Finger Robot Grippers, with screws for Gripper fixation	GRP-CPL-062	
ISO 9409-1-50-4-M6 coupling (coupling to controller)	ISO 9409-1-50-4-M6 coupling for 2-Finger Robot Grippers, with screws for Gripper fixation and 1 m pigtail cable	AGC-CPL-062-002	
ISO 9409-1-31.5-4-M5 coupling	ISO 9409-1-31.5-4-M5 coupling for Adaptive Robot Gripper 2-Finger, with screws for Gripper fixation and 1 m pigtail cable	AGC-CPL-063-002	
ISO 9409-1-40-4-M6 coupling	ISO 9409-1-40-4-M6 coupling for Adaptive Robot	AGC-CPL-064-002	



Item	Description	Ordering Number (2-Finger 85)	Ordering Number (2-Finger 140)
	GripperGripper 2-Finger, with screws for Gripper fixation and 1 m pigtail cable		
56-8M4-1D4 coupling	Coupling for 56 mm PCD ¹ with (8) M4 and (1) 4mm indexing pin, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-065-002	
56-6M4-1D6 coupling	Coupling for 56 mm PCD ¹ with (6) M4 and (1) 6mm indexing pin, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-066-002	
60-4Ø5-1D5 coupling	Coupling for 60 mmPCD ¹ with (4) M5 thread and (1) 5mm indexing pin, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-067-002	
63-6M6-2D6 coupling	Coupling for 63 mm PCD ¹ with (6) M6 and (2) 6mm indexing pins, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-068-002	
40-4M5-1D3 coupling	Coupling for 40 mm PCD ¹ with (4) M5 and (1) 3mm indexing pins, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-070-002	
31.5-4M4 coupling	Coupling for 31.5 mm PCD ¹ with (4) M4, with screws for 2-F Gripper fixation and 1 m pigtail cable	AGC-CPL-071-002	
Adapter plate to 63-4M6-71-2D3	Wrist adapter plate for use with AGC-CPL-064-002. Interface to 63 mm PCD ¹ with (4) M6 screws and 71 mm PCD ¹ with (2) M3 indexing pins	AGC-APL-151-002	
Adapter plate to 63-4M6-61_4-2D6	Wrist adapter plate for use with AGC-CPL-064-002. Interface to 63 mm PCD ¹ with (4) M6 screws and 61.4 mm PCD 1 with (2) M6 indexing pins	AGC-APL-152-002	
Adapter plate to 80-6M8-2D82D8	Wrist adapter plate for use with AGC-CPL-064-002. Interface to 80 mm PCD ¹ with (6) M8 screws and (2) M8 indexing pins	AGC-APL-153-002	
2-Finger Device Cable (5M)	5 m Robotiq device cable for power and communication. Straight M12 5-pins female on one side, single ended on the other, shielded	CBL-COM-2065-05	
2-Finger Device Cable (10M)	10 m Robotiq device cable for power and communication. Straight M12 5-pins female on one side, single ended on the other, shielded	CBL-COM-2065-10	
USB to RS485 adapter	USB to RS485 adapter, can be used with device cable for USB connection	ACC-ADT-USB-RS485	
Silicone fingertip ²	Flat silicone fingertip for 2-Finger Adaptive Gripper, included: - one (1) silicone fingertip. - two (2) dowel pins (dia. 2 x 6 mm)	AGC-TIP-204-085	AGC-TIP-420-140



Item	Description	Ordering Number (2-Finger 85)	Ordering Number (2-Finger 140)
	two (2) screws (M5-0.5 x 12 mm SHCS)		
Grooved fingertip ²	Grooved aluminium fingertip for 2-Finger Adaptive Gripper, included: - one (1) V-grooved fingertip - two (2) screws M5-0.5 x 12 mm SHCS - two (2) dowel pins dia. 2 x 6 mm	AGC-TIP-205-085	AGC-TIP-421-140
Replacement finger	Replacement finger for 2-Finger Adaptive Gripper, included: - One (1) finger - Four (4) screws M3-0.5 X 8 LHCS - One (1) proximal locking shaft - One (1) link/bar locking shaft	AGC-FIN-085	AGC-FIN-140
Finger Protector ²	Protector for 2-Finger Adaptive Gripper, included: - One (1) protector - Four (4) M3 screws	AGC-PRO-KIT-V4	N/A
Parallel locking shaft ²	Parallel mechanism locking shaft for 2-Finger Adaptive Gripper, included : - One (1) parallel locking shaft - Two (2) M3 screws	AGC-PARA-KIT	

¹Pitch Circle Diameter

²Part listed for two units, since two are required per Gripper.

Tip

For legacy part replacement visit the documentation archives and the appropriate manual or consult your Robotiq distributor.



9. Troubleshooting

When using 2-Finger with USB to RS485 serial converter:

Troubleshooting from the Gripper LED:

1. LED is not lit, Gripper not powered.
 - a. Check Gripper power supply and electrical setup
2. LED is solid blue and red, Gripper is in fault.
 - a. Disconnect the communication and power cycle the Gripper. The LED should turn solid red when powered.
 - b. While powered with red LED, connect the communication, LED should turn solid blue when connection is established.
3. Flashing blue and red LED, Gripper is in fault.
 - a. Gripper may be in auto-release, wait for auto-release to be completed, then turn off auto-release and initialize the Gripper.
 - b. Gripper may have a major fault, check the error from fault status and contact support.
4. Solid blue, no fault and communicating.
 - a. If you can control the Gripper from the GUI on the pendant, check your program structure.
 - b. If you can not control the Gripper from the pendant, contact support.
5. Solid red, no fault, but Gripper is not communicating.
 - a. USB-RS485 converter LEDs are :
 - Not lit, no USB communication.
 - i. Check USB connection.
 - ii. Re-install drivers.
 - iii. Contact support.
 - Red flashes at slow rate.
 - i. Check the DB-9 connector.
 - ii. Check the cable.
 - iii. Check the communication parameters from the Robotiq User Interface, see recovery procedure.
 - iv. Contact support.
 - Red and green flashing at high speed.
 - i. Check fault status, maximum operating temperature could be exceeded.
 - ii. Contact support.



When using 2-Finger with Universal Robots package:

Troubleshooting from the previous section (LED references) still applies.

From the UR teach pendant:

URCap install:

1. Verify that the latest URCap is installed from the **Robot Setup** menu, inside the **URCaps** tab
 - a. URCaps installation instructions are in the **Installing URCap Package** section
 - b. Inside the URCaps tab, if you select the appropriate URCap file, you can see the current software version
 - c. Gripper toolbar should appear if installed properly
2. URCap is installed but Gripper cannot be controlled :
 1. If Gripper LED is not blue, follow the steps in the first section above
 2. If Gripper LED is blue, URCap is installed with the latest available version, contact support@robotiq.com



Driver package install:

1. Driver checkup :

- a. Use the latest driver version available at support.robotiq.com
 - Use the installation instruction from the **UR Package without URCaps** section
- b. To check your driver version :
 - With the Gripper unplugged
 - From the UR teach pendant screen, click the "Gripper" button
 - i. Driver version will be shown
 - ii. If Gripper button is not displayed, driver package is not properly installed, see instruction from the **Control over Universal Robots without URCaps** section

2. Driver is installed but Gripper cannot be controlled :

- a. If Gripper LED is not blue, follow the steps in the first section above
- b. If Gripper LED is blue, driver is installed with the latest available version, contact support@robotiq.com



10. Warranty and Patent

Robotiq warrants the 2F-85 and 2F-140 Adaptive Robot Grippers against defects in material and workmanship for a period of one year from the date of reception when utilized as intended. Robotiq also warrants that this equipment will meet applicable specifications under normal use.

Warning

Warranty applies under the following conditions:

- Usage respects the operating and storage conditions specified in Section 3.3
- Proper installation of the Gripper specified in Section 3 and the following subsections.
- Usage under normal one-shift operation (40h a week)
 - Or until a 2 000 000 cycle count¹ has been reached.
- Usage respects maintenance specified in Section 7.
- Usage respects recommended payload and force specified in the **Payload and force** section.

¹Cycle count: One (1) cycle is defined as an object picking attempt, successful or not (open or closing onto an object, or closing on itself). It is calculated in the internal memory of the 2-Finger Adaptive Gripper and can be seen with the Robotiq User Interface.

During the warranty period, Robotiq will repair or replace any defective 2-Finger Adaptive Robot Gripper, as well as verify and adjust the Gripper free of charge if the equipment should need to be repaired or if the original adjustment is erroneous. If the equipment is sent back for verification during the warranty period and found to meet all published specifications, Robotiq will charge standard verification fees.

The unit is considered defective when at least one of the following conditions occurs :

- The Gripper fingers cannot close or open;
- The Gripper feedback necessary for the robot program is not accessible.

Parts that come into contact with the work piece and wearing parts such as the finger and palm pads are not covered by the warranty.

Caution

The warranty will become null and void if the :

- Unit has been tampered with, repaired or worked on by unauthorized individuals.
- Warranty sticker has been removed.
- Screws, other than as explained in this guide, have been removed.
- Unit has been opened other than as explained in this guide.
- Unit serial number has been altered, erased, or removed.
- Unit has been misused, neglected, or damaged by accident.

This warranty is in lieu of all other warranties expressed, implied, or statutory, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose. In no event shall Robotiq be liable for special, incidental, or consequential damages.



Robotiq shall not be liable for damages resulting from the use of the Robotiq 2-Finger Adaptive Robot Gripper, nor shall Robotiq be responsible for any failure in the performance of other items to which the 2-Finger Adaptive Robot Gripper is connected or the operation of any system of which the Gripper may be a part.



Exclusions

This warranty excludes failure resulting from: improper use or installation, normal wear and tear, accident, abuse, neglect, fire, water, lightning or other acts of nature, causes external to the Gripper or other factors beyond Robotiq's control.

Robotiq reserves the right to make changes in the design or construction of any of its products at any time without incurring any obligation to make any changes whatsoever on units already purchased.

Patent

This product incorporates technology developed by Lionel Birglen, professor at Polytechnique Montreal, and is used under license of Polyvalor LP.



11. Contact

www.robotiq.com

Contact Us

Phone

1-888-ROBOTIQ (762-6847)
(01) 418-380-2788 Outside US and Canada

Fax

1-418-800-0046

Technical support and engineering

option 3

Sales

option 2

Head office

Robotiq:
966, chemin Olivier
Suite 500
St-Nicolas, Québec
G7A 2N1
Canada



Where automation Pros come to share their know-how and get answers.

dof.robotiq.com



12. Harmonized Standards, Declarations and Certificates

12.1. Translation of original EC declaration of incorporation



Translation of original declaration of incorporation

In terms of the Directive 2006/42/EC, Annex II, part B of the European Parliament and of the Council on machinery,

We, the manufacturer, Robotiq Inc.
966, Chemin Olivier, suite 325
Lévis, Québec, Canada, G7A 2N1

Hereby declare that the following product:

Robotiq 2-Finger Adaptive Gripper – 85 / 140
Identified C-3001 and over

Meets the applicable basic requirements of the **Machinery Directive 2006/42/EC**

The incomplete machine may not be put into operation until conformity of the machine into which the incomplete machine is to be installed with the provisions of the Machinery Directive is confirmed. Compliance with all essential requirements of Machinery Directive relies on the specific robot application and overall risk assessment.

The manufacturer agrees to forward on demand of national authorities the relevant technical documents specified by Annex VII part B within the required time.

Additionally the product declares in conformity with the following directives, according to which the product is CE marked:

2004/108/EC Electromagnetic Compatibility Directive (EMC)

2011/65/EU Restriction of the use of certain hazardous substances (ROHS)

Person responsible for documentation: Mr. Étienne Samson, address: see manufacturer address

Lévis, September 2015
(place and date of emission)

A handwritten signature in black ink, appearing to read 'JP Jobin'.

Jean-Philippe Jobin
Chief Technical Officer
Robotiq Inc.



12.2. Applied standards

This section describes all applied harmonized standards for the design and production of the Robotiq 2-Finger Adaptive Gripper. Standards are applied were applicable, some points may not be applied if not applicable to this specific product. Conformity is not enforced by any laws, it is self-applied and the aim is to define normal safety and performance requirements for similar products.

Caution

Conformity of the product is only met if all instructions of the following manual are followed. Among others; installation, safety measure and normal usage must be respected.

The following standards have been applied:

NF EN ISO 14539	2000	Manipulating industrial robots – Object handling with grasp-type grippers – Vocabulary and presentation of characteristics
NF EN ISO 12100	2010	Safety of machinery – General principles for design – Risk assessment and risk reduction
NF EN IEC 60204-1	2006	Safety of machinery – Electrical equipment of machines – Part 1: General requirements

